



**Generation Interconnection
System Impact Study Report
for**

Queue Project AF1-238

SHERMAN AVE. 69 KV

51 MW Capacity / 127.5 MW Energy

August 2020

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is AEC.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

3 General

The Interconnection Customer (IC), has proposed a Storage generating facility located in Cumberland County, New Jersey. The installed facilities will have a total capability of 127.5 MW with 51 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is May 31, 2023. This study does not imply a TO commitment to this in-service date.

Queue Number	AF1-238
Project Name	SHERMAN AVE. 69 KV
State	New Jersey
County	Cumberland
Transmission Owner	AEC
MFO	127.5
MWE	127.5
MWC	51
Fuel	Storage
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF1-238 will interconnect with the Vineland transmission system at the Sherman Ave-West Vineland 69 kV line.

5 Cost Summary

The AF1-238 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$0
Allocation towards System Network Upgrade Costs*	\$434,733,997
Total Costs	\$434,733,997

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

There is no Atlantic City Electric attachment facility or direct connection work scope. The Interconnection Customer is responsible for contacting the Vineland Municipal Electric Utility directly for attachment facilities work scope and single line drawing.

7 Schedule

The Interconnection Customer is responsible for contacting the Vineland Municipal Electric Utility directly for schedule to construct the physical interconnection for the AF1-239 project.

8 Transmission Owner Analysis

None.

9 Interconnection Customer Requirements

The Interconnection Customer is responsible for contacting the Vineland Municipal Electric Utility for any additional Interconnection Customer requirements.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

11 Summer Peak Analysis

The Queue Project AF1-238 was evaluated as a 127.5 MW (Capacity 51.0 MW) injection at the Sherman Ave 138 kV substation in the AEC area. Project AF1-238 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-238 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
59580241	228716	S121TAP	69.0	AE	228714	CNTRL N	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	64.69	109.81	AC	50.98

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41042413	228401	MCKLTON	230.0	AE	213559	DELCOTAP	230.0	PECO	1	PS_P7-1_V2274+P2242_LT	tower	725.0	98.71	101.59	AC	17.47
41042049	228714	CNTRL N	69.0	AE	228504	SHLDLY T	69.0	AE	1	AE_P7-1 AE7TOWER	tower	143.0	82.9	102.29	AC	33.06

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42777536	206294	28LARRABEE	230.0	JCP&L	206318	28VANHISVL	230.0	JCP&L	1	JC-P7-1-JCC-230-11A	tower	869.0	106.39	107.59	AC	10.6
42777150	206297	28MANITOU	230.0	JCP&L	206319	28WHITINGS	230.0	JCP&L	1	JC-P7-1-JCC-230-12	tower	817.0	160.62	161.53	AC	8.57
41041986	206302	28OYSTERC	230.0	JCP&L	227955	CEDAR	230.0	AE	1	JC-P7-1-JCC-230-10A	tower	564.0	144.93	146.64	AC	9.67
41648392	206302	28OYSTERC	230.0	JCP&L	227955	CEDAR	230.0	AE	1	JC-P2-3-JCC-230-15E	breaker	564.0	126.55	128.75	AC	12.65
41648393	206302	28OYSTERC	230.0	JCP&L	227955	CEDAR	230.0	AE	1	JC-P2-3-JCC-230-17A	breaker	564.0	126.4	128.6	AC	12.65
49873384	206302	28OYSTERC	230.0	JCP&L	227955	CEDAR	230.0	AE	1	JC-P7-1-JCC-230-7A	tower	564.0	142.72	144.71	AC	11.42
42777172	206318	28VANHISVL	230.0	JCP&L	206294	28LARRABEE	230.0	JCP&L	1	JC-P7-1-JCC-230-12	tower	869.0	146.18	147.22	AC	10.98
42777173	206318	28VANHISVL	230.0	JCP&L	206294	28LARRABEE	230.0	JCP&L	1	JC-P7-1-JCC-230-11A	tower	869.0	126.4	127.4	AC	10.6

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS T LOAD %	POST PROJECTIONS T LOAD %	AC/D C	MW IMPACT
42020349	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	PECO_P2-2_CHI230B1/* \$ DELCO \$ CHI230B1 \$ B	bus	1180.0	135.17	137.11	AC	26.79
42251090	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	PECO_P4_CHICH045/* \$ DELCO \$ CHICH045 \$ STBK	breaker	1180.0	135.18	137.12	AC	26.79
42251091	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	PECO_P4_PEACH215/* \$ CHESCO \$ PEACH215 \$ STBK	breaker	1180.0	125.79	127.99	AC	30.44
42020389	214206	RICHRE29	230.0	PECO	213922	RICHMOND	230.0	PECO	1	PECO_P2-2_CHI230B1/* \$ DELCO \$ CHI230B1 \$ B	bus	1336.0	133.65	135.48	AC	28.46
42250938	214206	RICHRE29	230.0	PECO	213922	RICHMOND	230.0	PECO	1	PECO_P4_PEACH215/* \$ CHESCO \$ PEACH215 \$ STBK	breaker	1336.0	125.18	127.27	AC	32.43
42250940	214206	RICHRE29	230.0	PECO	213922	RICHMOND	230.0	PECO	1	PECO_P4_CHICH045/* \$ DELCO \$ CHICH045 \$ STBK	breaker	1336.0	133.66	135.49	AC	28.46
42250997	219100	NEWFRDM	230.0	PSE&G	219704	HILLTOP_3	230.0	PSE&G	1	PECO_P4_PEACH215/* \$ CHESCO \$ PEACH215 \$ STBK	breaker	886.0	106.4	107.35	AC	9.82
42777235	219100	NEWFRDM	230.0	PSE&G	219704	HILLTOP_3	230.0	PSE&G	1	PS_P7-1_V2274+P2242_LT	tower	886.0	113.98	115.42	AC	13.94
42777237	219100	NEWFRDM	230.0	PSE&G	219704	HILLTOP_3	230.0	PSE&G	1	PS_P7-1_W2275+O2241_LT	tower	886.0	108.85	110.29	AC	14.0
42251364	219108	CUTHBERT	230.0	PSE&G	219125	CAMDEN	230.0	PSE&G	1	PS_P2-3_CAMD_5-6_LT	breaker	771.0	113.47	115.53	AC	15.59
42251258	219110	GLOUCSTR	230.0	PSE&G	219125	CAMDEN	230.0	PSE&G	1	PS_P2-3_CUTB_1-4_LT	breaker	771.0	121.13	123.24	AC	15.74
42251259	219110	GLOUCSTR	230.0	PSE&G	219125	CAMDEN	230.0	PSE&G	1	PS_P2-3_CUTB_3-4_LT	breaker	771.0	119.45	121.53	AC	15.52
42251260	219110	GLOUCSTR	230.0	PSE&G	219125	CAMDEN	230.0	PSE&G	1	PS_P2-3_CUTB_1-2_LT	breaker	771.0	116.78	118.82	AC	15.2
42251261	219110	GLOUCSTR	230.0	PSE&G	219125	CAMDEN	230.0	PSE&G	1	PS_P2-3_CUTB_2-3_LT	breaker	771.0	114.92	116.92	AC	14.95
41648510	219125	CAMDEN	230.0	PSE&G	214206	RICHRE29	230.0	PECO	1	PECO_P4_PEACH215/* \$ CHESCO \$ PEACH215 \$ STBK	breaker	1336.0	125.18	127.27	AC	32.43
41648512	219125	CAMDEN	230.0	PSE&G	214206	RICHRE29	230.0	PECO	1	PECO_P4_CHICH045/* \$ DELCO \$ CHICH045 \$ STBK	breaker	1336.0	133.66	135.49	AC	28.46
41866184	219125	CAMDEN	230.0	PSE&G	214206	RICHRE29	230.0	PECO	1	PECO_P2-2_CHI230B1/* \$ DELCO \$ CHI230B1 \$ B	bus	1336.0	133.65	135.48	AC	28.46
42777311	219702	HILLTOP_1	230.0	PSE&G	219101	BEAVERBK_1	230.0	PSE&G	1	PS_P7-1_V2274+P2242_LT	tower	886.0	101.53	102.88	AC	13.11
42251517	219754	CUTHBERT_3	230.0	PSE&G	219125	CAMDEN	230.0	PSE&G	1	PS_P2-3_CAMD_5-6_LT	breaker	792.0	100.33	102.16	AC	14.17
42777679	219756	BEAVERBK_2	230.0	PSE&G	219110	GLOUCSTR	230.0	PSE&G	1	PS_P7-1_V2274+P2242_LT	tower	740.0	111.13	112.76	AC	13.11
42777680	219756	BEAVERBK_2	230.0	PSE&G	219110	GLOUCSTR	230.0	PSE&G	1	AE_P7-1_W2275_O2241	tower	740.0	108.62	110.25	AC	13.16
41041980	227900	CARDIFF	230.0	AE	219100	NEWFRDM	230.0	PSE&G	1	JC-P7-1-JCC-230-13	tower	692.0	263.11	265.21	DC	17.07
41041981	227900	CARDIFF	230.0	AE	219100	NEWFRDM	230.0	PSE&G	1	AE_P7-1_AE13TOWER	tower	692.0	231.72	233.36	AC	12.89
41648246	227900	CARDIFF	230.0	AE	219100	NEWFRDM	230.0	PSE&G	1	AE_P4-2 AE46	breaker	692.0	212.47	216.89	AC	31.31
41648247	227900	CARDIFF	230.0	AE	219100	NEWFRDM	230.0	PSE&G	1	AE_P4-2 AE45	breaker	692.0	197.81	199.77	AC	15.93

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS T LOADING %	POST PROJECTIONS T LOADING %	AC/D C	MW IMPACT
42250807	227900	CARDIFF	230.0	AE	227955	CEDAR	230.0	AE	1	AE_P4-2 AE7	breaker	805.0	163.26	164.71	AC	13.85
42250808	227900	CARDIFF	230.0	AE	227955	CEDAR	230.0	AE	1	AE_P4-2 AE6	breaker	805.0	163.25	164.7	AC	13.86
41648231	227905	SCULL#1	138.0	AE	227903	MILL #1	138.0	AE	1	AE_P4-2 AE29	breaker	306.0	127.37	131.7	AC	15.58
41648241	227906	SCULL#2	138.0	AE	227904	MILL #2	138.0	AE	1	AE_P4-2 AE28	breaker	306.0	128.7	133.61	AC	17.63
51072462	227913	CARDIFF	138.0	AE	227902	LEWIS #1	138.0	AE	1	AE_P4-2 AE9	breaker	478.0	121.08	123.42	AC	11.14
51072561	227934	CARDIFF2	138.0	AE	227945	LEWIS #2	138.0	AE	1	AE_P4-3 AE2	breaker	478.0	101.89	104.05	AC	10.32
51073203	227934	CARDIFF2	138.0	AE	227945	LEWIS #2	138.0	AE	1	AE_P7-1 AE3TOWER	tower	478.0	133.12	134.91	AC	8.54
41648251	227955	CEDAR	230.0	AE	206302	28OYSTER C	230.0	JCP&L	1	AE_P4-2 AE7	breaker	564.0	218.94	221.39	AC	16.13
41648252	227955	CEDAR	230.0	AE	206302	28OYSTER C	230.0	JCP&L	1	AE_P4-2 AE6	breaker	564.0	218.93	221.38	AC	16.13
41648226	228110	BLE	138.0	AE	227905	SCULL#1	138.0	AE	1	AE_P4-2 AE29	breaker	306.0	134.29	138.62	AC	15.58
41648236	228110	BLE	138.0	AE	227906	SCULL#2	138.0	AE	1	AE_P4-2 AE28	breaker	306.0	133.64	138.55	AC	17.63
42777361	228311	CHAMBERS	230.0	AE	228312	PEDRKTWN	230.0	AE	1	JC-P7-1-JCC-230-13	tower	552.0	114.11	117.01	AC	18.82
42777290	228312	PEDRKTWN	230.0	AE	228313	BRIDGPRT	230.0	AE	1	JC-P7-1-JCC-230-13	tower	552.0	130.58	133.48	AC	18.75
42777400	228313	BRIDGPRT	230.0	AE	228401	MCKLTON	230.0	AE	1	JC-P7-1-JCC-230-13	tower	804.0	114.74	116.73	AC	18.6
88617735	228502	MNOTLA 1	138.0	AE	228500	LANDIS	138.0	AE	1	AE_P4-2 AE46	breaker	478.0	146.12	147.94	AC	10.16
41649105	940000	AE1-240 TAP	69.0	AE	228226	SHRMAN# 2	69.0	AE	1	AE_P4-2 AE46	breaker	93.0	105.37	122.96	AC	17.19

11.4 Steady-State Voltage Requirements

To be determined

11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPAC T
42558357	200051	ROCKSPGS	500.0	PJM	200065	PCHBTM2S	500.0	PJM	1	PECO_P1-2_5038/* \$ CHESCO \$ 5038 \$ L	operation	2905.0	122.09	124.18	AC	44.13
42558154	206295	28LEISUR D	230.0	JCP&L	206323	28LAKEWOD	230.0	JCP&L	1	JC-P1-2-JCC-230-020	operation	869.0	116.04	116.9	AC	8.96
42558060	206296	28LEISUR U	230.0	JCP&L	206323	28LAKEWOD	230.0	JCP&L	1	JC-P1-2-JCC-230-019	operation	817.0	131.96	132.88	AC	8.92
42558032	206297	28MANITO U	230.0	JCP&L	206296	28LEISUR U	230.0	JCP&L	1	JC-P1-2-JCC-230-019	operation	817.0	144.2	145.13	AC	9.04
42558038	206297	28MANITO U	230.0	JCP&L	206295	28LEISUR D	230.0	JCP&L	1	JC-P1-2-JCC-230-020	operation	817.0	143.72	144.65	AC	9.04
41332322	206302	28OYSTER C	230.0	JCP&L	227955	CEDAR	230.0	AE	1	Base Case	operation	464.0	136.04	138.67	AC	12.44
41332323	206302	28OYSTER C	230.0	JCP&L	227955	CEDAR	230.0	AE	1	JC-P1-2-JCC-230-031	operation	564.0	125.79	127.99	AC	12.65
42557940	206302	28OYSTER C	230.0	JCP&L	206297	28MANITO U	230.0	JCP&L	1	JC-P1-2-JCC-230-022	operation	817.0	232.06	233.25	AC	11.36
42557961	206302	28OYSTER C	230.0	JCP&L	206297	28MANITO U	230.0	JCP&L	2	JC-P1-2-JCC-230-021	operation	869.0	218.2	219.32	AC	11.36
42558345	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	Base Case	operation	760.0	153.52	156.19	AC	23.69
42558347	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	PECO_P1-2_5014/* \$ CHESCO \$ 5014 \$ L	operation	1180.0	123.03	125.23	AC	30.44
42558297	214206	RICHRE29	230.0	PECO	213922	RICHMOND	230.0	PECO	1	PECO_P1-2_5014/* \$ CHESCO \$ 5014 \$ L	operation	1336.0	123.07	125.15	AC	32.43
42558298	214206	RICHRE29	230.0	PECO	213922	RICHMOND	230.0	PECO	1	Base Case	operation	1075.0	124.24	126.27	AC	25.33
42558350	219100	NEWFRDM	230.0	PSE&G	219704	HILLTOP_3	230.0	PSE&G	1	PECO_P1-2_5014/* \$ CHESCO \$ 5014 \$ L	operation	886.0	105.48	106.43	AC	9.82
42558799	219108	CUTHBERT	230.0	PSE&G	219125	CAMDEN	230.0	PSE&G	1	PS_P1-2_Y-2304	operation	771.0	112.43	114.47	AC	15.24
42558513	219110	GLOUCSTR	230.0	PSE&G	219755	CUTHBERT_4	230.0	PSE&G	1	PS_P1-2_C-2308	operation	758.0	131.22	133.39	AC	15.98
42558517	219110	GLOUCSTR	230.0	PSE&G	219755	CUTHBERT_4	230.0	PSE&G	1	Base Case	operation	550.0	117.52	119.47	AC	10.37
42558579	219110	GLOUCSTR	230.0	PSE&G	219125	CAMDEN	230.0	PSE&G	1	Base Case	operation	500.0	126.46	128.65	AC	10.59
42558580	219110	GLOUCSTR	230.0	PSE&G	219125	CAMDEN	230.0	PSE&G	1	PS_P1-2_D-2282	operation	771.0	111.12	113.03	AC	14.23

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPAC T
42558659	219110	GLOUCSTR	230.0	PSE&G	219753	CUTHBERT_2	230.0	PSE&G	1	PS_P1-2_D-2282	operation	771.0	122.5	124.53	AC	15.15
42558661	219110	GLOUCSTR	230.0	PSE&G	219753	CUTHBERT_2	230.0	PSE&G	1	Base Case	operation	500.0	113.33	115.21	AC	9.06
41332539	219125	CAMDEN	230.0	PSE&G	214206	RICHRE29	230.0	PECO	1	PECO_P1-2_5014/*\$CHESCO\$5014\$ L	operation	1336.0	123.07	125.15	AC	32.43
41332540	219125	CAMDEN	230.0	PSE&G	214206	RICHRE29	230.0	PECO	1	Base Case	operation	1075.0	124.24	126.27	AC	25.33
42558926	219754	CUTHBERT_3	230.0	PSE&G	219125	CAMDEN	230.0	PSE&G	1	PS_P1-2_Y-2304	operation	792.0	99.41	101.22	AC	13.85
41332128	227900	CARDIFF	230.0	AE	219100	NEWFRDM	230.0	PSE&G	1	JC-P1-2-JCC-230-002T	operation	692.0	192.83	194.99	AC	17.61
41332132	227900	CARDIFF	230.0	AE	219100	NEWFRDM	230.0	PSE&G	1	Base Case	operation	650.0	163.29	164.9	AC	12.26
42558055	227900	CARDIFF	230.0	AE	227955	CEDAR	230.0	AE	1	PS_P1-2_2310	operation	805.0	162.41	163.88	AC	14.02
42558056	227900	CARDIFF	230.0	AE	227955	CEDAR	230.0	AE	1	Base Case	operation	650.0	155.05	156.42	AC	10.46
41332140	227903	MILL #1	138.0	AE	227902	LEWIS #1	138.0	AE	1	AE_P1-2 BLE-ML-LEW2	operation	306.0	104.37	109.39	AC	17.99
41332112	227905	SCULL#1	138.0	AE	227903	MILL #1	138.0	AE	1	AE_P1-2 BLE-SC-ML2	operation	306.0	118.53	124.03	AC	19.78
41332102	227906	SCULL#2	138.0	AE	227904	MILL #2	138.0	AE	1	AE_P1-2 BLE-SC-ML1	operation	306.0	119.95	125.46	AC	19.81
51073056	227913	CARDIFF	138.0	AE	227902	LEWIS #1	138.0	AE	1	AE_P1-3 CARD 7 XFR	operation	478.0	103.58	105.77	AC	10.48
51073062	227934	CARDIFF2	138.0	AE	227945	LEWIS #2	138.0	AE	1	AE_P1-3 CARD 6 XFR	operation	478.0	101.95	104.11	AC	10.32
41332122	227955	CEDAR	230.0	AE	206302	28OYSTER C	230.0	JCP&L	1	PS_P1-2_2310	operation	564.0	218.23	220.7	AC	16.23
41332123	227955	CEDAR	230.0	AE	206302	28OYSTER C	230.0	JCP&L	1	Base Case	operation	464.0	206.63	208.9	AC	12.44
41332082	228110	BLE	138.0	AE	227905	SCULL#1	138.0	AE	1	AE_P1-2 BLE-SC-ML2	operation	306.0	125.44	130.94	AC	19.78
41332084	228110	BLE	138.0	AE	227905	SCULL#1	138.0	AE	1	Base Case	operation	218.0	101.44	105.79	AC	11.34
41332092	228110	BLE	138.0	AE	227906	SCULL#2	138.0	AE	1	AE_P1-2 BLE-SC-ML1	operation	306.0	124.88	130.4	AC	19.81
41332094	228110	BLE	138.0	AE	227906	SCULL#2	138.0	AE	1	Base Case	operation	218.0	100.13	104.49	AC	11.37
41332314	228111	MDLE TP	138.0	AE	228107	CORSON 2	138.0	AE	1	AE_P1-2 ENG-MERION	operation	315.0	99.04	105.3	AC	19.89
41333144	228210	UNION	138.0	AE	228108	CORSON 3	138.0	AE	1	AE_P1-2 CUMB-DENN	operation	400.0	96.94	102.77	AC	23.99
59580377	228256	SHRMAN#1	69.0	AE	945730	AF1-238 TAP	69.0	AE	1	AE_P1-2 CNT-V-BUT	operation	132.0	21.97	91.94	AC	95.59
42558303	228311	CHAMBERS	230.0	AE	228312	PEDRKTWN	230.0	AE	1	AE_P1-2 ORCHAR D XF	operation	552.0	107.69	113.09	AC	32.53

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPAC T
42558208	228312	PEDRKTWN	230.0	AE	228313	BRIDGPRT	230.0	AE	1	AE_P1-2 ORCHAR D XF	operation	552.0	123.21	128.72	AC	32.46
42558388	228313	BRIDGPRT	230.0	AE	228401	MCKLTON	230.0	AE	1	AE_P1-2 ORCHAR D XF	operation	804.0	111.53	115.56	AC	32.31
42558389	228313	BRIDGPRT	230.0	AE	228401	MCKLTON	230.0	AE	1	Base Case	operation	650.0	110.1	112.37	AC	17.29
59580341	228705	VINE G10	69.0	AE	228716	S121TAP	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	107.0	12.12	125.6	AC	127.48
59580344	228706	WEST VLD	69.0	AE	228705	VINE G10	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	107.0	12.13	125.61	AC	127.48
51072920	228716	S121TAP	69.0	AE	228714	CNTRL N	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	107.0	64.69	179.1	AC	127.45
51072810	940000	AE1-240 TAP	69.0	AE	228226	SHRMAN#2	69.0	AE	1	228228 SO MVLL 69.0 939500 AE1-179 TAP 69.0 1	operation	93.0	100.24	119.0	AC	18.03
59580293	945730	AF1-238 TAP	69.0	AE	228256	SHRMAN#1	69.0	AE	1	AE_P1-2 NCENT-S121	operation	132.0	52.44	144.44	AC	127.45
59580298	945730	AF1-238 TAP	69.0	AE	228256	SHRMAN#1	69.0	AE	1	Base Case	operation	119.0	15.84	89.05	AC	91.53
59580486	945730	AF1-238 TAP	69.0	AE	228706	WEST VLD	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	132.0	0.04	93.27	AC	127.5

11.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number											
41866184,41648510,41648512	13	CAMDEN 230.0 kV - RICHRE29 230.0 kV Ckt 1	ProjectId : n6850 Description : Replace down drops and disconnect switch at Camden 230kV Type : FAC Total Cost : \$200,000 Time Estimate : 9 Months Ratings : 1593/1770 Notes : PSEG Reinforcement 1. Per PJM cost allocation rules, AF1-238 presently does not receive cost allocation for this upgrade.	\$51,619,729	\$51,619,729	N6850 N6851 N6583											
			ProjectId : n6851 Description : Wreck and Rebuild (PSEG Portion). Upgrade Camden 230kV terminal equipment to achieve a SER of 1891MVA. Type : FAC Total Cost : \$51,619,729 Time Estimate : 45.0 Months Ratings : 1593/1891 Notes : PSEG Reinforcement 2														
			Queue				MW	Cost %	Cost \$	AF1-238	28.46	100%	\$51,619,729				
Queue	MW	Cost %	Cost \$														
AF1-238	28.46	100%	\$51,619,729														
ProjectId : n6583 Description : Reconductor PECO portion of line using 'Pecos' conductor on this bifurcated line (conductors run on parallel towers and remain tied together as they are today, which remains a 6-wire line) and also rebuild the Richmond 230 kV substation ring bus to increase bus rating (Does not include permitting or ROW acquisition). Cost & Ratings for PECO portion of tie line only. Type : FAC Total Cost : \$51,619,729 Time Estimate : 45.0 Months Ratings : 1593/1891 Notes : PECO Reinforcement 6	<table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AE2-251</td><td>44.17</td><td>36.83%</td><td>\$ 5,765,124.68</td></tr><tr><td>AF1-126</td><td>47.3</td><td>39.44%</td><td>\$ 6,173,396.29</td></tr><tr><td>AF1-238</td><td>28.46</td><td>23.73%</td><td>\$ 3,714,479.03</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-251	44.17	36.83%	\$ 5,765,124.68	AF1-126	47.3	39.44%	\$ 6,173,396.29	AF1-238	28.46	23.73%	\$ 3,714,479.03
Queue	MW	Cost %	Cost \$														
AE2-251	44.17	36.83%	\$ 5,765,124.68														
AF1-126	47.3	39.44%	\$ 6,173,396.29														
AF1-238	28.46	23.73%	\$ 3,714,479.03														
42777680,42777679	16	BEAVERBK_2 230.0 kV - GLOUCSTR 230.0 kV Ckt 1	Base case ratings discrepancy, Transmission Owner confirmed this facility is not an overload.	\$0	\$0	NV#RI											

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																												
42251260,42251261,42251259,42251258	12	GLOUCSTR 230.0 kV - CAMDEN 230.0 kV Ckt 1	ProjectId : n6408 Description : Expand Beaver Brook 230kV to accommodate new cable . Construct new U/G Cable from Beaver Brook 230kV to Camden 230kV. Type : CON Total Cost : \$269,682,900 Time Estimate : 45.0 Months Ratings : 550.0/750.0/ Notes : The UG cable will relive the overloads for Gloucester - Camden, Gloucester - Cuthbert, and Cuthbert - Camden <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-021</td><td>43.38</td><td>14.51%</td><td>\$39,134,815</td></tr><tr><td>AE2-022</td><td>31.24</td><td>10.45%</td><td>\$28,180,134</td></tr><tr><td>AE2-222</td><td>36.6</td><td>12.24%</td><td>\$33,018,309</td></tr><tr><td>AE2-251</td><td>136.42</td><td>45.64%</td><td>\$123,069,882</td></tr><tr><td>AF1-208</td><td>5.8</td><td>1.94%</td><td>\$5,232,410</td></tr><tr><td>AF1-238</td><td>45.5</td><td>15.22%</td><td>\$41,047,351</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-021	43.38	14.51%	\$39,134,815	AE2-022	31.24	10.45%	\$28,180,134	AE2-222	36.6	12.24%	\$33,018,309	AE2-251	136.42	45.64%	\$123,069,882	AF1-208	5.8	1.94%	\$5,232,410	AF1-238	45.5	15.22%	\$41,047,351	\$269,682,900	\$41,047,351	N6408
Queue	MW	Cost %	Cost \$																															
AE2-021	43.38	14.51%	\$39,134,815																															
AE2-022	31.24	10.45%	\$28,180,134																															
AE2-222	36.6	12.24%	\$33,018,309																															
AE2-251	136.42	45.64%	\$123,069,882																															
AF1-208	5.8	1.94%	\$5,232,410																															
AF1-238	45.5	15.22%	\$41,047,351																															
41042413	2	MCKLTON 230.0 kV - DELCOTAP 230.0 kV Ckt 1	ProjectId : N6564 Description : To mitigate the (ACE)Mickleton to Delco Tap 230 kV line (from bus 228401 to bus 213559 ckt 1) overload, it will require various terminal reinforcements at Mickleton. Type : FAC Total Cost : \$200,000 Time Estimate : 12-24 Months Ratings : 796/796/796 Notes : Per PJM cost allocation rules, AF1-238 presently does not receive cost allocation for this upgrade.	\$200,000	\$0	N6564																												
42251517	11	CUTHBERT 230.0 kV - CAMDEN 230.0 kV Ckt 1	ProjectId: n6408 Description : Expand Beaver Brook 230kV to accommodate new cable . Construct new U/G Cable from Beaver Brook 230kV to Camden 230kV. Type : CON Total Cost : \$269,682,900 Time Estimate : 45.0 Months Ratings : 550.0/750.0/ Notes : The UG cable will relive the overloads for Gloucester - Camden, Gloucester - Cuthbert, and Cuthbert - Camden	\$269,682,900	\$0	N6408																												
42251364	15	CUTHBERT_3 230.0 kV - CAMDEN 230.0 kV Ckt 1	<table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-021</td><td>43.38</td><td>14.51%</td><td>\$39,134,815</td></tr><tr><td>AE2-022</td><td>31.24</td><td>10.45%</td><td>\$28,180,134</td></tr><tr><td>AE2-222</td><td>36.6</td><td>12.24%</td><td>\$33,018,309</td></tr><tr><td>AE2-251</td><td>136.42</td><td>45.64%</td><td>\$123,069,882</td></tr><tr><td>AF1-208</td><td>5.8</td><td>1.94%</td><td>\$5,232,410</td></tr><tr><td>AF1-238</td><td>45.5</td><td>15.22%</td><td>\$41,047,351</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-021	43.38	14.51%	\$39,134,815	AE2-022	31.24	10.45%	\$28,180,134	AE2-222	36.6	12.24%	\$33,018,309	AE2-251	136.42	45.64%	\$123,069,882	AF1-208	5.8	1.94%	\$5,232,410	AF1-238	45.5	15.22%	\$41,047,351			
Queue	MW	Cost %	Cost \$																															
AE2-021	43.38	14.51%	\$39,134,815																															
AE2-022	31.24	10.45%	\$28,180,134																															
AE2-222	36.6	12.24%	\$33,018,309																															
AE2-251	136.42	45.64%	\$123,069,882																															
AF1-208	5.8	1.94%	\$5,232,410																															
AF1-238	45.5	15.22%	\$41,047,351																															

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																																				
42250997,42777237,42777235	10	NEWFRDM 230.0 kV - HILLTOP_3 230.0 kV Ckt 1	ProjectId : n6455 Description : Reconductor to 1080MVA SER Type : FAC Total Cost : \$29,523,000 Time Estimate : 33.0 Months Ratings : 934.0/1080.0/ <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-021</td><td>25.17</td><td>10.35%</td><td>\$3,055,359</td></tr><tr><td>AE2-022</td><td>37.1</td><td>15.25%</td><td>\$4,503,529</td></tr><tr><td>AE2-222</td><td>31.4</td><td>12.91%</td><td>\$3,811,612</td></tr><tr><td>AE2-251</td><td>126.0</td><td>51.81%</td><td>\$15,295,004</td></tr><tr><td>AE2-257</td><td>9.6</td><td>3.95%</td><td>\$1,165,334</td></tr><tr><td>AF1-238</td><td>13.94</td><td>5.73%</td><td>\$1,692,162</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-021	25.17	10.35%	\$3,055,359	AE2-022	37.1	15.25%	\$4,503,529	AE2-222	31.4	12.91%	\$3,811,612	AE2-251	126.0	51.81%	\$15,295,004	AE2-257	9.6	3.95%	\$1,165,334	AF1-238	13.94	5.73%	\$1,692,162	\$29,523,000	\$1,692,162	N6455								
Queue	MW	Cost %	Cost \$																																							
AE2-021	25.17	10.35%	\$3,055,359																																							
AE2-022	37.1	15.25%	\$4,503,529																																							
AE2-222	31.4	12.91%	\$3,811,612																																							
AE2-251	126.0	51.81%	\$15,295,004																																							
AE2-257	9.6	3.95%	\$1,165,334																																							
AF1-238	13.94	5.73%	\$1,692,162																																							
41648252,41648251	23	CEDAR 230.0 kV - 28OYSTER C 230.0 kV Ckt 1	ProjectId : N6512 Description : To mitigate the (ACE) Cedar Oyster Creek 230 kV line (from bus 227955 to bus 206302 ckt 1) overload, it will require increasing the emergency rating of the Cedar to Oyster Creek 230 kV line by rebuilding the circuit. The rebuild will include the installation of new poles, foundations, insulators, and conductor. In addition, various terminal reinforcements are required at Cedar. Type : FAC Total Cost : \$63,000,000 Time Estimate : 36-60 Months Ratings : 1195.0/1195.0/1195.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-000</td><td>184.75</td><td>17.72%</td><td>\$11,163,602</td></tr><tr><td>AE2-021</td><td>201.66</td><td>19.34%</td><td>\$12,185,397</td></tr><tr><td>AE2-022</td><td>117.64</td><td>11.28%</td><td>\$7,108,269</td></tr><tr><td>AE2-222</td><td>85.7</td><td>8.22%</td><td>\$5,178,461</td></tr><tr><td>AE2-251</td><td>400.1</td><td>38.37%</td><td>\$24,176,224</td></tr><tr><td>AE2-314</td><td>24.0</td><td>2.30%</td><td>\$1,450,211</td></tr><tr><td>AF1-126</td><td>12.63</td><td>1.21%</td><td>\$763,173</td></tr><tr><td>AF1-238</td><td>16.13</td><td>1.55%</td><td>\$974,663</td></tr></table> ProjectId : N6513 Description : To mitigate the (ACE) Cedar Oyster Creek 230 kV line (from bus 227955 to bus 206302 ckt 1) overload, it will require various terminal reinforcements at Cedar. Type : FAC Total Cost : \$700,000 Time Estimate : 24-48 Months Ratings : 1283.0/1509.0/1509.0 Notes : Per PJM cost allocation rules, AF1-238 presently does not receive cost allocation for this upgrade.	Queue	MW	Cost %	Cost \$	AE2-000	184.75	17.72%	\$11,163,602	AE2-021	201.66	19.34%	\$12,185,397	AE2-022	117.64	11.28%	\$7,108,269	AE2-222	85.7	8.22%	\$5,178,461	AE2-251	400.1	38.37%	\$24,176,224	AE2-314	24.0	2.30%	\$1,450,211	AF1-126	12.63	1.21%	\$763,173	AF1-238	16.13	1.55%	\$974,663	\$63,700,000	\$974,633	N6512 N6513
Queue	MW	Cost %	Cost \$																																							
AE2-000	184.75	17.72%	\$11,163,602																																							
AE2-021	201.66	19.34%	\$12,185,397																																							
AE2-022	117.64	11.28%	\$7,108,269																																							
AE2-222	85.7	8.22%	\$5,178,461																																							
AE2-251	400.1	38.37%	\$24,176,224																																							
AE2-314	24.0	2.30%	\$1,450,211																																							
AF1-126	12.63	1.21%	\$763,173																																							
AF1-238	16.13	1.55%	\$974,663																																							

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																				
41648226	24	BLE 138.0 kV - SCULL#1 138.0 kV Ckt 1	ProjectId : n6194 Description : To mitigate the (ACE) B.L. England Scull#1 138 kV line (from bus 228110 to bus 227905 ckt 1) overload, it will require increasing the emergency rating of the B.L. England to Scull#1 138 kV line by rebuilding the circuit. The rebuild will include the installation of new poles, foundations, insulators, and conductor. In addition, various terminal reinforcements are required at both B.L. England and Scull#1 Type : FAC Total Cost : \$5,900,000 Time Estimate : 36-60 Months Ratings : 549.0/621.0/621.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE1-104</td><td>231.27</td><td>88.60%</td><td>\$5,227,542</td></tr><tr><td>AE1-179</td><td>7.72</td><td>2.96%</td><td>\$174,500</td></tr><tr><td>AE1-240</td><td>6.45</td><td>2.47%</td><td>\$145,793</td></tr><tr><td>AF1-238</td><td>15.58</td><td>5.97%</td><td>\$352,165</td></tr></table>	Queue	MW	Cost %	Cost \$	AE1-104	231.27	88.60%	\$5,227,542	AE1-179	7.72	2.96%	\$174,500	AE1-240	6.45	2.47%	\$145,793	AF1-238	15.58	5.97%	\$352,165	\$5,900,000	\$352,165	N6194
Queue	MW	Cost %	Cost \$																							
AE1-104	231.27	88.60%	\$5,227,542																							
AE1-179	7.72	2.96%	\$174,500																							
AE1-240	6.45	2.47%	\$145,793																							
AF1-238	15.58	5.97%	\$352,165																							
41648231	19	SCULL#1 138.0 kV - MILL #1 138.0 kV Ckt 1	ProjectId: N6195 Description: To mitigate the (ACE) Scull#1-Mill#1 - 138 kV line (from bus 227905 to bus 227903 ckt 1) overload, it will require increasing the emergency rating of the Scull#1 to Mill#1 138 kV line by rebuilding the circuit. The rebuild will include the installation of new poles, foundations, insulators, and conductor. In addition, various terminal reinforcements at both, Mill#1 & Scull#1. Type : FAC Total Cost : \$12,800,000 Time Estimate : 36-60 Months Ratings : 549.0/621.0/621.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE1-104</td><td>224.77</td><td>88.31%</td><td>\$11,303,850</td></tr><tr><td>AE1-179</td><td>7.72</td><td>3.03%</td><td>\$388,245</td></tr><tr><td>AE1-240</td><td>6.45</td><td>2.53%</td><td>\$324,375</td></tr><tr><td>AF1-238</td><td>15.58</td><td>6.12%</td><td>\$783,530</td></tr></table>	Queue	MW	Cost %	Cost \$	AE1-104	224.77	88.31%	\$11,303,850	AE1-179	7.72	3.03%	\$388,245	AE1-240	6.45	2.53%	\$324,375	AF1-238	15.58	6.12%	\$783,530	\$12,800,000	\$783,530	N6195
Queue	MW	Cost %	Cost \$																							
AE1-104	224.77	88.31%	\$11,303,850																							
AE1-179	7.72	3.03%	\$388,245																							
AE1-240	6.45	2.53%	\$324,375																							
AF1-238	15.58	6.12%	\$783,530																							
41648236	25	BLE 138.0 kV - SCULL#2 138.0 kV Ckt 1	ProjectId : n6196 Description : To mitigate the (ACE) B.L. England Scull#2 138 kV line (from bus 228110 to bus 227906 ckt 1) overload, it will require increasing the emergency rating of the B.L. England to Scull#2 138 kV line by rebuilding the circuit. The rebuild will include the installation of new poles, foundations, insulators, and conductor. In addition, various terminal reinforcements are required at both B.L. England and Scull#2 Type : FAC Total Cost : \$6,000,000 Time Estimate : 36-60 Months Ratings : 578.0/578.0/578.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE1-104</td><td>205.79</td><td>85.76%</td><td>\$5,145,608</td></tr><tr><td>AE1-179</td><td>9.01</td><td>3.75%</td><td>\$225,288</td></tr><tr><td>AE1-240</td><td>7.53</td><td>3.14%</td><td>\$188,281</td></tr><tr><td>AF1-238</td><td>17.63</td><td>7.35%</td><td>\$440,823</td></tr></table>	Queue	MW	Cost %	Cost \$	AE1-104	205.79	85.76%	\$5,145,608	AE1-179	9.01	3.75%	\$225,288	AE1-240	7.53	3.14%	\$188,281	AF1-238	17.63	7.35%	\$440,823	\$6,000,000	\$440,823	N6196
Queue	MW	Cost %	Cost \$																							
AE1-104	205.79	85.76%	\$5,145,608																							
AE1-179	9.01	3.75%	\$225,288																							
AE1-240	7.53	3.14%	\$188,281																							
AF1-238	17.63	7.35%	\$440,823																							

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																														
42020349,4225 1091,42251090	8	RICHMOND 230.0 kV - WANEETA3 230.0 kV Ckt 1	ProjectId : N6482 Description : Rebuild the aerial portion of the Richmond-Waneeta 230 kV line with bundled Pecos conductors (Does not include permitting) Type : FAC Total Cost : \$15,855,000 Time Estimate : 84.0 Months Ratings : 1245.0/1387.0/1586.0	\$37,432,000	\$1,861,348	N6482 N6477 N6483 N6484																														
			<table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-021</td><td>23.89</td><td>5.40%</td><td>\$855,894</td></tr><tr><td>AE2-022</td><td>64.84</td><td>14.65%</td><td>\$2,322,988</td></tr><tr><td>AE2-222</td><td>56.6</td><td>12.79%</td><td>\$2,027,778</td></tr><tr><td>AE2-251</td><td>220.56</td><td>49.84%</td><td>\$7,901,884</td></tr><tr><td>AF1-126</td><td>49.87</td><td>11.27%</td><td>\$1,786,666</td></tr><tr><td>AF1-238</td><td>26.79</td><td>6.05%</td><td>\$959,791</td></tr></table>				Queue	MW	Cost %	Cost \$	AE2-021	23.89	5.40%	\$855,894	AE2-022	64.84	14.65%	\$2,322,988	AE2-222	56.6	12.79%	\$2,027,778	AE2-251	220.56	49.84%	\$7,901,884	AF1-126	49.87	11.27%	\$1,786,666	AF1-238	26.79	6.05%	\$959,791		
			Queue				MW	Cost %	Cost \$																											
			AE2-021				23.89	5.40%	\$855,894																											
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			AE2-251				220.56	49.84%	\$7,901,884																											
			AF1-126				49.87	11.27%	\$1,786,666																											
			AF1-238				26.79	6.05%	\$959,791																											
			ProjectId : N6477 Description : Add a second pair of underground cables in parallel with the existing underground portion of the Richmond - Waneeta 230 kV line (Does not include permitting) Type : FAC Total Cost : \$19,000,000 Time Estimate : 84.0 Months Ratings : 1026.0/1247.0/1337.0																																	
<table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-021</td><td>102.21</td><td>18.10%</td><td>\$3,439,646</td></tr><tr><td>AE2-022</td><td>64.84</td><td>11.48%</td><td>\$2,182,044</td></tr><tr><td>AE2-222</td><td>56.6</td><td>10.02%</td><td>\$1,904,745</td></tr><tr><td>AE2-251</td><td>220.56</td><td>39.07%</td><td>\$7,422,448</td></tr><tr><td>AF1-126</td><td>49.87</td><td>8.83%</td><td>\$1,678,262</td></tr><tr><td>AF1-222</td><td>33.04</td><td>5.85%</td><td>\$1,111,886</td></tr><tr><td>AF1-237</td><td>10.68</td><td>1.89%</td><td>\$359,411</td></tr><tr><td>AF1-238</td><td>26.79</td><td>4.75%</td><td>\$901,557</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-021	102.21	18.10%	\$3,439,646	AE2-022	64.84	11.48%	\$2,182,044	AE2-222	56.6	10.02%	\$1,904,745	AE2-251	220.56	39.07%	\$7,422,448	AF1-126	49.87	8.83%	\$1,678,262	AF1-222	33.04	5.85%	\$1,111,886	AF1-237	10.68	1.89%	\$359,411	AF1-238	26.79	4.75%	\$901,557
Queue	MW	Cost %	Cost \$																																	
AE2-021	102.21	18.10%	\$3,439,646																																	
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AF1-237	10.68	1.89%	\$359,411																																	
AF1-238	26.79	4.75%	\$901,557																																	
ProjectId : N6484 Description : Replace 7 disconnect switches on Richmond-Waneeta line Type : FAC Total Cost : \$1,448,000 Time Estimate : 36.0 Months Ratings : 1296.0/1645.0/1823.0 Notes: Per PJM Cost Allocation Rules, this reinforcement cost is driven by previous queue. This may change with future re-tools as projects withdraw.																																				
ProjectId : N6483 Description : Replace 3 circuit breakers on Richmond-Waneeta line Type : FAC Total Cost : \$1,129,000 Time Estimate : 36.0 Months Ratings : 1296.0/1557.0/1805.0 Notes: Per PJM Cost Allocation Rules, this reinforcement cost is driven by previous queue. This may change with future re-tools as projects withdraw.																																				

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																																												
42777290	27	PEDRKTWN 230.0 kV - BRIDGPRT 230.0 kV Ckt 1	Base case ratings discrepancy, Transmission Owner confirmed this facility is not an overload.	\$0	\$0	NV#RI																																												
42777400	28	BRIDGPRT 230.0 kV - MCKLTON 230.0 kV Ckt 1	ProjectId : n6791 Description : To mitigate the (ACE) Bridgeport to Mickleton 230 kV line (from bus 228313 to bus 228401 ckt 1) overload, it will require increasing the emergency rating of the Bridgeport to Mickleton 230 kV line by rebuilding the circuit. The rebuild will include the installation of new poles, foundations, insulators, and conductor. Type : FAC Total Cost : \$18,000,000 Time Estimate : 36-60 Months Ratings : 993.0/993.0/993.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-Queue</td><td>66.9</td><td>43.31%</td><td>\$7,795,184</td></tr><tr><td>AF1-101</td><td>59.48</td><td>38.50%</td><td>\$6,930,606</td></tr><tr><td>AF1-208</td><td>9.5</td><td>6.15%</td><td>\$1,106,939</td></tr><tr><td>AF1-238</td><td>18.6</td><td>12.04%</td><td>\$2,167,271</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-Queue	66.9	43.31%	\$7,795,184	AF1-101	59.48	38.50%	\$6,930,606	AF1-208	9.5	6.15%	\$1,106,939	AF1-238	18.6	12.04%	\$2,167,271	\$18,000,000	\$2,167,271	N6791																								
Queue	MW	Cost %	Cost \$																																															
AE2-Queue	66.9	43.31%	\$7,795,184																																															
AF1-101	59.48	38.50%	\$6,930,606																																															
AF1-208	9.5	6.15%	\$1,106,939																																															
AF1-238	18.6	12.04%	\$2,167,271																																															
42777150	5	28MANITOU 230.0 kV - 28WHITINGS 230.0 kV Ckt 1	ProjectId: N6413 Description: Replace substation conductor at Manitou and Whitings. Reconductor the Manitou - Whitings 230 kV line (~8.78 miles). Replace wave trap, circuit breaker and disconnect switch at Manitou. Replace relaying and current transformers at Manitou. Replace metering at Whitings. Type : FAC Total Cost : \$43,435,779 Time Estimate : 48.0 Months Ratings : 1355.0/1625.0/1625.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-021</td><td>2.28</td><td>0.27%</td><td>\$115,626</td></tr><tr><td>AE2-022</td><td>52.8</td><td>6.16%</td><td>\$2,677,655</td></tr><tr><td>AE2-023</td><td>50.01</td><td>5.84%</td><td>\$2,536,368</td></tr><tr><td>AE2-222</td><td>40.94</td><td>4.78%</td><td>\$2,075,944</td></tr><tr><td>AE2-232</td><td>203.31</td><td>23.74%</td><td>\$10,310,494</td></tr><tr><td>AE2-251</td><td>179.63</td><td>20.97%</td><td>\$9,109,607</td></tr><tr><td>AE2-314</td><td>10.78</td><td>1.26%</td><td>\$546,688</td></tr><tr><td>AF1-101</td><td>292.01</td><td>34.09%</td><td>\$14,808,753</td></tr><tr><td>AF1-126</td><td>16.17</td><td>1.89%</td><td>\$820,032</td></tr><tr><td>AF1-238</td><td>8.57</td><td>1.00%</td><td>\$434,612</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-021	2.28	0.27%	\$115,626	AE2-022	52.8	6.16%	\$2,677,655	AE2-023	50.01	5.84%	\$2,536,368	AE2-222	40.94	4.78%	\$2,075,944	AE2-232	203.31	23.74%	\$10,310,494	AE2-251	179.63	20.97%	\$9,109,607	AE2-314	10.78	1.26%	\$546,688	AF1-101	292.01	34.09%	\$14,808,753	AF1-126	16.17	1.89%	\$820,032	AF1-238	8.57	1.00%	\$434,612	\$43,435,779	\$434,612	N6413
Queue	MW	Cost %	Cost \$																																															
AE2-021	2.28	0.27%	\$115,626																																															
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ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																																																																
41648246,41648247,41041981,41041980	17	CARDIFF 230.0 kV - NEWFRDM 230.0 kV Ckt 1	ProjectId : N6342 Description : To mitigate the (ACE) Cardiff New Freedom 230 kV line (from bus 227900 to bus 219100 ckt 1) overload, it will require increasing the emergency rating of the Cardiff to New Freedom 230 kV line by rebuilding the circuit. The rebuild will include the installation of new poles, foundations, insulators, and conductor. Type : FAC Total Cost : \$105,000,000 Time Estimate : 36-60 Months Ratings : 796.0/932.0/932.0 Notes : Reinforcement #1 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-021</td><td>188.62</td><td>15.05%</td><td>\$15,804,758</td></tr><tr><td>AE2-022</td><td>149.8</td><td>11.95%</td><td>\$12,551,971</td></tr><tr><td>AE2-024</td><td>39.8</td><td>3.18%</td><td>\$3,334,903</td></tr><tr><td>AE2-025</td><td>20.09</td><td>1.60%</td><td>\$1,683,372</td></tr><tr><td>AE2-222</td><td>102.7</td><td>8.20%</td><td>\$8,605,390</td></tr><tr><td>AE2-251</td><td>509.6</td><td>40.67%</td><td>\$42,700,162</td></tr><tr><td>AF1-101</td><td>225.43</td><td>17.99%</td><td>\$18,889,124</td></tr><tr><td>AF1-238</td><td>17.07</td><td>1.36%</td><td>\$1,430,321</td></tr></table> ProjectId : n6569 Description : To mitigate the (ACE) Cardiff New Freedom 230 kV line (from bus 227900 to bus 219100 ckt 1) overload, it will require increasing the emergency rating of the Cardiff to New Freedom 230 kV line by rebuilding the circuit. The rebuild will include the installation of new poles, foundations, insulators, and conductor. Type : FAC Total Cost : \$160,000,000 Time Estimate : 60-72 Months Ratings : 2159.0/2622.0/2622.0 Notes : Reinforcement #6 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-251</td><td>330.72</td><td>57.70%</td><td>\$92,312,201</td></tr><tr><td>AF1-101</td><td>225.43</td><td>39.33%</td><td>\$62,923,136</td></tr><tr><td>AF1-238</td><td>17.07</td><td>2.98%</td><td>\$4,764,663</td></tr></table> ProjectId : n6794 Description : Upgrade New Freedom 230kV terminal equipment to achieve a SER of 1992 MVA Type : FAC Total Cost : \$25,493,511 Time Estimate : 27.0 Months Ratings : 1992.0/1992.0/ Notes : Reinforcement #8 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AF1-101</td><td>224.0</td><td>92.92%</td><td>\$23,688,333</td></tr><tr><td>AF1-238</td><td>17.07</td><td>7.08%</td><td>\$1,805,178</td></tr></table> 8 total Reinforcements: Only Reinforcements #1, #6 and #8 apply for AF1-238.	Queue	MW	Cost %	Cost \$	AE2-021	188.62	15.05%	\$15,804,758	AE2-022	149.8	11.95%	\$12,551,971	AE2-024	39.8	3.18%	\$3,334,903	AE2-025	20.09	1.60%	\$1,683,372	AE2-222	102.7	8.20%	\$8,605,390	AE2-251	509.6	40.67%	\$42,700,162	AF1-101	225.43	17.99%	\$18,889,124	AF1-238	17.07	1.36%	\$1,430,321	Queue	MW	Cost %	Cost \$	AE2-251	330.72	57.70%	\$92,312,201	AF1-101	225.43	39.33%	\$62,923,136	AF1-238	17.07	2.98%	\$4,764,663	Queue	MW	Cost %	Cost \$	AF1-101	224.0	92.92%	\$23,688,333	AF1-238	17.07	7.08%	\$1,805,178	\$293,726,649	\$290,493,511	N6342 N6569 N6794
Queue	MW	Cost %	Cost \$																																																																			
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AF1-238	17.07	7.08%	\$1,805,178																																																																			

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number
			Description: To mitigate the (ACE) Cardiff - New Freedom 230 kV line (from bus 227900 to bus 219100 ckt 1) overload, it will require various terminal reinforcements are required at Cardiff. Type: FAC Total Cost: \$400,000 Time Estimate: 24-36 Months Ratings: 993.0/993.0/993.0 Notes: For this reinforcement, Reinforcement #2. Per PJM Cost Allocation rules, this project does not receive Cost Allocation. Description: To mitigate the (ACE) Cardiff -New Freedom 230 kV line (from bus 227900 to bus 219100 ckt 1) overload, it will require various terminal reinforcements at Cardiff. Type: FAC Total Cost: \$600,000 Time Estimate: 24-36 Months Ratings: 1195.0/1195.0/1195.0 Notes: For this reinforcement, Reinforcement #3. Per PJM Cost Allocation rules, this project does not receive Cost Allocation. Description: To mitigate the (ACE) Cardiff- New Freedom 230 kV line (from bus 227900 to bus 219100 ckt 1) overload, it will require various terminal reinforcements at Cardiff. Type: FAC Total Cost: \$600,000 Time Estimate: 24-36 Months Ratings: 1189.0/1427.0/1427.0 Notes: For this reinforcement, Reinforcement #4. Per PJM Cost Allocation rules, this project does not receive Cost Allocation. Description: To mitigate the (ACE) Cardiff - New Freedom 230 kV line (from bus 227900 to bus 219100 ckt 1) overload, it will require various terminal reinforcements at Cardiff. Type: FAC Total Cost: \$600,000 Time Estimate: 24-36 Months Ratings: 1593.0/1593.0/1593.0 Notes: For this reinforcement, Reinforcement #5. Per PJM Cost Allocation rules, this project does not receive Cost Allocation. Description: Upgrade New Freedom 230kV terminal equipment to achieve a SER of 1593 MVA for PSEG Portion. Type: FAC Total Cost: \$1,033,138 Time Estimate: 13 Months Ratings: 1593.0/1593.0/ Notes: For this reinforcement, Reinforcement #7. Per PJM Cost Allocation rules, this project does not receive Cost Allocation.			

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																				
42777311	14	HILLTOP_1 230.0 kV - BEAVERBK_1 230.0 kV Ckt 1	ProjectId : n6502 Description : Reconductor to 1080MVA SER Type : FAC Total Cost : \$6,438,257 Time Estimate : 33.0 Months Ratings : 934.0/1080.0/ <table><thead><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr></thead><tbody><tr><td>AE2-222</td><td>0.16</td><td>10.28%</td><td>\$662,045</td></tr><tr><td>AE2-251</td><td>1.19</td><td>75.66%</td><td>\$4,870,966</td></tr><tr><td>AE2-257</td><td>0.09</td><td>5.72%</td><td>\$368,485</td></tr><tr><td>AF1-238</td><td>0.13</td><td>8.34%</td><td>\$536,760</td></tr></tbody></table>	Queue	MW	Cost %	Cost \$	AE2-222	0.16	10.28%	\$662,045	AE2-251	1.19	75.66%	\$4,870,966	AE2-257	0.09	5.72%	\$368,485	AF1-238	0.13	8.34%	\$536,760	\$6,438,257	\$536,760	N6502
Queue	MW	Cost %	Cost \$																							
AE2-222	0.16	10.28%	\$662,045																							
AE2-251	1.19	75.66%	\$4,870,966																							
AE2-257	0.09	5.72%	\$368,485																							
AF1-238	0.13	8.34%	\$536,760																							

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																																
88617735	29	MNOTLA 1 138.0 kV - LANDIS 138.0 kV Ckt 1	ProjectId : N6409 Description : To mitigate the (AE - AE) MNOTLA 1-LANDIS 138 kV line (from bus 228502 to bus 228500 ckt 1) overload will include rebuilding the Minotola to Landis 138 kV transmission line. Also, upgrades to the terminals at Minotola and Landis Substation will be required. Type : FAC Total Cost : \$12,800,000 Time Estimate : 36-48 Months Ratings : 478.0/559.0/559.0 Notes : Reinforcement #1	\$13,500,000	\$400,123	N6409																																
			<table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-020</td><td>8.82</td><td>2.71%</td><td>\$347,351</td></tr><tr><td>AE2-021</td><td>57.6</td><td>17.72%</td><td>\$2,268,414</td></tr><tr><td>AE2-022</td><td>33.6</td><td>10.34%</td><td>\$1,323,242</td></tr><tr><td>AE2-222</td><td>42.77</td><td>13.16%</td><td>\$1,684,376</td></tr><tr><td>AE2-251</td><td>114.35</td><td>35.18%</td><td>\$4,503,354</td></tr><tr><td>AF1-101</td><td>57.72</td><td>17.76%</td><td>\$2,273,140</td></tr><tr><td>AF1-238</td><td>10.16</td><td>3.13%</td><td>\$400,123</td></tr></table>				Queue	MW	Cost %	Cost \$	AE2-020	8.82	2.71%	\$347,351	AE2-021	57.6	17.72%	\$2,268,414	AE2-022	33.6	10.34%	\$1,323,242	AE2-222	42.77	13.16%	\$1,684,376	AE2-251	114.35	35.18%	\$4,503,354	AF1-101	57.72	17.76%	\$2,273,140	AF1-238	10.16	3.13%	\$400,123
			Queue				MW	Cost %	Cost \$																													
AE2-020	8.82	2.71%	\$347,351																																			
AE2-021	57.6	17.72%	\$2,268,414																																			
AE2-022	33.6	10.34%	\$1,323,242																																			
AE2-222	42.77	13.16%	\$1,684,376																																			
AE2-251	114.35	35.18%	\$4,503,354																																			
AF1-101	57.72	17.76%	\$2,273,140																																			
AF1-238	10.16	3.13%	\$400,123																																			
ProjectId : N6414 Description : To mitigate the (AE - AE) MNOTLA 1-LANDIS 138 kV line (from bus 228502 to bus 228500 ckt 1) overload, several terminal equipment reinforcement is required at both, Landis and Minotola substation. Type : FAC Total Cost : \$200,000 Time Estimate : 12-24 Months Ratings : 549.0/621.0/621.0 Notes : Reinforcement #2: Per PJM Cost Allocation Rules, this reinforcement cost is driven by previous queue. This may change with future re-tools as projects withdraw.																																						
			ProjectId : N6517 Description : To mitigate the (AE - AE) MNOTLA 1-LANDIS 138 kV line (from bus 228502 to bus 228500 ckt 1) overload, several terminal equipment reinforcement is required at both, Landis and Minotola substation. Type : FAC Total Cost : \$200,000 Time Estimate : 12-24 Months Ratings : 532.0/662.0/662.0 Notes : Reinforcement #3: Per PJM Cost Allocation Rules, this reinforcement cost is driven by previous queue. This may change with future re-tools as projects withdraw.																																			

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number
			ProjectId : N6515 Description : To mitigate the (AE - AE) MNOTLA 1-LANDIS 138 kV line (from bus 228502 to bus 228500 ckt 1) overload , several terminal equipment reinforcement is required at both, Landis and Minotola substation. Type : FAC Total Cost : \$300,000 Time Estimate : 12-24 Months Ratings : 717.0/717.0/717.0 Notes : Reinforcement #4: Per PJM Cost Allocation Rules, this reinforcement cost is driven by previous queue. This may change with future re-tools as projects withdraw. ProjectId : N6570 Description : To mitigate the (AE - AE) MNOTLA 1-LANDIS 138 kV line (from bus 228502 to bus 228500 ckt 1) overload , several terminal equipment reinforcement is required at both, Landis and Minotola substation. Type : FAC Total Cost : \$600,000 Time Estimate : 24-48 Months Ratings : 605/726/726 Notes: Reinforcement #5: Per PJM Cost Allocation Rules, this reinforcement cost is driven by previous queue. This may change with future re-tools as projects withdraw. ProjectId : N6571 Description : To mitigate the (AE - AE) MNOTLA 1-LANDIS 138 kV line (from bus 228502 to bus 228500 ckt 1) overload , several terminal equipment reinforcement is required at both, Landis and Minotola substation. Type : FAC Total Cost : \$400,000 Time Estimate : 12-24 Months Ratings : 717/838/838 Notes: Reinforcement #6: Per PJM Cost Allocation Rules, this reinforcement cost is driven by previous queue. This may change with future re-tools as projects withdraw.			
59580241	1	S121TAP 69.0 kV - CNTRL N 69.0 kV Ckt 1	This is a Light Load violation. R Light Load Section for this reinforcement.	N/A	N/A	N/A

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number												
42250940,42250938,42020389	9	RICHRE29 230.0 kV - RICHMOND 230.0 kV Ckt 1	ProjectId : N6583 Description : Reconductor PECO portion of line using 'Pecos' conductor on this bifurcated line (conductors run on parallel towers and remain tied together as they are today, which remains a 6-wire line) and also rebuild the Richmond 230 kV substation ring bus to increase bus rating (Does not include permitting or ROW acquisition). Cost & Ratings for PECO portion of tie line only. Type : FAC Total Cost : \$15,653,000 Time Estimate : 36.0 Months Ratings : 1612.0/2016.0/2257.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-251</td><td>44.17</td><td>60.82%</td><td>\$9,519,543</td></tr><tr><td>AF1-238</td><td>28.46</td><td>39.18%</td><td>\$6,133,457</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-251	44.17	60.82%	\$9,519,543	AF1-238	28.46	39.18%	\$6,133,457	\$15,653,000	\$6,133,457	N6583
Queue	MW	Cost %	Cost \$															
AE2-251	44.17	60.82%	\$9,519,543															
AF1-238	28.46	39.18%	\$6,133,457															
49873384,41041986,41648393,41648392	6	28OYSTER C 230.0 kV - CEDAR 230.0 kV Ckt 1	ProjectId : N6587.2 Description : Replace disconnect switch at Oyster Creek. Additionally, AE would need to replace their section of the limiting conductor and provide estimates for their replacement. Type : FAC Total Cost : \$402,742 Time Estimate : 12.0 Months Ratings : 1355.0/1625.0/1625.0 Notes: Per PJM Cost Allocation Rules, this reinforcement cost is driven by previous queue. This may change with future re-tools as projects withdraw. ProjectId : N6587 Description : Reconductor the Oyster Creek - Cedar 230 kV line (JCP&L portion only ~0.1 miles. AE portion ~14 miles) Type : FAC Total Cost : \$469,866 Time Estimate : 12.0 Months Ratings : 913/1147/1147 Notes: Per PJM Cost Allocation Rules, this reinforcement cost is driven by previous queue. This may change with future re-tools as projects withdraw.	\$872,608	\$0	N6587 N6587.2												

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																				
41648241	20	SCULL#2 138.0 kV - MILL #2 138.0 kV Ckt 1	ProjectId : N6189 Description : To mitigate the (ACE) Mill#2 Scull#2 138 kV line (from bus 228904 to bus 227906 ckt 1) overload, it will require increasing the emergency rating of the Mill#2 to Scull#2 138 kV line by rebuilding the circuit. The rebuild will include the installation of new poles, foundations, insulators, and conductor. In addition, various terminal reinforcements are required at both Mill#2 to Scull#2. Type : FAC Total Cost : \$12,800,000 Time Estimate : 36-60 Months Ratings : 478.0/559.0/559.0 <table><thead><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr></thead><tbody><tr><td>AE1-104</td><td>190.16</td><td>84.77%</td><td>\$10,850,301</td></tr><tr><td>AE1-179</td><td>9.01</td><td>4.02%</td><td>\$514,100</td></tr><tr><td>AE1-240</td><td>7.53</td><td>3.36%</td><td>\$429,653</td></tr><tr><td>AF1-238</td><td>17.63</td><td>7.86%</td><td>\$1,005,947</td></tr></tbody></table>	Queue	MW	Cost %	Cost \$	AE1-104	190.16	84.77%	\$10,850,301	AE1-179	9.01	4.02%	\$514,100	AE1-240	7.53	3.36%	\$429,653	AF1-238	17.63	7.86%	\$1,005,947	\$12,800,000	\$1,005,947	N6189
Queue	MW	Cost %	Cost \$																							
AE1-104	190.16	84.77%	\$10,850,301																							
AE1-179	9.01	4.02%	\$514,100																							
AE1-240	7.53	3.36%	\$429,653																							
AF1-238	17.63	7.86%	\$1,005,947																							

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number								
51073203,51072561	22	CARDIFF2 138.0 kV - LEWIS #2 138.0 kV Ckt 1	ProjectId : N6586 Description : To mitigate the (ACE) Cardiff to Lewis #2 138 kV line (from bus 227901 to bus 227945 ckt 1) overload, terminal reinforcement is required at both Cardiff & Lewis #2 substations. Type : FAC Total Cost : \$300,000 Time Estimate : 12-24 Months Ratings : 549.0/621.0/621.0 <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AF1-238</td><td>8.54</td><td>100.00%</td><td>\$300,000</td></tr></table>	Queue	MW	Cost %	Cost \$	AF1-238	8.54	100.00%	\$300,000	\$10,600,000	\$1,542,189	N6585 N6586 N6587
			Queue	MW	Cost %	Cost \$								
			AF1-238	8.54	100.00%	\$300,000								
			ProjectId : N6585 Description : To mitigate the (ACE) Cardiff to Lewis #2 138 kV line (from bus 227934 to bus 227945 ckt 1) overload, it will require increasing the emergency rating of the Cardiff to Lewis 138 kV line by rebuilding the circuit. The rebuild will include the installation of new poles, foundations, insulators, and conductor. Type : FAC Total Cost : \$10,000,000 Time Estimate : 48-60 Months Ratings : 478.0/559.0/559.0 <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AE2-251</td><td>82.1</td><td>90.58%</td><td>\$9,057,811</td></tr><tr><td>AF1-238</td><td>8.54</td><td>9.42%</td><td>\$942,189</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-251	82.1	90.58%	\$9,057,811			
Queue	MW	Cost %	Cost \$											
AE2-251	82.1	90.58%	\$9,057,811											
AF1-238	8.54	9.42%	\$942,189											
ProjectId : N6847 Description : To mitigate the (ACE) Cardiff to Lewis #2 138 kV line (from bus 227901 to bus 227945 ckt 1) overload, terminal reinforcement is required at both Cardiff & Lewis #2 substations, including strand buses and disc switches. Type : FAC Total Cost : \$300,000 Time Estimate : 12 to 24 Months Ratings : 780.0/965.0/965.0 <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AF1-238</td><td>8.54</td><td>100.00%</td><td>\$300,000</td></tr></table>	Queue	MW	Cost %	Cost \$	AF1-238	8.54	100.00%	\$300,000						
Queue	MW	Cost %	Cost \$											
AF1-238	8.54	100.00%	\$300,000											

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number												
41042049	3	CNTRL N 69.0 kV - SHLDLY T 69.0 kV Ckt 1	Project Id: N6800 Description: Line Upgrade CN-Metering POI, including the installation of new poles, circuit breaker, relaying, foundations, insulators, and conductor. Type : FAC Total Cost : \$4,000,000 Time Estimate : 18.0 Months Ratings : 200.0/286.0/286.0 <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AF1-238</td><td>33.06</td><td>80.95%</td><td>\$3,238,002</td></tr><tr><td>AF1-239</td><td>7.78</td><td>19.05%</td><td>\$761,998</td></tr></table>	Queue	MW	Cost %	Cost \$	AF1-238	33.06	80.95%	\$3,238,002	AF1-239	7.78	19.05%	\$761,998	\$11,605,000	\$5,730,657	N6349 N6800 N6801
			Queue	MW	Cost %	Cost \$												
			AF1-238	33.06	80.95%	\$3,238,002												
			AF1-239	7.78	19.05%	\$761,998												
Project Id: N6801 Description: Ring Bus for AF1-238 & AF1-239- New 69kV class substation and associated relaying, fiber, etc. Type : CON Total Cost : \$7,405,000 Time Estimate : 34.0 Months Ratings : 200.0/286.0/286.0 <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AF1-238</td><td>3.9</td><td>33.42%</td><td>\$2,474,441</td></tr><tr><td>AF1-239</td><td>7.78</td><td>66.58%</td><td>\$4,930,559</td></tr></table>	Queue	MW	Cost %	Cost \$	AF1-238	3.9	33.42%	\$2,474,441	AF1-239	7.78	66.58%	\$4,930,559						
Queue	MW	Cost %	Cost \$															
AF1-238	3.9	33.42%	\$2,474,441															
AF1-239	7.78	66.58%	\$4,930,559															
Project Id: N6349 Description: To mitigate the (ACE) Sheildalloy Tap Central Tap 69 kV line (from bus 228504 to bus 228714 ckt 1) overload, terminal reinforcement is required at Central North (Vineland). Type : FAC Total Cost : \$200,000 Time Estimate : 24-36 Months Ratings : 122.0/156.0/156.0 <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AF1-238</td><td>33.06</td><td>80.95%</td><td>\$161,900</td></tr><tr><td>AF1-239</td><td>7.78</td><td>19.05%</td><td>\$38,100</td></tr></table>	Queue	MW	Cost %	Cost \$	AF1-238	33.06	80.95%	\$161,900	AF1-239	7.78	19.05%	\$38,100						
Queue	MW	Cost %	Cost \$															
AF1-238	33.06	80.95%	\$161,900															
AF1-239	7.78	19.05%	\$38,100															

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																												
41649105	30	AE1-240 TAP 69.0 kV - SHRMAN#2 69.0 kV Ckt 1	Project Id : N6802 Description : To mitigate the AE1-240 tap to Sherman 69 kV line section overload a 600 amp disc switch must be upgraded at Sherman Type : FAC Total Cost : \$20,000 Time Estimate : 6 to 14 Months Ratings : 122.0/157.0/177.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AF1-208</td><td>13.52</td><td>44.02%</td><td>\$8,805</td></tr><tr><td>AF1-238</td><td>17.19</td><td>55.98%</td><td>\$11,195</td></tr></table>	Queue	MW	Cost %	Cost \$	AF1-208	13.52	44.02%	\$8,805	AF1-238	17.19	55.98%	\$11,195	\$20,000	\$11,195	N6802																
Queue	MW	Cost %	Cost \$																															
AF1-208	13.52	44.02%	\$8,805																															
AF1-238	17.19	55.98%	\$11,195																															
42250807,42250808	18	CARDIFF 230.0 kV - CEDAR 230.0 kV Ckt 1	ProjectId : N6412 Description : To mitigate the (ACE) Cardiff Cedar 230 kV line (from bus 940360 to bus 227955 ckt 1) overload, it will require increasing the emergency rating of the Cardiff to Cedar 230 kV line by rebuilding the circuit. The rebuild will include the installation of new poles, foundations, insulators, and conductor. Type : FAC Total Cost : \$120,000,000 Time Estimate : 48-60 Months Ratings : 1003.0/1003.0/1003.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-021</td><td>38.64</td><td>5.35%</td><td>\$6,415,141</td></tr><tr><td>AE2-022</td><td>110.8</td><td>15.33%</td><td>\$18,395,385</td></tr><tr><td>AE2-222</td><td>66.1</td><td>9.15%</td><td>\$10,974,142</td></tr><tr><td>AE2-251</td><td>377.0</td><td>52.16%</td><td>\$62,590,794</td></tr><tr><td>AF1-101</td><td>116.4</td><td>16.10%</td><td>\$19,325,115</td></tr><tr><td>AF1-238</td><td>13.85</td><td>1.92%</td><td>\$2,299,423</td></tr></table> ProjectId : N6535 Description : To mitigate the (ACE) AE2-020 TAP Cedar 230 kV line (from bus 940360 to bus 227955 ckt 1) overload, various terminal reinforcements are required at Cedar. Type : FAC Total Cost : \$300,000 Time Estimate : 24-36 Months Ratings : 1252.0/1519.0/1519.0 Notes : Per PJM cost allocation rules, [Queue#] presently does not receive cost allocation for this upgrade.	Queue	MW	Cost %	Cost \$	AE2-021	38.64	5.35%	\$6,415,141	AE2-022	110.8	15.33%	\$18,395,385	AE2-222	66.1	9.15%	\$10,974,142	AE2-251	377.0	52.16%	\$62,590,794	AF1-101	116.4	16.10%	\$19,325,115	AF1-238	13.85	1.92%	\$2,299,423	\$120,000,000	\$2,299,423	N6412
Queue	MW	Cost %	Cost \$																															
AE2-021	38.64	5.35%	\$6,415,141																															
AE2-022	110.8	15.33%	\$18,395,385																															
AE2-222	66.1	9.15%	\$10,974,142																															
AE2-251	377.0	52.16%	\$62,590,794																															
AF1-101	116.4	16.10%	\$19,325,115																															
AF1-238	13.85	1.92%	\$2,299,423																															

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																				
42777361	26	CHAMBERS 230.0 kV - PEDRKTWN 230.0 kV Ckt 1	ProjectId : n6792 Description To mitigate the (ACE) Chambers to Pedrickton 230 kV line (from bus 228311 to bus 228312 ckt 1) overload, it will require reinforcement of terminal equipment at both substations. Type : FAC Total Cost : \$1,000,000 Time Estimate : 12-32 Months Ratings : 796.0/932.0/932.0 Notes : Per PJM Cost Allocation Rules, this reinforcement is in the previous AE2 Queue. This may change with future re-tools as projects withdraw.	\$1,000,000	\$0	N6792																				
51072462	21	CARDIFF 138.0 kV - LEWIS #1 138.0 kV Ckt 1	ProjectId : N6577 Description : To mitigate the (ACE) Cardiff to Lewis#1 138 kV line (from bus 227913 to bus 227902 ckt 1) overload, it will require increasing the emergency rating of the Cardiff to Lewis#1 138 kV line by rebuilding the circuit. The rebuild will include the installation of new poles, foundations, insulators, and conductor. Type : FAC Total Cost : \$9,000,000 Time Estimate : 36-60 Months Ratings : 478.0/559.0/559.0 <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AE2-251</td><td>5.3</td><td>32.24%</td><td>\$2,901,460</td></tr><tr><td>AF1-238</td><td>11.14</td><td>67.76%</td><td>\$6,098,540</td></tr></table> ProjectId : N6848 Description : To mitigate the (ACE) Cardiff- Lewis#1 138 kV line (from bus 227913 to bus 227902 ckt 1) overload,will require substation reinforcements at both substations. Type : FAC Total Cost : \$900,000 Time Estimate : 24-32 Months Ratings : 780.0/965.0/965.0 Notes : AF1-238 <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AF1-238</td><td>11.14</td><td>100.00%</td><td>\$900,000</td></tr></table> ProjectId : N6576 Description : To mitigate the (ACE) Cardiff- Lewis#1 138 kV line (from bus 227913 to bus 227902 ckt 1) overload, will require substation reinforcements at both substations. Type : FAC Total Cost : \$900,000 Time Estimate : 24-32 Months Ratings : 390.0/482.0/482.0 Notes : Per PJM Cost Allocation Rules, this reinforcement is in the previous Queue.	Queue	MW	Cost %	Cost \$	AE2-251	5.3	32.24%	\$2,901,460	AF1-238	11.14	67.76%	\$6,098,540	Queue	MW	Cost %	Cost \$	AF1-238	11.14	100.00%	\$900,000	\$10,800,000	\$6,998,540	N6577 N6848 N6576
Queue	MW	Cost %	Cost \$																							
AE2-251	5.3	32.24%	\$2,901,460																							
AF1-238	11.14	67.76%	\$6,098,540																							
Queue	MW	Cost %	Cost \$																							
AF1-238	11.14	100.00%	\$900,000																							

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																																			
42777173,42777172 42777536	7	28VANHISVL 230.0 kV - 28LARRABEE 230.0 kV Ckt 1	ProjectId : N6537 Description : Reconnector the Larrabee - Van Hiseville 230 kV line (~7.5 miles). Replace wave trap at Larrabee Type : FAC Total Cost : \$35,407,778 Time Estimate : 42.0 Months Ratings : 956.0/956.0/956.0	\$37,186,557	\$490,750	N6537 N6537.2																																			
	4		<table><thead><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr></thead><tbody><tr><td>AE2-023</td><td>45.81</td><td>5.78%</td><td>\$2,047,501</td></tr><tr><td>AE2-222</td><td>39.3</td><td>4.96%</td><td>\$1,756,510</td></tr><tr><td>AE2-232</td><td>148.5</td><td>18.75%</td><td>\$6,637,193</td></tr><tr><td>AE2-251</td><td>167.62</td><td>21.16%</td><td>\$7,491,760</td></tr><tr><td>AE2-314</td><td>10.06</td><td>1.27%</td><td>\$449,631</td></tr><tr><td>AF1-101</td><td>213.84</td><td>26.99%</td><td>\$9,557,558</td></tr><tr><td>AF1-126</td><td>99.57</td><td>12.57%</td><td>\$4,450,272</td></tr><tr><td>AF1-222</td><td>56.53</td><td>7.14%</td><td>\$2,526,603</td></tr><tr><td>AF1-238</td><td>10.98</td><td>1.39%</td><td>\$490,750</td></tr></tbody></table> ProjectId: N6537.2 Description : Replace relaying, metering, current transformers, and fault switch at Larrabee. Upgrade terminal equipment at Larrabee as necessary. Type : FAC Total Cost : \$1,778,779 Time Estimate : 12.0 Months Ratings : 1355/1434/1434 Notes: Per PJM Cost Allocation Rules, this reinforcement is in the previous Queue. This may change with future re-tools as projects withdraw.				Queue	MW	Cost %	Cost \$	AE2-023	45.81	5.78%	\$2,047,501	AE2-222	39.3	4.96%	\$1,756,510	AE2-232	148.5	18.75%	\$6,637,193	AE2-251	167.62	21.16%	\$7,491,760	AE2-314	10.06	1.27%	\$449,631	AF1-101	213.84	26.99%	\$9,557,558	AF1-126	99.57	12.57%	\$4,450,272	AF1-222	56.53	7.14%
Queue	MW	Cost %	Cost \$																																						
AE2-023	45.81	5.78%	\$2,047,501																																						
AE2-222	39.3	4.96%	\$1,756,510																																						
AE2-232	148.5	18.75%	\$6,637,193																																						
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AF1-101	213.84	26.99%	\$9,557,558																																						
AF1-126	99.57	12.57%	\$4,450,272																																						
AF1-222	56.53	7.14%	\$2,526,603																																						
AF1-238	10.98	1.39%	\$490,750																																						
			TOTAL COST	\$1,035,308,922	\$416,525,426																																				

Note: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

11.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

11.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
59580241	228716	S121TAP	AE	228714	CNTRL N	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	64.69	109.81	AC	50.98

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
228702	WEST CT	4.3197	80/20	4.3197
228717	S121	10.4452	80/20	10.4452
945731	AF1-238 C	50.9781	80/20	50.9781
945741	AF1-239 C	11.9948	80/20	11.9948
NEWTON	NEWTON	0.0462	Confirmed LTF	0.0462
FARMERCITY	FARMERCITY	0.0024	Confirmed LTF	0.0024
NY	NY	0.0243	Confirmed LTF	0.0243
PRAIRIE	PRAIRIE	0.1111	Confirmed LTF	0.1111
COFFEEN	COFFEEN	0.0086	Confirmed LTF	0.0086
CHEOAH	CHEOAH	0.0215	Confirmed LTF	0.0215
EDWARDS	EDWARDS	0.0150	Confirmed LTF	0.0150
TILTON	TILTON	0.0271	Confirmed LTF	0.0271
GIBSON	GIBSON	0.0235	Confirmed LTF	0.0235
CALDERWOOD	CALDERWOOD	0.0214	Confirmed LTF	0.0214
BLUEG	BLUEG	0.0746	Confirmed LTF	0.0746
TRIMBLE	TRIMBLE	0.0239	Confirmed LTF	0.0239
CATAWBA	CATAWBA	0.0150	Confirmed LTF	0.0150

11.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41042413	228401	MCKLTON	AE	213559	DELCOTAP	PECO	1	PS_P7-1_V2274+P2242_LT	tower	725.0	98.71	101.59	AC	17.47

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
219229	EAGLEPT_G3	2.3889	50/50	2.3889
219230	EAGLEPT_G1	4.5062	50/50	4.5062
219231	EAGLEPT_G2	4.5062	50/50	4.5062
219235	EAGLEPT_ST2	1.6288	50/50	1.6288
219258	KINSLEYDEP C	0.1032	50/50	0.1032
219259	KINSLEYDEP E	0.9503	50/50	0.9503
219384	DEPTFDSP1_C	0.0014	50/50	0.0014
219385	DEPTFDSP1_E	0.0126	50/50	0.0126
219388	DEPTFDSP2_C	0.0014	50/50	0.0014
219390	DEPTFDSP2_E	0.0126	50/50	0.0126
219393	DEPTFDSP3_C	0.0034	50/50	0.0034
219394	DEPTFDSP3_E	0.0316	50/50	0.0316
219408	DEPTFDSP4_C	0.0014	50/50	0.0014
219416	DEPTFDSP4_E	0.0126	50/50	0.0126
219662	THOROFSP1_C	0.0056	50/50	0.0056
219664	THOROFSP1_E	0.0515	50/50	0.0515
219665	THOROFSP2_C	0.0056	50/50	0.0056
219666	THOROFSP2_E	0.0515	50/50	0.0515
219683	THOSOLAR C	0.0309	50/50	0.0309
219684	THOSOLAR E	0.2851	50/50	0.2851
227881	GRENWCHG	0.4313	50/50	0.4313
227928	V4-067E	0.1387	Adder	0.16
228102	BLE#2 ST (Deactivation : 30/04/2019)	14.1992	Adder	16.7
228261	V4-054E	0.8439	Adder	0.99
228304	LOGAN	10.1100	50/50	10.1100
228306	PCLP STM	2.5774	50/50	2.5774
228307	PCLP GT	2.5774	50/50	2.5774
228309	CCLP NUG	8.8168	50/50	8.8168
228334	MANNMILG	1.4296	Adder	1.68
228357	V2-046E	2.1834	Adder	2.57
228400	MICK 1CT	2.2239	50/50	2.2239
228423	Q-090 2	39.7134	50/50	39.7134
228471	VALERO1	0.7526	50/50	0.7526
228472	VALERO2	0.4964	50/50	0.4964
228473	VALERO3	0.4964	50/50	0.4964
228484	VALERO4	0.4427	50/50	0.4427
228712	V2-041E	0.2673	Adder	0.31
228721	V2-035E	0.1795	Adder	0.21
228733	AB1-119 E	0.0679	Adder	0.08
902092	W1-130E	0.7191	Adder	0.85
902432	W2-030 E	0.5342	Adder	0.63

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
917471	Z2-083	4.0175	50/50	4.0175
919901	AB1-000 1	-0.6116	Adder	-0.72
919911	AB1-000 2	-0.6116	Adder	-0.72
919921	AB1-000 3	-0.6116	Adder	-0.72
923153	AB1-116 E	0.0921	Adder	0.11
923532	AB1-169AC	69.1807	Adder	81.39
924051	AB2-049 C	0.3942	Adder	0.46
924052	AB2-049 E	0.6432	Adder	0.76
924531	AB2-102 C	23.1546	Adder	27.24
924532	AB2-102 E	0.5145	Adder	0.61
924701	AB2-122 C	0.0599	Adder	0.07
924702	AB2-122 E	0.1027	Adder	0.12
930001	AB1-001 C	0.0733	Adder	0.09
930002	AB1-001 E	0.1204	Adder	0.14
933962	AD1-019 E	0.6769	Adder	0.8
936491	AD2-064 C (Withdrawn : 06/16/2020)	0.0493	Adder	0.06
936492	AD2-064 E (Withdrawn : 06/16/2020)	0.0679	Adder	0.08
936501	AD2-065 C	0.0916	Adder	0.11
936502	AD2-065 E	0.1264	Adder	0.15
937011	AD2-135 C	0.0530	Adder	0.06
937012	AD2-135 E	0.0902	Adder	0.11
938421	AE1-061 C	0.3470	Adder	0.41
938422	AE1-061 E	0.3470	Adder	0.41
938431	AE1-062 C	0.8218	Adder	0.97
938432	AE1-062 E	0.8218	Adder	0.97
938781	AE1-104 C O1	11.4613	Adder	13.48
938782	AE1-104 E O1	29.3236	Adder	34.5
938871	AE1-115 C	2.0458	50/50	2.0458
938872	AE1-115 E	2.0458	50/50	2.0458
939301	AE1-161 C	2.7151	Adder	3.19
939302	AE1-161 E	4.0726	Adder	4.79
939501	AE1-179 C O1	4.2763	Adder	5.03
939502	AE1-179 E O1	3.0178	Adder	3.55
939821	AE1-218 C O1	0.1055	Adder	0.12
939822	AE1-218 E O1	0.1582	Adder	0.19
939831	AE1-219 C O1	0.2373	Adder	0.28
939832	AE1-219 E O1	0.3427	Adder	0.4
939931	AE1-229 C O1	13.6253	Adder	16.03
939932	AE1-229 E O1	9.2315	Adder	10.86
940001	AE1-240 C O1	3.5597	Adder	4.19
940002	AE1-240 E O1	2.5409	Adder	2.99
940271	AE2-010	-0.2477	Adder	-0.29
940361	AE2-020 C	8.5787	Adder	10.09
940362	AE2-020 E	40.1663	Adder	47.25
940371	AE2-021 C	8.5787	Adder	10.09
940372	AE2-021 E	40.1663	Adder	47.25
940381	AE2-022 C	5.0043	Adder	5.89
940382	AE2-022 E	23.4304	Adder	27.57
940781	AE2-065 C	0.0380	50/50	0.0380
940782	AE2-065 E	0.3372	50/50	0.3372
942101	AE2-222 C O1	7.6099	Adder	8.95

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
942102	AE2-222 E O1	19.1166	Adder	22.49
942381	AE2-251 C	27.1773	Adder	31.97
942382	AE2-251 E	69.5391	Adder	81.81
943071	AE2-334 C	3.7883	Adder	4.46
943072	AE2-334 E	2.0195	Adder	2.38
943732	AF1-041 E	0.2122	Adder	0.25
944951	AF1-160 C	0.8218	Adder	0.97
944952	AF1-160 E	0.8629	Adder	1.02
945431	AF1-208 C O1	3.9339	Adder	4.63
945432	AF1-208 E O1	2.6226	Adder	3.09
945731	AF1-238 C	5.9389	Adder	6.99
945732	AF1-238 E	8.9084	Adder	10.48
945741	AF1-239 C	1.3974	Adder	1.64
945742	AF1-239 E	2.0961	Adder	2.47
945971	AF1-262	0.1302	Adder	0.15
999905	MARINGEN 2	0.3563	Adder	0.42
999906	PVILLEG 2	0.1515	Adder	0.18
NEWTON	NEWTON	0.3041	Confirmed LTF	0.3041
FARMERCITY	FARMERCITY	0.0159	Confirmed LTF	0.0159
G-007A	G-007A	4.2220	Confirmed LTF	4.2220
VFT	VFT	7.4885	Confirmed LTF	7.4885
PRAIRIE	PRAIRIE	0.7336	Confirmed LTF	0.7336
COFFEEN	COFFEEN	0.0565	Confirmed LTF	0.0565
CHEOAH	CHEOAH	0.1461	Confirmed LTF	0.1461
EDWARDS	EDWARDS	0.0983	Confirmed LTF	0.0983
TILTON	TILTON	0.1770	Confirmed LTF	0.1770
GIBSON	GIBSON	0.1540	Confirmed LTF	0.1540
CALDERWOOD	CALDERWOOD	0.1451	Confirmed LTF	0.1451
BLUEG	BLUEG	0.4896	Confirmed LTF	0.4896
TRIMBLE	TRIMBLE	0.1569	Confirmed LTF	0.1569
CATAWBA	CATAWBA	0.1050	Confirmed LTF	0.1050

11.7.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41042049	228714	CNTRL N	AE	228504	SHLDLY T	AE	1	AE_P7-1 AE7TOWER	tower	143.0	82.9	102.29	AC	33.06

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
228102	BLE#2 ST (Deactivation : 30/04/2019)	13.6878	Adder	16.1
228200	CARL#1CT	0.7518	50/50	0.7518
228201	CARL#2CT	0.7958	50/50	0.7958
228206	SHRMN CT	1.8936	50/50	1.8936
228251	CARLLS#4	0.0847	50/50	0.0847
228260	V4-054C	0.2124	50/50	0.2124
228261	V4-054E	0.9785	50/50	0.9785
228702	WEST CT	1.3019	50/50	1.3019
228711	V2-041C	0.0412	50/50	0.0412
228712	V2-041E	0.3796	50/50	0.3796
228717	S121	3.3775	50/50	3.3775
228727	W2-039G	2.7378	50/50	2.7378
924531	AB2-102 C	18.8458	Adder	22.17
924532	AB2-102 E	0.4188	Adder	0.49
938781	AE1-104 C O1	11.0486	Adder	13.0
938782	AE1-104 E O1	28.2676	Adder	33.26
939501	AE1-179 C O1	7.4862	50/50	7.4862
939502	AE1-179 E O1	5.2831	50/50	5.2831
940001	AE1-240 C O1	5.8206	50/50	5.8206
940002	AE1-240 E O1	4.1547	50/50	4.1547
945431	AF1-208 C O1	2.8024	Adder	3.3
945432	AF1-208 E O1	1.8683	Adder	2.2
945731	AF1-238 C	13.2248	50/50	13.2248
945732	AF1-238 E	19.8372	50/50	19.8372
945741	AF1-239 C	3.1117	50/50	3.1117
945742	AF1-239 E	4.6676	50/50	4.6676
NEWTON	NEWTON	0.0527	Confirmed LTF	0.0527
FARMERCITY	FARMERCITY	0.0027	Confirmed LTF	0.0027
NY	NY	0.0426	Confirmed LTF	0.0426
PRAIRIE	PRAIRIE	0.1266	Confirmed LTF	0.1266
O-066	O-066	0.5779	Confirmed LTF	0.5779
COFFEEN	COFFEEN	0.0098	Confirmed LTF	0.0098
CHEOAH	CHEOAH	0.0245	Confirmed LTF	0.0245
EDWARDS	EDWARDS	0.0175	Confirmed LTF	0.0175
TILTON	TILTON	0.0309	Confirmed LTF	0.0309
G-007	G-007	0.0666	Confirmed LTF	0.0666
MADISON	MADISON	0.0020	Confirmed LTF	0.0020
GIBSON	GIBSON	0.0268	Confirmed LTF	0.0268
CALDERWOOD	CALDERWOOD	0.0244	Confirmed LTF	0.0244
BLUEG	BLUEG	0.0851	Confirmed LTF	0.0851
TRIMBLE	TRIMBLE	0.0273	Confirmed LTF	0.0273

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
CATAWBA	CATAWBA	0.0168	Confirmed LTF	0.0168

11.7.4 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42777536	206294	28LARRABEE	JCP&L	206318	28VANHISVL	JCP&L	1	JC-P7-1-JCC-230-11A	tower	869.0	106.39	107.59	AC	10.6

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206201	28JERSVGEN	0.0435	50/50	0.0435
206271	28MCRC/REC	0.9134	50/50	0.9134
206306	28LKWD G1	2.9436	50/50	2.9436
206308	28LKWD G2	2.9436	50/50	2.9436
206312	28LKWD G3	2.1580	50/50	2.1580
206366	28LKWD CT1	5.9331	50/50	5.9331
206367	28LKWD CT2	5.9331	50/50	5.9331
206368	28MDLSEXCO	0.3827	50/50	0.3827
206370	LKWD CT3	39.8560	50/50	39.8560
206412	28R11	37.6805	Adder	44.33
206414	28X4-005E	1.5739	Adder	1.85
206416	28MLH_W1-032	0.0230	50/50	0.0230
207143	28BYD_X4-031	0.2663	50/50	0.2663
207144	28HOW_X1-037	0.1440	50/50	0.1440
207145	28FRN_Y2-051	0.0353	50/50	0.0353
207148	28MDF_W3-079	0.0569	50/50	0.0569
207169	28W4-060COL	0.0775	50/50	0.0775
207204	28HOL_W1-112	0.0359	50/50	0.0359
207205	28MAN_W1-024	0.0302	50/50	0.0302
207206	28TIN_W1-124	0.1623	50/50	0.1623
207401	AA1-060 BAT	4.8964	50/50	4.8964
207419	AB2-139 E	0.0341	Adder	0.04
207420	AC1-207 C	0.4228	Adder	0.5
227928	V4-067E	-0.1854	Adder	-0.22
901032	W1-024E OP1	0.2808	50/50	0.2808
901112	W1-032 E OP1	0.2191	50/50	0.2191
901422	W1-113 E	0.3593	Adder	0.42
901912	W1-112E OP1	0.3307	50/50	0.3307
902032	W1-124E	1.5037	50/50	1.5037
902322	W2-019 E	-1.0042	Adder	-1.18
902912	W2-078 E	0.4023	Adder	0.47
903982	W3-079 E	0.5318	50/50	0.5318
905502	W4-060 E OP1	0.7139	50/50	0.7139
907082	X1-037 E	1.3265	50/50	1.3265
914092	Y2-051 E	0.3250	50/50	0.3250
919662	AA2-048 E	0.8461	50/50	0.8461
920732	AA2-184 E	1.6501	50/50	1.6501
923791	AB2-014	0.2859	50/50	0.2859
924701	AB2-122 C	-0.0801	Adder	-0.09
927132	AC1-207 E	0.7076	Adder	0.83
934351	AD1-059	0.5259	50/50	0.5259

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
934841	AD1-113 (Suspended)	5.9020	Adder	6.94
938423	AE1-061 BAT	0.4930	Merchant Transmission	0.4930
939121	AE1-142 C O1	-2.7307	Adder	-3.21
939303	AE1-161 BAT	3.9545	Merchant Transmission	3.9545
939981	AE1-238 C	35.0865	50/50	35.0865
939982	AE1-238 E	92.1605	50/50	92.1605
940401	AE2-024 C O1	30.9389	50/50	30.9389
940402	AE2-024 E O1	144.8525	50/50	144.8525
940411	AE2-025 C	15.6179	50/50	15.6179
940412	AE2-025 E	73.1149	50/50	73.1149
940691	AE2-056 C	0.2009	50/50	0.2009
940692	AE2-056 E	0.1758	50/50	0.1758
940701	AE2-057 C	0.0613	50/50	0.0613
940702	AE2-057 E	0.0613	50/50	0.0613
940921	AE2-081 C	0.2135	50/50	0.2135
940922	AE2-081 E	0.2197	50/50	0.2197
940931	AE2-082 C	0.2983	Adder	0.35
940932	AE2-082 E	0.3159	Adder	0.37
943521	AF1-023	1.1949	50/50	1.1949
944613	AF1-126 BAT	99.9225	50/50	99.9225
945201	AF1-185 1	0.3541	Adder	0.42
945211	AF1-185 2	0.0787	Adder	0.09
945571	AF1-222 C	21.8706	50/50	21.8706
945572	AF1-222 E	57.6588	50/50	57.6588
945733	AF1-238 BAT	10.5991	Merchant Transmission	10.5991
945743	AF1-239 BAT	2.4939	Merchant Transmission	2.4939
945951	AF1-260	0.1209	50/50	0.1209
999905	MARINGEN 2	-0.4480	Adder	-0.53
999906	PVILLEG 2	-0.1926	Adder	-0.23
NEWTON	NEWTON	0.0484	Confirmed LTF	0.0484
FARMERCITY	FARMERCITY	0.0027	Confirmed LTF	0.0027
G-007A	G-007A	22.1961	Confirmed LTF	22.1961
VFT	VFT	23.1749	Confirmed LTF	23.1749
DUCKCREEK /* 35% REVERSE 3023630 4273103 TURNED OFF IN CASE FOR DEACTIVATION	DUCKCREEK /* 35% REVERSE 3023630 4273103 TURNED OFF IN CASE FOR DEACTIVATION	0.0000	LTF	0.0000
PRAIRIE	PRAIRIE	0.1266	Confirmed LTF	0.1266
COFFEEN	COFFEEN	0.0090	Confirmed LTF	0.0090
CHEOAH	CHEOAH	0.0375	Confirmed LTF	0.0375
EDWARDS	EDWARDS	0.0144	Confirmed LTF	0.0144
TILTON	TILTON	0.0258	Confirmed LTF	0.0258
GIBSON	GIBSON	0.0240	Confirmed LTF	0.0240
CALDERWOOD	CALDERWOOD	0.0368	Confirmed LTF	0.0368
BLUEG	BLUEG	0.0764	Confirmed LTF	0.0764
TRIMBLE	TRIMBLE	0.0239	Confirmed LTF	0.0239
CATAWBA	CATAWBA	0.0346	Confirmed LTF	0.0346

11.7.5 Index 5

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42777150	206297	28MANITOU	JCP&L	206319	28WHITINGS	JCP&L	1	JC-P7-1-JCC-230-12	tower	817.0	160.62	161.53	AC	8.57

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	280 C GEN (Deactivation : 17/09/2018)	121.0661	50/50	121.0661
206360	280 CRK C1	3.2431	50/50	3.2431
206361	280 CRK C2	2.1704	50/50	2.1704
227842	MARINGEN	0.0979	50/50	0.0979
227927	V4-067C	0.0261	50/50	0.0261
227928	V4-067E	0.2433	50/50	0.2433
228014	PVILLEG	0.0340	50/50	0.0340
228102	BLE#2 ST (Deactivation : 30/04/2019)	18.2751	50/50	18.2751
228261	V4-054E	0.3066	Adder	0.36
228712	V2-041E	0.1648	Adder	0.19
228729	W2-030 C	0.0923	50/50	0.0923
228731	V3-036	0.1424	50/50	0.1424
228732	V1-021 C	0.0363	50/50	0.0363
902092	W1-130E	0.3559	Adder	0.42
902322	W2-019 E	-0.1801	Adder	-0.21
902432	W2-030 E	0.8571	50/50	0.8571
923293	AB1-138 E	0.3486	Adder	0.41
923532	AB1-169AC	34.6745	Adder	40.79
924531	AB2-102 C	15.0609	Adder	17.72
924532	AB2-102 E	0.3347	Adder	0.39
924701	AB2-122 C	0.1051	50/50	0.1051
924702	AB2-122 E	0.1802	50/50	0.1802
930001	AB1-001 C	0.1591	50/50	0.1591
930002	AB1-001 E	0.2616	50/50	0.2616
933962	AD1-019 E	1.1110	50/50	1.1110
938421	AE1-061 C	0.1717	Adder	0.2
938422	AE1-061 E	0.1717	Adder	0.2
938781	AE1-104 C O1	14.7513	50/50	14.7513
938782	AE1-104 E O1	37.7410	50/50	37.7410
939121	AE1-142 C O1	3.8891	50/50	3.8891
939122	AE1-142 E O1	5.5965	50/50	5.5965
939501	AE1-179 C O1	1.9427	Adder	2.29
939502	AE1-179 E O1	1.3710	Adder	1.61
940001	AE1-240 C O1	1.6143	Adder	1.9
940002	AE1-240 E O1	1.1523	Adder	1.36
940161	AE2-000 C O1	123.4895	50/50	123.4895
940162	AE2-000 E O1	315.9673	50/50	315.9673
940361	AE2-020 C	17.0432	50/50	17.0432
940362	AE2-020 E	79.7974	50/50	79.7974
940371	AE2-021 C	17.0432	50/50	17.0432

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
940372	AE2-021 E	79.7974	50/50	79.7974
940381	AE2-022 C	9.9418	50/50	9.9418
940382	AE2-022 E	46.5485	50/50	46.5485
942101	AE2-222 C O1	12.5636	50/50	12.5636
942102	AE2-222 E O1	31.5604	50/50	31.5604
942381	AE2-251 C	53.9925	50/50	53.9925
942382	AE2-251 E	138.1515	50/50	138.1515
943732	AF1-041 E	0.3565	50/50	0.3565
944332	AF1-101 E O1	309.7740	50/50	309.7740
944613	AF1-126 BAT	16.1750	Merchant Transmission	16.1750
945731	AF1-238 C	2.9153	Adder	3.43
945732	AF1-238 E	4.3729	Adder	5.14
945741	AF1-239 C	0.6859	Adder	0.81
945742	AF1-239 E	1.0289	Adder	1.21
999905	MARINGEN 2	0.5821	50/50	0.5821
999906	PVILLEG 2	0.2513	50/50	0.2513
NEWTON	NEWTON	0.0064	Confirmed LTF	0.0064
CPL	CPL	0.0146	Confirmed LTF	0.0146
FARMERCITY	FARMERCITY	0.0003	Confirmed LTF	0.0003
NY	NY	0.4933	Confirmed LTF	0.4933
PRAIRIE	PRAIRIE	0.0129	Confirmed LTF	0.0129
TVA	TVA	0.0014	Confirmed LTF	0.0014
O-066	O-066	8.9040	Confirmed LTF	8.9040
COFFEEN	COFFEEN	0.0014	Confirmed LTF	0.0014
EDWARDS	EDWARDS	0.0028	Confirmed LTF	0.0028
CBM-S2	CBM-S2	0.0867	Confirmed LTF	0.0867
TILTON	TILTON	0.0050	Confirmed LTF	0.0050
G-007	G-007	3.1678	Confirmed LTF	3.1678
MADISON	MADISON	0.0060	Confirmed LTF	0.0060
GIBSON	GIBSON	0.0038	Confirmed LTF	0.0038
BLUEG	BLUEG	0.0122	Confirmed LTF	0.0122
TRIMBLE	TRIMBLE	0.0039	Confirmed LTF	0.0039

11.7.6 Index 6

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41041986	206302	28OYSTER C	JCP&L	227955	CEDAR	AE	1	JC-P7-1-JCC-230-10A	tower	564.0	144.93	146.64	AC	9.67

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206306	28LKWD G1	5.0724	50/50	5.0724
206308	28LKWD G2	5.0724	50/50	5.0724
206312	28LKWD G3	3.7187	50/50	3.7187
206325	28O C GEN (Deactivation : 17/09/2018)	90.7518	50/50	90.7518
206360	28O CRK C1	2.4310	50/50	2.4310
206361	28O CRK C2	1.6269	50/50	1.6269
206366	28LKWD CT1	10.2239	50/50	10.2239
206367	28LKWD CT2	10.2239	50/50	10.2239
206370	LKWD CT3	68.6800	50/50	68.6800
207143	28BYD_X4-031	0.1134	Adder	0.13
207400	28AA1-060 E	1.1351	Adder	1.34
227928	V4-067E	-0.2306	Adder	-0.27
228261	V4-054E	-0.3471	Adder	-0.41
901982	W1-119 E	0.5528	Adder	0.65
901992	W1-120E	0.5528	Adder	0.65
902082	W1-129E	0.2365	Adder	0.28
902322	W2-019 E	0.2690	Adder	0.32
902952	W2-082 E OP1	0.8464	Adder	1.0
903982	W3-079 E	0.3398	Adder	0.4
905252	W4-025 E	0.3357	Adder	0.39
907082	X1-037 E	0.5834	Adder	0.69
907272	X1-085 E	0.1767	Adder	0.21
912102	X4-015 E	0.2421	Adder	0.28
914092	Y2-051 E	0.2861	Adder	0.34
917612	Z2-102 E	0.8116	Adder	0.95
919662	AA2-048 E	0.5406	Adder	0.64
923293	AB1-138 E	1.2022	Adder	1.41
923463	AB1-163 E	0.6333	Adder	0.75
923791	AB2-014	0.4926	50/50	0.4926
924701	AB2-122 C	-0.0996	Adder	-0.12
934351	AD1-059	0.9062	50/50	0.9062
938423	AE1-061 BAT	0.4551	Merchant Transmission	0.4551
939121	AE1-142 C O1	2.7174	50/50	2.7174
939122	AE1-142 E O1	3.9104	50/50	3.9104
940161	AE2-000 C O1	92.5684	50/50	92.5684
940162	AE2-000 E O1	236.8508	50/50	236.8508
940401	AE2-024 C O1	8.9129	Adder	10.49
940402	AE2-024 E O1	41.7293	Adder	49.09
940411	AE2-025 C	4.4992	Adder	5.29
940412	AE2-025 E	21.0630	Adder	24.78
940691	AE2-056 C	0.0885	Adder	0.1

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
940692	AE2-056 E	0.0774	Adder	0.09
940921	AE2-081 C	0.0940	Adder	0.11
940922	AE2-081 E	0.0968	Adder	0.11
943561	AF1-027	0.1261	Adder	0.15
944332	AF1-101 E O1	232.2082	50/50	232.2082
944611	AF1-126 C O1	2.8245	Adder	3.32
944612	AF1-126 E O1	11.2982	Adder	13.29
945733	AF1-238 BAT	9.6747	Merchant Transmission	9.6747
945743	AF1-239 BAT	2.2764	Merchant Transmission	2.2764
945951	AF1-260	0.0772	Adder	0.09
999905	MARINGEN 2	-0.5519	Adder	-0.65
999906	PVILLEG 2	-0.2382	Adder	-0.28
NEWTON	NEWTON	0.1289	Confirmed LTF	0.1289
FARMERCITY	FARMERCITY	0.0068	Confirmed LTF	0.0068
G-007A	G-007A	7.9765	Confirmed LTF	7.9765
VFT	VFT	8.7204	Confirmed LTF	8.7204
PRAIRIE	PRAIRIE	0.3125	Confirmed LTF	0.3125
COFFEEN	COFFEEN	0.0237	Confirmed LTF	0.0237
CHEOAH	CHEOAH	0.0661	Confirmed LTF	0.0661
EDWARDS	EDWARDS	0.0413	Confirmed LTF	0.0413
TILTON	TILTON	0.0743	Confirmed LTF	0.0743
GIBSON	GIBSON	0.0650	Confirmed LTF	0.0650
CALDERWOOD	CALDERWOOD	0.0656	Confirmed LTF	0.0656
BLUEG	BLUEG	0.2066	Confirmed LTF	0.2066
TRIMBLE	TRIMBLE	0.0662	Confirmed LTF	0.0662
CATAWBA	CATAWBA	0.0500	Confirmed LTF	0.0500

11.7.7 Index 7

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42777172	206318	28VANHISVL	JCP&L	206294	28LARRABEE	JCP&L	1	JC-P7-1-JCC-230-12	tower	869.0	146.18	147.22	AC	10.98

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206280	28LAKEHURS	0.4963	50/50	0.4963
206325	28O C GEN (Deactivation : 17/09/2018)	88.3262	50/50	88.3262
206360	28O CRK C1	2.3660	50/50	2.3660
206361	28O CRK C2	1.5834	50/50	1.5834
206414	28X4-005E	-1.6091	Adder	-1.89
207149	28UPF_X1-085	0.0636	50/50	0.0636
207150	28HAN_X4-015	0.1074	50/50	0.1074
207157	28PEM_W1-119	0.1977	50/50	0.1977
207158	28PEM_W1-120	0.1977	50/50	0.1977
207160	28EGP_W2-019	0.1286	50/50	0.1286
207164	28NHV_W2-082	0.3693	50/50	0.3693
207171	28DIX_Z2-102	0.2920	50/50	0.2920
207207	28JCB_W1-129	0.1070	50/50	0.1070
207208	28JCB_W4-025	0.1464	50/50	0.1464
207400	28AA1-060 E	4.3463	50/50	4.3463
207419	AB2-139 E	-0.0351	Adder	-0.04
207420	AC1-207 C	-0.4341	Adder	-0.51
218660	EHAMPSOLAR C	0.0236	50/50	0.0236
218661	EHAMPSOLAR E	0.2183	50/50	0.2183
219272	CLEANLIGHT C	0.0202	50/50	0.0202
219273	CLEANLIGHT E	0.1876	50/50	0.1876
219287	REEVES ST C	0.1064	50/50	0.1064
219288	REEVES ST E	0.9802	50/50	0.9802
219291	REEVESSO C	0.0080	50/50	0.0080
219292	REEVESSO E	0.0736	50/50	0.0736
219579	COXSCOSP1_C	0.0010	50/50	0.0010
219582	COXSCOSP1_E	0.0087	50/50	0.0087
219609	COXSCOSP2_C	0.0010	50/50	0.0010
219614	COXSCOSP2_E	0.0087	50/50	0.0087
219628	LUMB23SP1_C	0.0033	50/50	0.0033
219652	LUMB23SP1_E	0.0299	50/50	0.0299
219653	LUMB23SP2_C	0.0033	50/50	0.0033
219656	LUMB23SP2_E	0.0299	50/50	0.0299
219686	LUMSOLAR_N C	0.0331	50/50	0.0331
219687	LUMSOLAR_N E	0.3018	50/50	0.3018
219688	LUMSOLAR_S C	0.0237	50/50	0.0237
219689	LUMSOLAR_S E	0.2163	50/50	0.2163
227842	MARINGEN	0.0928	50/50	0.0928
227927	V4-067C	0.0245	50/50	0.0245
227928	V4-067E	0.2283	50/50	0.2283
228014	PVILLEG	0.0321	50/50	0.0321

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
228102	BLE#2 ST (Deactivation : 30/04/2019)	18.2210	50/50	18.2210
228712	V2-041E	0.1931	Adder	0.23
228729	W2-030 C	0.0879	50/50	0.0879
228731	V3-036	0.1348	50/50	0.1348
228732	V1-021 C	0.0381	50/50	0.0381
901982	W1-119 E	1.8243	50/50	1.8243
901992	W1-120E	1.8243	50/50	1.8243
902082	W1-129E	0.9856	50/50	0.9856
902092	W1-130E	0.4501	Adder	0.53
902322	W2-019 E	1.1678	50/50	1.1678
902432	W2-030 E	0.8158	50/50	0.8158
902691	W2-056 C	0.0925	50/50	0.0925
902692	W2-056 E	0.1510	50/50	0.1510
902952	W2-082 E OP1	3.3679	50/50	3.3679
905252	W4-025 E	1.3989	50/50	1.3989
907272	X1-085 E	0.5859	50/50	0.5859
912102	X4-015 E	0.9892	50/50	0.9892
917612	Z2-102 E	2.6902	50/50	2.6902
919663	AA2-048 BAT	1.0035	50/50	1.0035
923292	AB1-138 C	0.1267	50/50	0.1267
923293	AB1-138 E	1.1925	50/50	1.1925
923463	AB1-163 E	0.5686	50/50	0.5686
923532	AB1-169AC	43.6458	Adder	51.35
924531	AB2-102 C	17.3406	Adder	20.4
924532	AB2-102 E	0.3853	Adder	0.45
924701	AB2-122 C	0.0986	50/50	0.0986
924702	AB2-122 E	0.1691	50/50	0.1691
930001	AB1-001 C	0.1412	50/50	0.1412
930002	AB1-001 E	0.2320	50/50	0.2320
932361	AC2-050 C O1	0.4066	50/50	0.4066
932362	AC2-050 E O1	0.6635	50/50	0.6635
933962	AD1-019 E	1.0513	50/50	1.0513
938421	AE1-061 C	0.2172	Adder	0.26
938422	AE1-061 E	0.2172	Adder	0.26
938781	AE1-104 C O1	14.7076	50/50	14.7076
938782	AE1-104 E O1	37.6292	50/50	37.6292
939121	AE1-142 C O1	3.2835	50/50	3.2835
939122	AE1-142 E O1	4.7251	50/50	4.7251
939501	AE1-179 C O1	2.5192	Adder	2.96
939502	AE1-179 E O1	1.7779	Adder	2.09
940001	AE1-240 C O1	2.0874	Adder	2.46
940002	AE1-240 E O1	1.4899	Adder	1.75
940161	AE2-000 C O1	90.0943	50/50	90.0943
940162	AE2-000 E O1	230.5203	50/50	230.5203
940361	AE2-020 C	15.7095	50/50	15.7095
940362	AE2-020 E	73.5529	50/50	73.5529
940371	AE2-021 C	15.7095	50/50	15.7095
940372	AE2-021 E	73.5529	50/50	73.5529
940381	AE2-022 C	9.1639	50/50	9.1639
940382	AE2-022 E	42.9059	50/50	42.9059
940691	AE2-056 C	-0.1905	Adder	-0.22
940701	AE2-057 C	-0.0567	Adder	-0.07

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
940921	AE2-081 C	-0.2024	Adder	-0.24
942101	AE2-222 C O1	11.8700	50/50	11.8700
942102	AE2-222 E O1	29.8180	50/50	29.8180
942381	AE2-251 C	49.7673	50/50	49.7673
942382	AE2-251 E	127.3407	50/50	127.3407
943561	AF1-027	0.4829	50/50	0.4829
943732	AF1-041 E	0.3359	50/50	0.3359
944332	AF1-101 E O1	226.0018	50/50	226.0018
944611	AF1-126 C O1	20.0345	50/50	20.0345
944612	AF1-126 E O1	80.1380	50/50	80.1380
945201	AF1-185 1	-0.3620	Adder	-0.43
945211	AF1-185 2	-0.0805	Adder	-0.09
945731	AF1-238 C	3.7342	Adder	4.39
945732	AF1-238 E	5.6013	Adder	6.59
945741	AF1-239 C	0.8786	Adder	1.03
945742	AF1-239 E	1.3179	Adder	1.55
999905	MARINGEN 2	0.5514	50/50	0.5514
999906	PVILLEG 2	0.2371	50/50	0.2371
WEC	WEC	0.0126	Confirmed LTF	0.0126
LGEE	LGEE	0.0257	Confirmed LTF	0.0257
CPL	CPL	0.0800	Confirmed LTF	0.0800
CBM-W2	CBM-W2	0.4586	Confirmed LTF	0.4586
NY	NY	1.3676	Confirmed LTF	1.3676
TVA	TVA	0.0924	Confirmed LTF	0.0924
O-066	O-066	25.1194	Confirmed LTF	25.1194
CBM-S2	CBM-S2	0.6011	Confirmed LTF	0.6011
CBM-S1	CBM-S1	0.5282	Confirmed LTF	0.5282
G-007	G-007	9.8436	Confirmed LTF	9.8436
MADISON	MADISON	0.0141	Confirmed LTF	0.0141
MEC	MEC	0.0731	Confirmed LTF	0.0731
CBM-W1	CBM-W1	0.4378	Confirmed LTF	0.4378

11.7.8 Index 8

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC D C	MW IMPAC T
42251090	213922	RICHMOND	PECO	214012	WANEETA3	PECO	1	PECO_P4_CHICH045/ * \$ DELCO \$ CHICH045 \$ STBK	breaker	1180.0	135.18	137.12	AC	26.79

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
213918	RICHMD91	4.0860	50/50	4.0860
213919	RICHMD92	4.0860	50/50	4.0860
214225	Y1-057 BAT	0.3380	Merchant Transmission	0.3380
218661	EHAMPSOLAR E	0.3680	Adder	0.43
219124	CAMDEN_STG	4.0490	50/50	4.0490
219126	CAMDEN_CTG	4.9743	50/50	4.9743
219128	GLOUCSTR_26	2.0555	50/50	2.0555
219229	EAGLEPT_G3	2.6152	50/50	2.6152
219230	EAGLEPT_G1	4.9345	50/50	4.9345
219231	EAGLEPT_G2	4.9318	50/50	4.9318
219235	EAGLEPT_ST2	1.7837	50/50	1.7837
219236	SOUTHERNHQ C	0.0204	50/50	0.0204
219237	SOUTHERNHQ E	0.1909	50/50	0.1909
219240	MANTUACREK E	0.7013	50/50	0.7013
219241	CAMDEN1	0.3007	50/50	0.3007
219242	CAMDEN2	0.0325	50/50	0.0325
219258	KINSLEYDEP C	0.1124	50/50	0.1124
219259	KINSLEYDEP E	1.0354	50/50	1.0354
219273	CLEANLIGHT E	0.3812	Adder	0.45
219288	REEVES ST E	1.3713	Adder	1.61
219292	REEVESSO E	0.1264	Adder	0.15
219305	NAMERICA E	0.9503	Adder	1.12
219340	CUTHBSP1_C	0.0065	50/50	0.0065
219344	CUTHBSP1_E	0.0592	50/50	0.0592
219345	CUTHBSP2_C	0.0065	50/50	0.0065
219347	CUTHBSP2_E	0.0592	50/50	0.0592
219349	LDSOLAR E	0.8320	Adder	0.98
219350	CUTHBSP3_C	0.0065	50/50	0.0065
219351	CUTHBSP3_E	0.0592	50/50	0.0592
219357	CUTHBSP4_C	0.0052	50/50	0.0052
219379	CUTHBSP4_E	0.0476	50/50	0.0476
219384	DEPTFDSP1_C	0.0015	50/50	0.0015
219385	DEPTFDSP1_E	0.0137	50/50	0.0137
219388	DEPTFDSP2_C	0.0015	50/50	0.0015
219390	DEPTFDSP2_E	0.0137	50/50	0.0137
219393	DEPTFDSP3_C	0.0037	50/50	0.0037
219394	DEPTFDSP3_E	0.0344	50/50	0.0344
219408	DEPTFDSP4_C	0.0015	50/50	0.0015
219416	DEPTFDSP4_E	0.0137	50/50	0.0137
219566	BURLINSP1_E	0.1141	Adder	0.13
219574	CAMDENSP1_C	0.0361	50/50	0.0361

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
219575	CAMDENSP1_E	0.3329	50/50	0.3329
219582	COXSCOSP1_E	0.0158	Adder	0.02
219608	WPEMNERT E	0.9503	Adder	1.12
219611	FLORENCE E	2.7151	Adder	3.19
219614	COXSCOSP2_E	0.0158	Adder	0.02
219619	MILLCREEK E	0.5310	Adder	0.62
219623	KINSLEYBEA C	0.0822	50/50	0.0822
219624	KINSLEYBEA E	0.7825	50/50	0.7825
219647	COOPERSOLA E	0.1391	50/50	0.1391
219652	LUMB23SP1_E	0.0418	Adder	0.05
219656	LUMB23SP2_E	0.0418	Adder	0.05
219662	THOROFSP1_C	0.0057	50/50	0.0057
219664	THOROFSP1_E	0.0524	50/50	0.0524
219665	THOROFSP2_C	0.0057	50/50	0.0057
219666	THOROFSP2_E	0.0527	50/50	0.0527
219683	THOSOLAR C	0.0315	50/50	0.0315
219684	THOSOLAR E	0.2902	50/50	0.2902
219687	LUMSOLAR_N E	0.4223	Adder	0.5
219689	LUMSOLAR_S E	0.4406	Adder	0.52
219706	BEAVBRSP1_C	0.0059	50/50	0.0059
219707	BEAVBRSP1_E	0.0547	50/50	0.0547
219708	BEAVBRSP2_C	0.0059	50/50	0.0059
219709	BEAVBRSP2_E	0.0546	50/50	0.0546
219710	CINNAMSP1_C	0.0034	50/50	0.0034
219711	CINNAMSP1_E	0.0314	50/50	0.0314
219712	CINNAMSP2_C	0.0020	50/50	0.0020
219713	CINNAMSP2_E	0.0188	50/50	0.0188
219714	CINNAMSP3_C	0.0034	50/50	0.0034
219715	CINNAMSP3_E	0.0314	50/50	0.0314
219716	CINNAMSP4_C	0.0034	50/50	0.0034
219717	CINNAMSP4_E	0.0313	50/50	0.0313
219718	GLOUCSP1_C	0.0111	50/50	0.0111
219719	GLOUCSP1_E	0.1027	50/50	0.1027
219726	BAT.STORG E	0.3410	50/50	0.3410
219765	PARADISE E	1.1220	50/50	1.1220
219766	PARADISE C	0.1180	50/50	0.1180
227881	GRENWCHG	0.3890	Adder	0.46
227928	V4-067E	0.2592	Adder	0.3
228102	BLE#2 ST (Deactivation : 30/04/2019)	24.6885	Adder	29.05
228261	V4-054E	1.1656	Adder	1.37
228334	MANNMILG	1.7280	Adder	2.03
228357	V2-046E	2.6554	Adder	3.12
228712	V2-041E	0.4239	Adder	0.5
228721	V2-035E	0.2442	Adder	0.29
228733	AB1-119 E	0.1417	Adder	0.17
902092	W1-130E	1.0760	Adder	1.27
902432	W2-030 E	0.9706	Adder	1.14
910862	OWENSCORIN E	1.0088	50/50	1.0088
918622	AA1-079 E	7.0461	Adder	8.29
918632	AA1-079 E	7.0435	Adder	8.29
923153	AB1-116 E	0.1561	Adder	0.18
923532	AB1-169AC	103.6681	Adder	121.96

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
924051	AB2-049 C	0.7462	Adder	0.88
924052	AB2-049 E	1.2175	Adder	1.43
924531	AB2-102 C	37.4315	Adder	44.04
924532	AB2-102 E	0.8318	Adder	0.98
924701	AB2-122 C	0.1120	Adder	0.13
924702	AB2-122 E	0.1920	Adder	0.23
925391	AC1-010 C (Suspended)	0.7714	50/50	0.7714
925392	AC1-010 E (Suspended)	1.3324	50/50	1.3324
925441	AC1-016 C (Withdrawn : Unavailable)	0.7313	Adder	0.86
925442	AC1-016 E (Withdrawn : Unavailable)	1.1933	Adder	1.4
925452	AC1-017 E	0.5652	Adder	0.66
925561	AC1-030 C	0.0555	50/50	0.0555
925562	AC1-030 E	0.4951	50/50	0.4951
930001	AB1-001 C	0.1359	Adder	0.16
930002	AB1-001 E	0.2233	Adder	0.26
930242	AB1-063 E	0.1391	50/50	0.1391
932361	AC2-050 C O1	0.7645	Adder	0.9
932362	AC2-050 E O1	1.2474	Adder	1.47
933962	AD1-019 E	1.2298	Adder	1.45
936211	AD2-027 C	0.2470	50/50	0.2470
936212	AD2-027 E	2.2756	50/50	2.2756
936321	AD2-042 C	0.3250	50/50	0.3250
936322	AD2-042 E	3.0097	50/50	3.0097
936491	AD2-064 C (Withdrawn : 06/16/2020)	0.1029	Adder	0.12
936492	AD2-064 E (Withdrawn : 06/16/2020)	0.1417	Adder	0.17
936501	AD2-065 C	0.2105	Adder	0.25
936502	AD2-065 E	0.2904	Adder	0.34
937011	AD2-135 C	0.1143	Adder	0.13
937012	AD2-135 E	0.1942	Adder	0.23
938421	AE1-061 C	0.5191	Adder	0.61
938422	AE1-061 E	0.5191	Adder	0.61
938431	AE1-062 C	1.9392	Adder	2.28
938432	AE1-062 E	1.9392	Adder	2.28
938781	AE1-104 C O1	19.9281	Adder	23.44
938782	AE1-104 E O1	50.9856	Adder	59.98
938871	AE1-115 C	2.0522	Adder	2.41
938872	AE1-115 E	2.0522	Adder	2.41
939301	AE1-161 C	3.9136	Adder	4.6
939302	AE1-161 E	5.8704	Adder	6.91
939501	AE1-179 C O1	6.3341	Adder	7.45
939502	AE1-179 E O1	4.4700	Adder	5.26
939821	AE1-218 C O1	0.1626	Adder	0.19
939822	AE1-218 E O1	0.2439	Adder	0.29
939831	AE1-219 C O1	0.3658	Adder	0.43
939832	AE1-219 E O1	0.5283	Adder	0.62
939931	AE1-229 C O1	17.6325	Adder	20.74
939932	AE1-229 E O1	11.9465	Adder	14.05
940001	AE1-240 C O1	5.2490	Adder	6.18
940002	AE1-240 E O1	3.7467	Adder	4.41

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
940271	AE2-010	-0.5760	Adder	-0.68
940361	AE2-020 C	16.7612	Adder	19.72
940362	AE2-020 E	78.4772	Adder	92.33
940371	AE2-021 C	16.7612	Adder	19.72
940372	AE2-021 E	78.4772	Adder	92.33
940381	AE2-022 C	9.7774	Adder	11.5
940382	AE2-022 E	45.7784	Adder	53.86
940771	AE2-064 C	0.0499	50/50	0.0499
940772	AE2-064 E	0.3813	50/50	0.3813
940781	AE2-065 C	0.0414	50/50	0.0414
940782	AE2-065 E	0.3674	50/50	0.3674
942101	AE2-222 C O1	13.8236	Adder	16.26
942102	AE2-222 E O1	34.7258	Adder	40.85
942381	AE2-251 C	53.0992	Adder	62.47
942382	AE2-251 E	135.8660	Adder	159.84
943071	AE2-334 C	5.7726	Adder	6.79
943072	AE2-334 E	3.0774	Adder	3.62
943732	AF1-041 E	0.3875	Adder	0.46
944951	AF1-160 C	1.9392	Adder	2.28
944952	AF1-160 E	2.0361	Adder	2.4
945431	AF1-208 C O1	5.1984	Adder	6.12
945432	AF1-208 E O1	3.4656	Adder	4.08
945731	AF1-238 C	9.1091	Adder	10.72
945732	AF1-238 E	13.6637	Adder	16.07
945741	AF1-239 C	2.1433	Adder	2.52
945742	AF1-239 E	3.2150	Adder	3.78
945961	AF1-261	0.1971	Adder	0.23
945971	AF1-262	0.1772	Adder	0.21
945991	AF1-264	0.4372	50/50	0.4372
999905	MARINGEN 2	0.6473	Adder	0.76
999906	PVILLEG 2	0.2751	Adder	0.32
NEWTON	NEWTON	1.2851	Confirmed LTF	1.2851
FARMERCITY	FARMERCITY	0.0669	Confirmed LTF	0.0669
G-007A	G-007A	12.9225	Confirmed LTF	12.9225
VFT	VFT	23.4909	Confirmed LTF	23.4909
DUCKCREEK /* 35% REVERSE 3023630 4273103 TURNED OFF IN CASE FOR DEACTIVATION	DUCKCREEK /* 35% REVERSE 3023630 4273103 TURNED OFF IN CASE FOR DEACTIVATION	0.0000	LTF	0.0000
PRAIRIE	PRAIRIE	3.0867	Confirmed LTF	3.0867
COFFEEN	COFFEEN	0.2386	Confirmed LTF	0.2386
CHEOAH	CHEOAH	0.5951	Confirmed LTF	0.5951
EDWARDS	EDWARDS	0.4190	Confirmed LTF	0.4190
TILTON	TILTON	0.7541	Confirmed LTF	0.7541
MADISON	MADISON	0.0040	Confirmed LTF	0.0040
GIBSON	GIBSON	0.6530	Confirmed LTF	0.6530
CALDERWOOD	CALDERWOOD	0.5914	Confirmed LTF	0.5914
BLUEG	BLUEG	2.0763	Confirmed LTF	2.0763
TRIMBLE	TRIMBLE	0.6656	Confirmed LTF	0.6656
CATAWBA	CATAWBA	0.4144	Confirmed LTF	0.4144

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	ACID C	MW IMPACT
42250940	214206	RICHRE29	PECO	213922	RICHMOND	PECO	1	PECO_P4_CHICH045/ * \$ DELCO \$ CHICH045 \$ STBK	breaker	1336.0	133.66	135.49	AC	28.46

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
218661	EHAMPSOLAR E	0.3427	Adder	0.4
219124	CAMDEN_STG	4.5219	50/50	4.5219
219126	CAMDEN_CTG	5.5552	50/50	5.5552
219128	GLOUCSTR_26	2.2956	50/50	2.2956
219229	EAGLEPT_G3	2.9118	50/50	2.9118
219230	EAGLEPT_G1	5.4944	50/50	5.4944
219231	EAGLEPT_G2	5.4912	50/50	5.4912
219235	EAGLEPT_ST2	1.9861	50/50	1.9861
219236	SOUTHERNHQ C	0.0204	50/50	0.0204
219237	SOUTHERNHQ E	0.1912	50/50	0.1912
219240	MANTUACREK E	0.7832	50/50	0.7832
219241	CAMDEN1	0.3371	50/50	0.3371
219242	CAMDEN2	0.0364	50/50	0.0364
219258	KINSLEYDEP C	0.1251	50/50	0.1251
219259	KINSLEYDEP E	1.1525	50/50	1.1525
219273	CLEANLIGHT E	0.3710	Adder	0.44
219288	REEVES ST E	1.2933	Adder	1.52
219292	REEVESSO E	0.1184	Adder	0.14
219305	NAMERICA E	0.6477	Adder	0.76
219340	CUTHBSP1_C	0.0072	50/50	0.0072
219344	CUTHBSP1_E	0.0663	50/50	0.0663
219345	CUTHBSP2_C	0.0072	50/50	0.0072
219347	CUTHBSP2_E	0.0662	50/50	0.0662
219349	LDSOLAR E	0.5671	Adder	0.67
219350	CUTHBSP3_C	0.0072	50/50	0.0072
219351	CUTHBSP3_E	0.0663	50/50	0.0663
219357	CUTHBSP4_C	0.0058	50/50	0.0058
219379	CUTHBSP4_E	0.0533	50/50	0.0533
219384	DEPTFDSP1_C	0.0016	50/50	0.0016
219385	DEPTFDSP1_E	0.0152	50/50	0.0152
219388	DEPTFDSP2_C	0.0016	50/50	0.0016
219390	DEPTFDSP2_E	0.0152	50/50	0.0152
219393	DEPTFDSP3_C	0.0041	50/50	0.0041
219394	DEPTFDSP3_E	0.0382	50/50	0.0382
219408	DEPTFDSP4_C	0.0016	50/50	0.0016
219416	DEPTFDSP4_E	0.0152	50/50	0.0152
219566	BURLINSP1_E	0.0778	Adder	0.09
219574	CAMDENSP1_C	0.0405	50/50	0.0405
219575	CAMDENSP1_E	0.3732	50/50	0.3732
219582	COXSCOSP1_E	0.0149	Adder	0.02
219608	WPEMNERT E	0.6477	Adder	0.76

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
219611	FLORENCE E	1.8505	Adder	2.18
219614	COXSCOSP2_E	0.0149	Adder	0.02
219619	MILLCREEK E	0.4441	Adder	0.52
219623	KINSLEYBEA C	0.0912	50/50	0.0912
219624	KINSLEYBEA E	0.8684	50/50	0.8684
219647	COOPERSOLA E	0.1530	50/50	0.1530
219652	LUMB23SP1_E	0.0394	Adder	0.05
219656	LUMB23SP2_E	0.0394	Adder	0.05
219662	THOROFSP1_C	0.0063	50/50	0.0063
219664	THOROFSP1_E	0.0581	50/50	0.0581
219665	THOROFSP2_C	0.0063	50/50	0.0063
219666	THOROFSP2_E	0.0584	50/50	0.0584
219683	THOSOLAR C	0.0349	50/50	0.0349
219684	THOSOLAR E	0.3215	50/50	0.3215
219687	LUMSOLAR_N E	0.3982	Adder	0.47
219689	LUMSOLAR_S E	0.3950	Adder	0.46
219706	BEAVBRSP1_C	0.0066	50/50	0.0066
219707	BEAVBRSP1_E	0.0607	50/50	0.0607
219708	BEAVBRSP2_C	0.0066	50/50	0.0066
219709	BEAVBRSP2_E	0.0606	50/50	0.0606
219710	CINNAMSP1_C	0.0034	50/50	0.0034
219711	CINNAMSP1_E	0.0315	50/50	0.0315
219712	CINNAMSP2_C	0.0020	50/50	0.0020
219713	CINNAMSP2_E	0.0188	50/50	0.0188
219714	CINNAMSP3_C	0.0034	50/50	0.0034
219715	CINNAMSP3_E	0.0315	50/50	0.0315
219716	CINNAMSP4_C	0.0034	50/50	0.0034
219717	CINNAMSP4_E	0.0313	50/50	0.0313
219718	GLOUCSP1_C	0.0124	50/50	0.0124
219719	GLOUCSP1_E	0.1147	50/50	0.1147
219726	BAT.STORG E	0.3772	50/50	0.3772
219765	PARADISE E	1.2531	50/50	1.2531
219766	PARADISE C	0.1318	50/50	0.1318
227881	GRENWCHG	0.4249	Adder	0.5
227928	V4-067E	0.2701	Adder	0.32
228102	BLE#2 ST (Deactivation : 30/04/2019)	25.9093	Adder	30.48
228261	V4-054E	1.2471	Adder	1.47
228334	MANNMILG	1.8680	Adder	2.2
228357	V2-046E	2.8690	Adder	3.38
228400	MICK 1CT	2.5152	50/50	2.5152
228423	Q-090 2	41.9172	50/50	41.9172
228471	VALERO1	0.8589	50/50	0.8589
228472	VALERO2	0.5666	50/50	0.5666
228473	VALERO3	0.5666	50/50	0.5666
228484	VALERO4	0.5053	50/50	0.5053
228712	V2-041E	0.4493	Adder	0.53
228721	V2-035E	0.2618	Adder	0.31
228733	AB1-119 E	0.1471	Adder	0.17
902092	W1-130E	1.1442	Adder	1.35
902432	W2-030 E	1.0135	Adder	1.19
910862	OWENSCORIN E	1.1159	50/50	1.1159
917471	Z2-083	4.2404	50/50	4.2404

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
919901	AB1-000 1	0.9070	Adder	1.07
919911	AB1-000 2	0.9070	Adder	1.07
919921	AB1-000 3	0.9070	Adder	1.07
923153	AB1-116 E	0.1646	Adder	0.19
923532	AB1-169AC	110.2318	Adder	129.68
924051	AB2-049 C	0.7810	Adder	0.92
924052	AB2-049 E	1.2742	Adder	1.5
924531	AB2-102 C	39.6461	Adder	46.64
924532	AB2-102 E	0.8810	Adder	1.04
924701	AB2-122 C	0.1167	Adder	0.14
924702	AB2-122 E	0.2001	Adder	0.24
925391	AC1-010 C (Suspended)	0.8615	50/50	0.8615
925392	AC1-010 E (Suspended)	1.4880	50/50	1.4880
925441	AC1-016 C (Withdrawn : Unavailable)	0.4984	Adder	0.59
925442	AC1-016 E (Withdrawn : Unavailable)	0.8133	Adder	0.96
925452	AC1-017 E	0.4728	Adder	0.56
925561	AC1-030 C	0.0558	50/50	0.0558
925562	AC1-030 E	0.4976	50/50	0.4976
930001	AB1-001 C	0.1410	Adder	0.17
930002	AB1-001 E	0.2317	Adder	0.27
930242	AB1-063 E	0.1530	50/50	0.1530
932361	AC2-050 C O1	0.7355	Adder	0.87
932362	AC2-050 E O1	1.2001	Adder	1.41
933962	AD1-019 E	1.2837	Adder	1.51
934661	AD1-097 1	1.4633	Adder	1.72
934671	AD1-097 2	1.4633	Adder	1.72
934681	AD1-097 3	1.4633	Adder	1.72
934691	AD1-097 4	0.8662	Adder	1.02
936211	AD2-027 C	0.2769	50/50	0.2769
936212	AD2-027 E	2.5512	50/50	2.5512
936321	AD2-042 C	0.3644	50/50	0.3644
936322	AD2-042 E	3.3741	50/50	3.3741
936491	AD2-064 C (Withdrawn : 06/16/2020)	0.1068	Adder	0.13
936492	AD2-064 E (Withdrawn : 06/16/2020)	0.1471	Adder	0.17
936501	AD2-065 C	0.2157	Adder	0.25
936502	AD2-065 E	0.2975	Adder	0.35
937011	AD2-135 C	0.1193	Adder	0.14
937012	AD2-135 E	0.2029	Adder	0.24
938421	AE1-061 C	0.5520	Adder	0.65
938422	AE1-061 E	0.5520	Adder	0.65
938431	AE1-062 C	1.9904	Adder	2.34
938432	AE1-062 E	1.9904	Adder	2.34
938781	AE1-104 C O1	20.9135	Adder	24.6
938782	AE1-104 E O1	53.5069	Adder	62.95
938871	AE1-115 C	2.2232	Adder	2.62
938872	AE1-115 E	2.2232	Adder	2.62
939301	AE1-161 C	4.1781	Adder	4.92
939302	AE1-161 E	6.2671	Adder	7.37
939501	AE1-179 C O1	6.7443	Adder	7.93

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
939502	AE1-179 E O1	4.7596	Adder	5.6
939821	AE1-218 C O1	0.1726	Adder	0.2
939822	AE1-218 E O1	0.2589	Adder	0.3
939831	AE1-219 C O1	0.3884	Adder	0.46
939832	AE1-219 E O1	0.5610	Adder	0.66
939931	AE1-229 C O1	18.9715	Adder	22.32
939932	AE1-229 E O1	12.8537	Adder	15.12
940001	AE1-240 C O1	5.5909	Adder	6.58
940002	AE1-240 E O1	3.9907	Adder	4.69
940271	AE2-010	-0.5403	Adder	-0.64
940361	AE2-020 C	17.4045	Adder	20.48
940362	AE2-020 E	81.4891	Adder	95.87
940371	AE2-021 C	17.4045	Adder	20.48
940372	AE2-021 E	81.4891	Adder	95.87
940381	AE2-022 C	10.1526	Adder	11.94
940382	AE2-022 E	47.5353	Adder	55.92
940771	AE2-064 C	0.0554	50/50	0.0554
940772	AE2-064 E	0.4232	50/50	0.4232
940781	AE2-065 C	0.0461	50/50	0.0461
940782	AE2-065 E	0.4090	50/50	0.4090
942101	AE2-222 C O1	14.4270	Adder	16.97
942102	AE2-222 E O1	36.2415	Adder	42.64
942381	AE2-251 C	55.1371	Adder	64.87
942382	AE2-251 E	141.0803	Adder	165.98
943071	AE2-334 C	6.1390	Adder	7.22
943072	AE2-334 E	3.2727	Adder	3.85
943732	AF1-041 E	0.4042	Adder	0.48
944951	AF1-160 C	1.9904	Adder	2.34
944952	AF1-160 E	2.0899	Adder	2.46
945431	AF1-208 C O1	5.5808	Adder	6.57
945432	AF1-208 E O1	3.7205	Adder	4.38
945731	AF1-238 C	9.6770	Adder	11.38
945732	AF1-238 E	14.5155	Adder	17.08
945741	AF1-239 C	2.2769	Adder	2.68
945742	AF1-239 E	3.4154	Adder	4.02
945961	AF1-261	0.1343	Adder	0.16
945971	AF1-262	0.1900	Adder	0.22
945991	AF1-264	0.4836	50/50	0.4836
999905	MARINGEN 2	0.6756	Adder	0.79
999906	PVILLEG 2	0.2871	Adder	0.34
NEWTON	NEWTON	1.0831	Confirmed LTF	1.0831
FARMERCITY	FARMERCITY	0.0563	Confirmed LTF	0.0563
G-007A	G-007A	10.1510	Confirmed LTF	10.1510
VFT	VFT	15.1511	Confirmed LTF	15.1511
PRAIRIE	PRAIRIE	2.5959	Confirmed LTF	2.5959
COFFEEN	COFFEEN	0.2011	Confirmed LTF	0.2011
CHEOAH	CHEOAH	0.4965	Confirmed LTF	0.4965
EDWARDS	EDWARDS	0.3535	Confirmed LTF	0.3535
TILTON	TILTON	0.6363	Confirmed LTF	0.6363
MADISON	MADISON	0.0081	Confirmed LTF	0.0081
GIBSON	GIBSON	0.5504	Confirmed LTF	0.5504
CALDERWOOD	CALDERWOOD	0.4930	Confirmed LTF	0.4930

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
BLUEG	BLUEG	1.7499	Confirmed LTF	1.7499
TRIMBLE	TRIMBLE	0.5610	Confirmed LTF	0.5610
CATAWBA	CATAWBA	0.3426	Confirmed LTF	0.3426

11.7.10 Index 10

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42777235	219100	NEWFRDM	PSE&G	219704	HILLTOP_3	PSE&G	1	PS_P7-1_V2274+P2242_LT	tower	886.0	113.98	115.42	AC	13.94

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200062	SALEM G3	2.6292	Adder	3.09
219259	KINSLEYDEP E	0.2604	Adder	0.31
219385	DEPTFDSP1_E	0.0034	Adder	0.0
219390	DEPTFDSP2_E	0.0034	Adder	0.0
219394	DEPTFDSP3_E	0.0087	Adder	0.01
219416	DEPTFDSP4_E	0.0034	Adder	0.0
219664	THOROFSP1_E	0.0141	Adder	0.02
219666	THOROFSP2_E	0.0141	Adder	0.02
219684	THOSOLAR E	0.0781	Adder	0.09
219727	Z1-082 BAT	0.2597	50/50	0.2597
227881	GRENWCHG	0.1501	Adder	0.18
227928	V4-067E	0.1447	Adder	0.17
228102	BLE#2 ST (Deactivation : 30/04/2019)	13.4309	Adder	15.8
228201	CARL#2CT	0.5565	50/50	0.5565
228251	CARLLS#4	0.0591	50/50	0.0591
228260	V4-054C	0.1449	50/50	0.1449
228261	V4-054E	0.6675	50/50	0.6675
228334	MANNMILG	0.7611	Adder	0.9
228357	V2-046E	1.1775	Adder	1.39
228702	WEST CT	0.4785	50/50	0.4785
228712	V2-041E	0.2223	Adder	0.26
228717	S121	1.1644	50/50	1.1644
228720	V2-035C	0.0156	50/50	0.0156
228721	V2-035E	0.1438	50/50	0.1438
228727	W2-039G	1.2162	50/50	1.2162
228728	W1-130C	0.0734	50/50	0.0734
228733	AB1-119 E	0.0944	50/50	0.0944
902092	W1-130E	0.6745	50/50	0.6745
902432	W2-030 E	0.5358	Adder	0.63
918622	AA1-079 E	-3.1610	Adder	-3.72
918632	AA1-079 E	-3.1601	Adder	-3.72
923153	AB1-116 E	0.1010	50/50	0.1010
923532	AB1-169AC	65.0100	50/50	65.0100
924051	AB2-049 C	0.5007	50/50	0.5007
924052	AB2-049 E	0.8170	50/50	0.8170
924531	AB2-102 C	19.7121	Adder	23.19
924532	AB2-102 E	0.4380	Adder	0.52
924701	AB2-122 C	0.0625	Adder	0.07
924702	AB2-122 E	0.1072	Adder	0.13
925441	AC1-016 C (Withdrawn : Unavailable)	-0.2952	Adder	-0.35

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
930001	AB1-001 C	0.0727	Adder	0.09
930002	AB1-001 E	0.1195	Adder	0.14
930242	AB1-063 E	-0.0739	Adder	-0.09
933962	AD1-019 E	0.6775	Adder	0.8
936212	AD2-027 E	-1.0106	Adder	-1.19
936322	AD2-042 E	-1.3366	Adder	-1.57
936491	AD2-064 C (Withdrawn : 06/16/2020)	0.0686	50/50	0.0686
936492	AD2-064 E (Withdrawn : 06/16/2020)	0.0944	50/50	0.0944
936501	AD2-065 C	0.1251	50/50	0.1251
936502	AD2-065 E	0.1726	50/50	0.1726
937011	AD2-135 C	0.0872	50/50	0.0872
937012	AD2-135 E	0.1482	50/50	0.1482
938421	AE1-061 C	0.3254	50/50	0.3254
938422	AE1-061 E	0.3254	50/50	0.3254
938431	AE1-062 C	1.2027	50/50	1.2027
938432	AE1-062 E	1.2027	50/50	1.2027
938781	AE1-104 C O1	10.8411	Adder	12.75
938782	AE1-104 E O1	27.7369	Adder	32.63
938871	AE1-115 C	0.8792	Adder	1.03
938872	AE1-115 E	0.8792	Adder	1.03
939301	AE1-161 C	2.4332	50/50	2.4332
939302	AE1-161 E	3.6498	50/50	3.6498
939501	AE1-179 C O1	3.7933	50/50	3.7933
939502	AE1-179 E O1	2.6770	50/50	2.6770
939821	AE1-218 C O1	0.1026	50/50	0.1026
939822	AE1-218 E O1	0.1539	50/50	0.1539
939831	AE1-219 C O1	0.2309	50/50	0.2309
939832	AE1-219 E O1	0.3335	50/50	0.3335
939931	AE1-229 C O1	9.8541	50/50	9.8541
939932	AE1-229 E O1	6.6764	50/50	6.6764
940001	AE1-240 C O1	3.1323	50/50	3.1323
940002	AE1-240 E O1	2.2358	50/50	2.2358
940361	AE2-020 C	9.5215	Adder	11.2
940362	AE2-020 E	44.5803	Adder	52.45
940371	AE2-021 C	9.5215	Adder	11.2
940372	AE2-021 E	44.5803	Adder	52.45
940381	AE2-022 C	5.5542	Adder	6.53
940382	AE2-022 E	26.0052	Adder	30.59
940782	AE2-065 E	0.0924	Adder	0.11
942101	AE2-222 C O1	7.6099	Adder	8.95
942102	AE2-222 E O1	19.1166	Adder	22.49
942381	AE2-251 C	30.1639	Adder	35.49
942382	AE2-251 E	77.1809	Adder	90.8
943071	AE2-334 C	3.6693	50/50	3.6693
943072	AE2-334 E	1.9561	50/50	1.9561
943732	AF1-041 E	0.2138	Adder	0.25
944951	AF1-160 C	1.2027	50/50	1.2027
944952	AF1-160 E	1.2628	50/50	1.2628
945431	AF1-208 C O1	2.4511	Adder	2.88
945432	AF1-208 E O1	1.6340	Adder	1.92
945731	AF1-238 C	5.5779	50/50	5.5779

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945732	AF1-238 E	8.3668	50/50	8.3668
945741	AF1-239 C	1.3124	50/50	1.3124
945742	AF1-239 E	1.9687	50/50	1.9687
945971	AF1-262	0.1044	50/50	0.1044
999905	MARINGEN 2	0.3567	Adder	0.42
999906	PVILLEG 2	0.1514	Adder	0.18
NEWTON	NEWTON	0.0967	Confirmed LTF	0.0967
FARMERCITY	FARMERCITY	0.0049	Confirmed LTF	0.0049
NY	NY	0.3954	Confirmed LTF	0.3954
PRAIRIE	PRAIRIE	0.2247	Confirmed LTF	0.2247
O-066	O-066	5.8128	Confirmed LTF	5.8128
COFFEEN	COFFEEN	0.0180	Confirmed LTF	0.0180
CHEOAH	CHEOAH	0.0350	Confirmed LTF	0.0350
EDWARDS	EDWARDS	0.0325	Confirmed LTF	0.0325
TILTON	TILTON	0.0580	Confirmed LTF	0.0580
G-007	G-007	0.1238	Confirmed LTF	0.1238
MADISON	MADISON	0.0101	Confirmed LTF	0.0101
GIBSON	GIBSON	0.0497	Confirmed LTF	0.0497
CALDERWOOD	CALDERWOOD	0.0353	Confirmed LTF	0.0353
BLUEG	BLUEG	0.1580	Confirmed LTF	0.1580
TRIMBLE	TRIMBLE	0.0506	Confirmed LTF	0.0506
CATAWBA	CATAWBA	0.0189	Confirmed LTF	0.0189

11.7.11 Index 11

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42251364	219108	CUTHBERT	PSE&G	219125	CAMDEN	PSE&G	1	PS_P2-3_CAMD_5-6_LT	breaker	771.0	113.47	115.53	AC	15.59

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
219124	CAMDEN_STG	3.4765	50/50	3.4765
219126	CAMDEN_CTG	4.2709	50/50	4.2709
219128	GLOUCSTR_26	1.7649	50/50	1.7649
219229	EAGLEPT_G3	2.1771	50/50	2.1771
219230	EAGLEPT_G1	4.1088	50/50	4.1088
219231	EAGLEPT_G2	4.1049	50/50	4.1049
219235	EAGLEPT_ST2	1.4854	50/50	1.4854
219240	MANTUACREK E	0.6021	50/50	0.6021
219258	KINSLEYDEP C	0.0933	50/50	0.0933
219259	KINSLEYDEP E	0.8591	50/50	0.8591
219340	CUTHBSP1_C	0.0062	50/50	0.0062
219344	CUTHBSP1_E	0.0570	50/50	0.0570
219345	CUTHBSP2_C	0.0059	50/50	0.0059
219347	CUTHBSP2_E	0.0537	50/50	0.0537
219350	CUTHBSP3_C	0.0055	50/50	0.0055
219351	CUTHBSP3_E	0.0505	50/50	0.0505
219357	CUTHBSP4_C	0.0047	50/50	0.0047
219379	CUTHBSP4_E	0.0432	50/50	0.0432
219384	DEPTFDSP1_C	0.0012	50/50	0.0012
219385	DEPTFDSP1_E	0.0113	50/50	0.0113
219388	DEPTFDSP2_C	0.0012	50/50	0.0012
219390	DEPTFDSP2_E	0.0114	50/50	0.0114
219393	DEPTFDSP3_C	0.0031	50/50	0.0031
219394	DEPTFDSP3_E	0.0285	50/50	0.0285
219408	DEPTFDSP4_C	0.0012	50/50	0.0012
219416	DEPTFDSP4_E	0.0114	50/50	0.0114
219623	KINSLEYBEA C	0.0674	50/50	0.0674
219624	KINSLEYBEA E	0.6411	50/50	0.6411
219662	THOROFSP1_C	0.0045	50/50	0.0045
219664	THOROFSP1_E	0.0412	50/50	0.0412
219665	THOROFSP2_C	0.0045	50/50	0.0045
219666	THOROFSP2_E	0.0416	50/50	0.0416
219683	THOSOLAR C	0.0248	50/50	0.0248
219684	THOSOLAR E	0.2283	50/50	0.2283
219706	BEAVBRSP1_C	0.0049	50/50	0.0049
219707	BEAVBRSP1_E	0.0448	50/50	0.0448
219708	BEAVBRSP2_C	0.0048	50/50	0.0048
219709	BEAVBRSP2_E	0.0447	50/50	0.0447
219718	GLOUCSP1_C	0.0095	50/50	0.0095
219719	GLOUCSP1_E	0.0882	50/50	0.0882
219726	BAT.STORG E	0.1238	Adder	0.15

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
219765	PARADISE E	0.9634	50/50	0.9634
219766	PARADISE C	0.1013	50/50	0.1013
227881	GRENWCHG	0.2737	Adder	0.32
227928	V4-067E	0.1340	Adder	0.16
228102	BLE#2 ST (Deactivation : 30/04/2019)	13.3337	Adder	15.69
228261	V4-054E	0.7140	Adder	0.84
228334	MANNMILG	1.1348	Adder	1.34
228357	V2-046E	1.7381	Adder	2.04
228400	MICK 1CT	1.6402	50/50	1.6402
228423	Q-090 2	28.3473	50/50	28.3473
228471	VALERO1	0.5576	50/50	0.5576
228472	VALERO2	0.3678	50/50	0.3678
228473	VALERO3	0.3678	50/50	0.3678
228484	VALERO4	0.3280	50/50	0.3280
228712	V2-041E	0.2424	Adder	0.29
228721	V2-035E	0.1519	Adder	0.18
228733	AB1-119 E	0.0690	Adder	0.08
902092	W1-130E	0.6370	Adder	0.75
902432	W2-030 E	0.5094	Adder	0.6
910862	OWENSCORIN E	0.4378	Adder	0.52
917471	Z2-083	2.8677	50/50	2.8677
918622	AA1-079 E	-2.8053	Adder	-3.3
918632	AA1-079 E	-2.8044	Adder	-3.3
923153	AB1-116 E	0.0859	Adder	0.1
923532	AB1-169AC	61.3079	Adder	72.13
924051	AB2-049 C	0.3880	Adder	0.46
924052	AB2-049 E	0.6330	Adder	0.74
924531	AB2-102 C	21.2326	Adder	24.98
924532	AB2-102 E	0.4718	Adder	0.56
924701	AB2-122 C	0.0579	Adder	0.07
924702	AB2-122 E	0.0993	Adder	0.12
925391	AC1-010 C (Suspended)	0.6623	50/50	0.6623
925392	AC1-010 E (Suspended)	1.1440	50/50	1.1440
925441	AC1-016 C (Withdrawn : Unavailable)	-0.2597	Adder	-0.31
930001	AB1-001 C	0.0689	Adder	0.08
930002	AB1-001 E	0.1132	Adder	0.13
933962	AD1-019 E	0.6443	Adder	0.76
936212	AD2-027 E	-0.9505	Adder	-1.12
936322	AD2-042 E	-1.2572	Adder	-1.48
936491	AD2-064 C (Withdrawn : 06/16/2020)	0.0501	Adder	0.06
936492	AD2-064 E (Withdrawn : 06/16/2020)	0.0690	Adder	0.08
936501	AD2-065 C	0.0915	Adder	0.11
936502	AD2-065 E	0.1262	Adder	0.15
937011	AD2-135 C	0.0584	Adder	0.07
937012	AD2-135 E	0.0993	Adder	0.12
938421	AE1-061 C	0.3073	Adder	0.36
938422	AE1-061 E	0.3073	Adder	0.36
938431	AE1-062 C	0.8496	Adder	1.0
938432	AE1-062 E	0.8496	Adder	1.0

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
938781	AE1-104 C O1	10.7627	Adder	12.66
938782	AE1-104 E O1	27.5362	Adder	32.4
938871	AE1-115 C	1.3650	Adder	1.61
938872	AE1-115 E	1.3650	Adder	1.61
939301	AE1-161 C	2.3732	Adder	2.79
939302	AE1-161 E	3.5598	Adder	4.19
939501	AE1-179 C O1	3.7482	Adder	4.41
939502	AE1-179 E O1	2.6452	Adder	3.11
939821	AE1-218 C O1	0.0945	Adder	0.11
939822	AE1-218 E O1	0.1417	Adder	0.17
939831	AE1-219 C O1	0.2125	Adder	0.25
939832	AE1-219 E O1	0.3070	Adder	0.36
939931	AE1-229 C O1	11.2265	Adder	13.21
939932	AE1-229 E O1	7.6062	Adder	8.95
940001	AE1-240 C O1	3.1126	Adder	3.66
940002	AE1-240 E O1	2.2217	Adder	2.61
940361	AE2-020 C	8.4720	Adder	9.97
940362	AE2-020 E	39.6665	Adder	46.67
940371	AE2-021 C	8.4720	Adder	9.97
940372	AE2-021 E	39.6665	Adder	46.67
940381	AE2-022 C	4.9420	Adder	5.81
940382	AE2-022 E	23.1388	Adder	27.22
940771	AE2-064 C	0.0409	50/50	0.0409
940772	AE2-064 E	0.3124	50/50	0.3124
940781	AE2-065 C	0.0344	50/50	0.0344
940782	AE2-065 E	0.3048	50/50	0.3048
942101	AE2-222 C O1	7.2382	Adder	8.52
942102	AE2-222 E O1	18.1828	Adder	21.39
942381	AE2-251 C	26.8391	Adder	31.58
942382	AE2-251 E	68.6737	Adder	80.79
943071	AE2-334 C	3.3926	Adder	3.99
943072	AE2-334 E	1.8086	Adder	2.13
943732	AF1-041 E	0.2023	Adder	0.24
944951	AF1-160 C	0.8496	Adder	1.0
944952	AF1-160 E	0.8921	Adder	1.05
945431	AF1-208 C O1	3.2571	Adder	3.83
945432	AF1-208 E O1	2.1714	Adder	2.55
945731	AF1-238 C	5.3021	Adder	6.24
945732	AF1-238 E	7.9532	Adder	9.36
945741	AF1-239 C	1.2476	Adder	1.47
945742	AF1-239 E	1.8713	Adder	2.2
945971	AF1-262	0.1103	Adder	0.13
945991	AF1-264	0.1897	Adder	0.22
999905	MARINGEN 2	0.3392	Adder	0.4
999906	PVILLEG 2	0.1440	Adder	0.17
NEWTON	NEWTON	0.2869	Confirmed LTF	0.2869
FARMERCITY	FARMERCITY	0.0148	Confirmed LTF	0.0148
NY	NY	0.4839	Confirmed LTF	0.4839
PRAIRIE	PRAIRIE	0.6819	Confirmed LTF	0.6819
O-066	O-066	6.8880	Confirmed LTF	6.8880
COFFEEN	COFFEEN	0.0533	Confirmed LTF	0.0533
CHEOAH	CHEOAH	0.1241	Confirmed LTF	0.1241

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
EDWARDS	EDWARDS	0.0941	Confirmed LTF	0.0941
TILTON	TILTON	0.1695	Confirmed LTF	0.1695
G-007	G-007	0.2891	Confirmed LTF	0.2891
MADISON	MADISON	0.0101	Confirmed LTF	0.0101
GIBSON	GIBSON	0.1458	Confirmed LTF	0.1458
CALDERWOOD	CALDERWOOD	0.1238	Confirmed LTF	0.1238
BLUEG	BLUEG	0.4635	Confirmed LTF	0.4635
TRIMBLE	TRIMBLE	0.1486	Confirmed LTF	0.1486
CATAWBA	CATAWBA	0.0819	Confirmed LTF	0.0819

11.7.12 Index 12

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42251258	219110	GLOUCSTR	PSE&G	219125	CAMDEN	PSE&G	1	PS_P2-3_CUTB_1-4_LT	breaker	771.0	121.13	123.24	AC	15.74

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
219124	CAMDEN_STG	3.7476	50/50	3.7476
219126	CAMDEN_CTG	4.6040	50/50	4.6040
219128	GLOUCSTR_26	1.9025	50/50	1.9025
219229	EAGLEPT_G3	2.3412	50/50	2.3412
219230	EAGLEPT_G1	4.4186	50/50	4.4186
219231	EAGLEPT_G2	4.4142	50/50	4.4142
219235	EAGLEPT_ST2	1.5974	50/50	1.5974
219240	MANTUACREK E	0.6491	50/50	0.6491
219258	KINSLEYDEP C	0.1003	50/50	0.1003
219259	KINSLEYDEP E	0.9236	50/50	0.9236
219384	DEPTFDSP1_C	0.0013	50/50	0.0013
219385	DEPTFDSP1_E	0.0122	50/50	0.0122
219388	DEPTFDSP2_C	0.0013	50/50	0.0013
219390	DEPTFDSP2_E	0.0122	50/50	0.0122
219393	DEPTFDSP3_C	0.0033	50/50	0.0033
219394	DEPTFDSP3_E	0.0306	50/50	0.0306
219408	DEPTFDSP4_C	0.0013	50/50	0.0013
219416	DEPTFDSP4_E	0.0122	50/50	0.0122
219623	KINSLEYBEA C	0.0722	50/50	0.0722
219624	KINSLEYBEA E	0.6869	50/50	0.6869
219662	THOROFSP1_C	0.0048	50/50	0.0048
219664	THOROFSP1_E	0.0441	50/50	0.0441
219665	THOROFSP2_C	0.0048	50/50	0.0048
219666	THOROFSP2_E	0.0446	50/50	0.0446
219683	THOSOLAR C	0.0265	50/50	0.0265
219684	THOSOLAR E	0.2444	50/50	0.2444
219706	BEAVBRSP1_C	0.0052	50/50	0.0052
219707	BEAVBRSP1_E	0.0480	50/50	0.0480
219708	BEAVBRSP2_C	0.0052	50/50	0.0052
219709	BEAVBRSP2_E	0.0478	50/50	0.0478
219718	GLOUCSP1_C	0.0103	50/50	0.0103
219719	GLOUCSP1_E	0.0951	50/50	0.0951
219726	BAT.STORG E	0.1380	Adder	0.16
219765	PARADISE E	1.0385	50/50	1.0385
219766	PARADISE C	0.1092	50/50	0.1092
227881	GRENWCHG	0.2881	Adder	0.34
227928	V4-067E	0.1313	Adder	0.15
228102	BLE#2 ST (Deactivation : 30/04/2019)	13.2263	Adder	15.56
228261	V4-054E	0.7300	Adder	0.86
228334	MANNMILG	1.1781	Adder	1.39
228357	V2-046E	1.8028	Adder	2.12

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
228400	MICK 1CT	1.7320	50/50	1.7320
228423	Q-090 2	30.1974	50/50	30.1974
228471	VALERO1	0.5881	50/50	0.5881
228472	VALERO2	0.3879	50/50	0.3879
228473	VALERO3	0.3879	50/50	0.3879
228484	VALERO4	0.3460	50/50	0.3460
228712	V2-041E	0.2440	Adder	0.29
228721	V2-035E	0.1556	Adder	0.18
228733	AB1-119 E	0.0654	Adder	0.08
902092	W1-130E	0.6442	Adder	0.76
902432	W2-030 E	0.5013	Adder	0.59
910862	OWENSCORIN E	0.4807	Adder	0.57
917471	Z2-083	3.0548	50/50	3.0548
918622	AA1-079 E	-2.8695	Adder	-3.38
918632	AA1-079 E	-2.8689	Adder	-3.38
923153	AB1-116 E	0.0846	Adder	0.1
923532	AB1-169AC	61.9952	Adder	72.94
924051	AB2-049 C	0.3751	Adder	0.44
924052	AB2-049 E	0.6120	Adder	0.72
924531	AB2-102 C	21.3588	Adder	25.13
924532	AB2-102 E	0.4746	Adder	0.56
924701	AB2-122 C	0.0567	Adder	0.07
924702	AB2-122 E	0.0973	Adder	0.11
925391	AC1-010 C (Suspended)	0.7140	50/50	0.7140
925392	AC1-010 E (Suspended)	1.2332	50/50	1.2332
925441	AC1-016 C (Withdrawn : Unavailable)	-0.2703	Adder	-0.32
930001	AB1-001 C	0.0673	Adder	0.08
930002	AB1-001 E	0.1106	Adder	0.13
933962	AD1-019 E	0.6336	Adder	0.75
936212	AD2-027 E	-0.8924	Adder	-1.05
936322	AD2-042 E	-1.1802	Adder	-1.39
936491	AD2-064 C (Withdrawn : 06/16/2020)	0.0475	Adder	0.06
936492	AD2-064 E (Withdrawn : 06/16/2020)	0.0654	Adder	0.08
936501	AD2-065 C	0.0832	Adder	0.1
936502	AD2-065 E	0.1148	Adder	0.14
937011	AD2-135 C	0.0562	Adder	0.07
937012	AD2-135 E	0.0955	Adder	0.11
938421	AE1-061 C	0.3108	Adder	0.37
938422	AE1-061 E	0.3108	Adder	0.37
938431	AE1-062 C	0.7684	Adder	0.9
938432	AE1-062 E	0.7684	Adder	0.9
938781	AE1-104 C O1	10.6760	Adder	12.56
938782	AE1-104 E O1	27.3145	Adder	32.13
938871	AE1-115 C	1.4223	Adder	1.67
938872	AE1-115 E	1.4223	Adder	1.67
939301	AE1-161 C	2.4123	Adder	2.84
939302	AE1-161 E	3.6185	Adder	4.26
939501	AE1-179 C O1	3.8006	Adder	4.47
939502	AE1-179 E O1	2.6821	Adder	3.16
939821	AE1-218 C O1	0.0946	Adder	0.11

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
939822	AE1-218 E O1	0.1420	Adder	0.17
939831	AE1-219 C O1	0.2130	Adder	0.25
939832	AE1-219 E O1	0.3076	Adder	0.36
939931	AE1-229 C O1	11.5722	Adder	13.61
939932	AE1-229 E O1	7.8405	Adder	9.22
940001	AE1-240 C O1	3.1579	Adder	3.72
940002	AE1-240 E O1	2.2541	Adder	2.65
940361	AE2-020 C	8.2458	Adder	9.7
940362	AE2-020 E	38.6074	Adder	45.42
940371	AE2-021 C	8.2458	Adder	9.7
940372	AE2-021 E	38.6074	Adder	45.42
940381	AE2-022 C	4.8101	Adder	5.66
940382	AE2-022 E	22.5210	Adder	26.5
940771	AE2-064 C	0.0438	50/50	0.0438
940772	AE2-064 E	0.3347	50/50	0.3347
940781	AE2-065 C	0.0369	50/50	0.0369
940782	AE2-065 E	0.3277	50/50	0.3277
942101	AE2-222 C O1	7.1169	Adder	8.37
942102	AE2-222 E O1	17.8782	Adder	21.03
942381	AE2-251 C	26.1225	Adder	30.73
942382	AE2-251 E	66.8403	Adder	78.64
943071	AE2-334 C	3.4124	Adder	4.01
943072	AE2-334 E	1.8191	Adder	2.14
943732	AF1-041 E	0.1988	Adder	0.23
944951	AF1-160 C	0.7684	Adder	0.9
944952	AF1-160 E	0.8068	Adder	0.95
945431	AF1-208 C O1	3.3477	Adder	3.94
945432	AF1-208 E O1	2.2318	Adder	2.63
945731	AF1-238 C	5.3533	Adder	6.3
945732	AF1-238 E	8.0299	Adder	9.45
945741	AF1-239 C	1.2596	Adder	1.48
945742	AF1-239 E	1.8894	Adder	2.22
945971	AF1-262	0.1129	Adder	0.13
945991	AF1-264	0.2083	Adder	0.25
999905	MARINGEN 2	0.3336	Adder	0.39
999906	PVILLEG 2	0.1416	Adder	0.17
NEWTON	NEWTON	0.2880	Confirmed LTF	0.2880
FARMERCITY	FARMERCITY	0.0149	Confirmed LTF	0.0149
NY	NY	0.5591	Confirmed LTF	0.5591
PRAIRIE	PRAIRIE	0.6845	Confirmed LTF	0.6845
O-066	O-066	8.2925	Confirmed LTF	8.2925
COFFEEN	COFFEEN	0.0535	Confirmed LTF	0.0535
CHEOAH	CHEOAH	0.1246	Confirmed LTF	0.1246
EDWARDS	EDWARDS	0.0948	Confirmed LTF	0.0948
TILTON	TILTON	0.1707	Confirmed LTF	0.1707
G-007	G-007	0.7155	Confirmed LTF	0.7155
MADISON	MADISON	0.0101	Confirmed LTF	0.0101
GIBSON	GIBSON	0.1469	Confirmed LTF	0.1469
CALDERWOOD	CALDERWOOD	0.1242	Confirmed LTF	0.1242
BLUEG	BLUEG	0.4670	Confirmed LTF	0.4670
TRIMBLE	TRIMBLE	0.1497	Confirmed LTF	0.1497
CATAWBA	CATAWBA	0.0816	Confirmed LTF	0.0816

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41866184	219125	CAMDEN	PSE&G	214206	RICHRE29	PECO	1	PECO_P2-2_CHI230B1/* \$ DELCO \$ CHI230B1 \$ B	bus	1336.0	133.65	135.48	AC	28.46

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
218661	EHAMPSOLAR E	0.3427	Adder	0.4
219124	CAMDEN_STG	4.5219	50/50	4.5219
219126	CAMDEN_CTG	5.5552	50/50	5.5552
219128	GLOUCSTR_26	2.2956	50/50	2.2956
219229	EAGLEPT_G3	2.9118	50/50	2.9118
219230	EAGLEPT_G1	5.4944	50/50	5.4944
219231	EAGLEPT_G2	5.4912	50/50	5.4912
219235	EAGLEPT_ST2	1.9861	50/50	1.9861
219236	SOUTHERNHQ C	0.0204	50/50	0.0204
219237	SOUTHERNHQ E	0.1912	50/50	0.1912
219240	MANTUACREK E	0.7832	50/50	0.7832
219241	CAMDEN1	0.3371	50/50	0.3371
219242	CAMDEN2	0.0364	50/50	0.0364
219258	KINSLEYDEP C	0.1251	50/50	0.1251
219259	KINSLEYDEP E	1.1525	50/50	1.1525
219273	CLEANLIGHT E	0.3710	Adder	0.44
219288	REEVES ST E	1.2933	Adder	1.52
219292	REEVESSO E	0.1184	Adder	0.14
219305	NAMERICA E	0.6477	Adder	0.76
219340	CUTHBSP1_C	0.0072	50/50	0.0072
219344	CUTHBSP1_E	0.0663	50/50	0.0663
219345	CUTHBSP2_C	0.0072	50/50	0.0072
219347	CUTHBSP2_E	0.0662	50/50	0.0662
219349	LDSOLAR E	0.5671	Adder	0.67
219350	CUTHBSP3_C	0.0072	50/50	0.0072
219351	CUTHBSP3_E	0.0663	50/50	0.0663
219357	CUTHBSP4_C	0.0058	50/50	0.0058
219379	CUTHBSP4_E	0.0533	50/50	0.0533
219384	DEPTFDSP1_C	0.0016	50/50	0.0016
219385	DEPTFDSP1_E	0.0152	50/50	0.0152
219388	DEPTFDSP2_C	0.0016	50/50	0.0016
219390	DEPTFDSP2_E	0.0152	50/50	0.0152
219393	DEPTFDSP3_C	0.0041	50/50	0.0041
219394	DEPTFDSP3_E	0.0382	50/50	0.0382
219408	DEPTFDSP4_C	0.0016	50/50	0.0016
219416	DEPTFDSP4_E	0.0152	50/50	0.0152
219566	BURLINSP1_E	0.0778	Adder	0.09
219574	CAMDENSP1_C	0.0405	50/50	0.0405
219575	CAMDENSP1_E	0.3732	50/50	0.3732
219582	COXSCOSP1_E	0.0149	Adder	0.02
219608	WPEMNERT E	0.6477	Adder	0.76

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
219611	FLORENCE E	1.8505	Adder	2.18
219614	COXSCOSP2_E	0.0149	Adder	0.02
219619	MILLCREEK E	0.4441	Adder	0.52
219623	KINSLEYBEA C	0.0912	50/50	0.0912
219624	KINSLEYBEA E	0.8684	50/50	0.8684
219647	COOPERSOLA E	0.1530	50/50	0.1530
219652	LUMB23SP1_E	0.0394	Adder	0.05
219656	LUMB23SP2_E	0.0394	Adder	0.05
219662	THOROFSP1_C	0.0063	50/50	0.0063
219664	THOROFSP1_E	0.0581	50/50	0.0581
219665	THOROFSP2_C	0.0063	50/50	0.0063
219666	THOROFSP2_E	0.0584	50/50	0.0584
219683	THOSOLAR C	0.0349	50/50	0.0349
219684	THOSOLAR E	0.3215	50/50	0.3215
219687	LUMSOLAR_N E	0.3982	Adder	0.47
219689	LUMSOLAR_S E	0.3950	Adder	0.46
219706	BEAVBRSP1_C	0.0066	50/50	0.0066
219707	BEAVBRSP1_E	0.0607	50/50	0.0607
219708	BEAVBRSP2_C	0.0066	50/50	0.0066
219709	BEAVBRSP2_E	0.0606	50/50	0.0606
219710	CINNAMSP1_C	0.0034	50/50	0.0034
219711	CINNAMSP1_E	0.0315	50/50	0.0315
219712	CINNAMSP2_C	0.0020	50/50	0.0020
219713	CINNAMSP2_E	0.0188	50/50	0.0188
219714	CINNAMSP3_C	0.0034	50/50	0.0034
219715	CINNAMSP3_E	0.0315	50/50	0.0315
219716	CINNAMSP4_C	0.0034	50/50	0.0034
219717	CINNAMSP4_E	0.0313	50/50	0.0313
219718	GLOUCSP1_C	0.0124	50/50	0.0124
219719	GLOUCSP1_E	0.1147	50/50	0.1147
219726	BAT.STORG E	0.3772	50/50	0.3772
219765	PARADISE E	1.2531	50/50	1.2531
219766	PARADISE C	0.1318	50/50	0.1318
227881	GRENWCHG	0.4249	Adder	0.5
227928	V4-067E	0.2701	Adder	0.32
228102	BLE#2 ST (Deactivation : 30/04/2019)	25.9093	Adder	30.48
228261	V4-054E	1.2471	Adder	1.47
228334	MANNMILG	1.8680	Adder	2.2
228357	V2-046E	2.8690	Adder	3.38
228400	MICK 1CT	2.5152	50/50	2.5152
228423	Q-090 2	41.9172	50/50	41.9172
228471	VALERO1	0.8589	50/50	0.8589
228472	VALERO2	0.5666	50/50	0.5666
228473	VALERO3	0.5666	50/50	0.5666
228484	VALERO4	0.5053	50/50	0.5053
228712	V2-041E	0.4493	Adder	0.53
228721	V2-035E	0.2618	Adder	0.31
228733	AB1-119 E	0.1471	Adder	0.17
902092	W1-130E	1.1442	Adder	1.35
902432	W2-030 E	1.0135	Adder	1.19
910862	OWENSCORIN E	1.1159	50/50	1.1159
917471	Z2-083	4.2404	50/50	4.2404

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
919901	AB1-000 1	0.9070	Adder	1.07
919911	AB1-000 2	0.9070	Adder	1.07
919921	AB1-000 3	0.9070	Adder	1.07
923153	AB1-116 E	0.1646	Adder	0.19
923532	AB1-169AC	110.2318	Adder	129.68
924051	AB2-049 C	0.7810	Adder	0.92
924052	AB2-049 E	1.2742	Adder	1.5
924531	AB2-102 C	39.6461	Adder	46.64
924532	AB2-102 E	0.8810	Adder	1.04
924701	AB2-122 C	0.1167	Adder	0.14
924702	AB2-122 E	0.2001	Adder	0.24
925391	AC1-010 C (Suspended)	0.8615	50/50	0.8615
925392	AC1-010 E (Suspended)	1.4880	50/50	1.4880
925441	AC1-016 C (Withdrawn : Unavailable)	0.4984	Adder	0.59
925442	AC1-016 E (Withdrawn : Unavailable)	0.8133	Adder	0.96
925452	AC1-017 E	0.4728	Adder	0.56
925561	AC1-030 C	0.0558	50/50	0.0558
925562	AC1-030 E	0.4976	50/50	0.4976
930001	AB1-001 C	0.1410	Adder	0.17
930002	AB1-001 E	0.2317	Adder	0.27
930242	AB1-063 E	0.1530	50/50	0.1530
932361	AC2-050 C O1	0.7355	Adder	0.87
932362	AC2-050 E O1	1.2001	Adder	1.41
933962	AD1-019 E	1.2837	Adder	1.51
934661	AD1-097 1	1.4633	Adder	1.72
934671	AD1-097 2	1.4633	Adder	1.72
934681	AD1-097 3	1.4633	Adder	1.72
934691	AD1-097 4	0.8662	Adder	1.02
936211	AD2-027 C	0.2769	50/50	0.2769
936212	AD2-027 E	2.5512	50/50	2.5512
936321	AD2-042 C	0.3644	50/50	0.3644
936322	AD2-042 E	3.3741	50/50	3.3741
936491	AD2-064 C (Withdrawn : 06/16/2020)	0.1068	Adder	0.13
936492	AD2-064 E (Withdrawn : 06/16/2020)	0.1471	Adder	0.17
936501	AD2-065 C	0.2157	Adder	0.25
936502	AD2-065 E	0.2975	Adder	0.35
937011	AD2-135 C	0.1193	Adder	0.14
937012	AD2-135 E	0.2029	Adder	0.24
938421	AE1-061 C	0.5520	Adder	0.65
938422	AE1-061 E	0.5520	Adder	0.65
938431	AE1-062 C	1.9904	Adder	2.34
938432	AE1-062 E	1.9904	Adder	2.34
938781	AE1-104 C O1	20.9135	Adder	24.6
938782	AE1-104 E O1	53.5069	Adder	62.95
938871	AE1-115 C	2.2232	Adder	2.62
938872	AE1-115 E	2.2232	Adder	2.62
939301	AE1-161 C	4.1781	Adder	4.92
939302	AE1-161 E	6.2671	Adder	7.37
939501	AE1-179 C O1	6.7443	Adder	7.93

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
939502	AE1-179 E O1	4.7596	Adder	5.6
939821	AE1-218 C O1	0.1726	Adder	0.2
939822	AE1-218 E O1	0.2589	Adder	0.3
939831	AE1-219 C O1	0.3884	Adder	0.46
939832	AE1-219 E O1	0.5610	Adder	0.66
939931	AE1-229 C O1	18.9715	Adder	22.32
939932	AE1-229 E O1	12.8537	Adder	15.12
940001	AE1-240 C O1	5.5909	Adder	6.58
940002	AE1-240 E O1	3.9907	Adder	4.69
940271	AE2-010	-0.5403	Adder	-0.64
940361	AE2-020 C	17.4045	Adder	20.48
940362	AE2-020 E	81.4891	Adder	95.87
940371	AE2-021 C	17.4045	Adder	20.48
940372	AE2-021 E	81.4891	Adder	95.87
940381	AE2-022 C	10.1526	Adder	11.94
940382	AE2-022 E	47.5353	Adder	55.92
940771	AE2-064 C	0.0554	50/50	0.0554
940772	AE2-064 E	0.4232	50/50	0.4232
940781	AE2-065 C	0.0461	50/50	0.0461
940782	AE2-065 E	0.4090	50/50	0.4090
942101	AE2-222 C O1	14.4270	Adder	16.97
942102	AE2-222 E O1	36.2415	Adder	42.64
942381	AE2-251 C	55.1371	Adder	64.87
942382	AE2-251 E	141.0803	Adder	165.98
943071	AE2-334 C	6.1390	Adder	7.22
943072	AE2-334 E	3.2727	Adder	3.85
943732	AF1-041 E	0.4042	Adder	0.48
944951	AF1-160 C	1.9904	Adder	2.34
944952	AF1-160 E	2.0899	Adder	2.46
945431	AF1-208 C O1	5.5808	Adder	6.57
945432	AF1-208 E O1	3.7205	Adder	4.38
945731	AF1-238 C	9.6770	Adder	11.38
945732	AF1-238 E	14.5155	Adder	17.08
945741	AF1-239 C	2.2769	Adder	2.68
945742	AF1-239 E	3.4154	Adder	4.02
945961	AF1-261	0.1343	Adder	0.16
945971	AF1-262	0.1900	Adder	0.22
945991	AF1-264	0.4836	50/50	0.4836
999905	MARINGEN 2	0.6756	Adder	0.79
999906	PVILLEG 2	0.2871	Adder	0.34
NEWTON	NEWTON	1.0831	Confirmed LTF	1.0831
FARMERCITY	FARMERCITY	0.0563	Confirmed LTF	0.0563
G-007A	G-007A	10.1510	Confirmed LTF	10.1510
VFT	VFT	15.1511	Confirmed LTF	15.1511
PRAIRIE	PRAIRIE	2.5959	Confirmed LTF	2.5959
COFFEEN	COFFEEN	0.2011	Confirmed LTF	0.2011
CHEOAH	CHEOAH	0.4965	Confirmed LTF	0.4965
EDWARDS	EDWARDS	0.3535	Confirmed LTF	0.3535
TILTON	TILTON	0.6363	Confirmed LTF	0.6363
MADISON	MADISON	0.0081	Confirmed LTF	0.0081
GIBSON	GIBSON	0.5504	Confirmed LTF	0.5504
CALDERWOOD	CALDERWOOD	0.4930	Confirmed LTF	0.4930

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
BLUEG	BLUEG	1.7499	Confirmed LTF	1.7499
TRIMBLE	TRIMBLE	0.5610	Confirmed LTF	0.5610
CATAWBA	CATAWBA	0.3426	Confirmed LTF	0.3426

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	ACID C	MW IMPACT
4277731 1	21970 2	HILLTOP_ 1	PSE& G	21910 1	BEAVERBK_ 1	PSE& G	1	PS_P7- 1_V2274+P2242_L T	tower	886.0	101.53	102.88	AC	13.11

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200062	SALEM G3	2.4715	Adder	2.91
219259	KINSLEYDEP E	0.2448	Adder	0.29
219385	DEPTFDSP1_E	0.0032	Adder	0.0
219390	DEPTFDSP2_E	0.0032	Adder	0.0
219394	DEPTFDSP3_E	0.0081	Adder	0.01
219416	DEPTFDSP4_E	0.0032	Adder	0.0
219664	THOROFSP1_E	0.0133	Adder	0.02
219666	THOROFSP2_E	0.0133	Adder	0.02
219684	THOSOLAR E	0.0734	Adder	0.09
227881	GRENWCHG	0.1411	Adder	0.17
227928	V4-067E	0.1361	Adder	0.16
228102	BLE#2 ST (Deactivation : 30/04/2019)	12.6306	Adder	14.86
228201	CARL#2CT	0.5234	50/50	0.5234
228260	V4-054C	0.1363	50/50	0.1363
228261	V4-054E	0.6276	50/50	0.6276
228334	MANNMILG	0.7156	Adder	0.84
228357	V2-046E	1.1072	Adder	1.3
228702	WEST CT	0.4500	50/50	0.4500
228712	V2-041E	0.2091	Adder	0.25
228717	S121	1.0949	50/50	1.0949
228720	V2-035C	0.0147	50/50	0.0147
228721	V2-035E	0.1352	50/50	0.1352
228727	W2-039G	1.1437	50/50	1.1437
228728	W1-130C	0.0690	50/50	0.0690
228733	AB1-119 E	0.0889	50/50	0.0889
902092	W1-130E	0.6342	50/50	0.6342
902432	W2-030 E	0.5038	Adder	0.59
918622	AA1-079 E	-2.9594	Adder	-3.48
918632	AA1-079 E	-2.9588	Adder	-3.48
923153	AB1-116 E	0.0951	50/50	0.0951
923532	AB1-169AC	61.1270	50/50	61.1270
924051	AB2-049 C	0.4712	50/50	0.4712
924052	AB2-049 E	0.7688	50/50	0.7688
924531	AB2-102 C	18.5359	Adder	21.81
924532	AB2-102 E	0.4119	Adder	0.48
924701	AB2-122 C	0.0588	Adder	0.07
924702	AB2-122 E	0.1008	Adder	0.12
925441	AC1-016 C (Withdrawn : Unavailable)	-0.2760	Adder	-0.32
930001	AB1-001 C	0.0684	Adder	0.08
930002	AB1-001 E	0.1124	Adder	0.13

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
930242	AB1-063 E	-0.0375	Adder	-0.04
933962	AD1-019 E	0.6371	Adder	0.75
936212	AD2-027 E	-0.9528	Adder	-1.12
936322	AD2-042 E	-1.2601	Adder	-1.48
936491	AD2-064 C (Withdrawn : 06/16/2020)	0.0646	50/50	0.0646
936492	AD2-064 E (Withdrawn : 06/16/2020)	0.0889	50/50	0.0889
936501	AD2-065 C	0.1181	50/50	0.1181
936502	AD2-065 E	0.1629	50/50	0.1629
937011	AD2-135 C	0.0820	50/50	0.0820
937012	AD2-135 E	0.1394	50/50	0.1394
938421	AE1-061 C	0.3060	50/50	0.3060
938422	AE1-061 E	0.3060	50/50	0.3060
938431	AE1-062 C	1.1330	50/50	1.1330
938432	AE1-062 E	1.1330	50/50	1.1330
938781	AE1-104 C O1	10.1952	Adder	11.99
938782	AE1-104 E O1	26.0842	Adder	30.69
938871	AE1-115 C	0.8266	Adder	0.97
938872	AE1-115 E	0.8266	Adder	0.97
939301	AE1-161 C	2.2878	50/50	2.2878
939302	AE1-161 E	3.4317	50/50	3.4317
939501	AE1-179 C O1	3.5669	50/50	3.5669
939502	AE1-179 E O1	2.5172	50/50	2.5172
939821	AE1-218 C O1	0.0965	50/50	0.0965
939822	AE1-218 E O1	0.1448	50/50	0.1448
939831	AE1-219 C O1	0.2172	50/50	0.2172
939832	AE1-219 E O1	0.3137	50/50	0.3137
939931	AE1-229 C O1	9.2649	50/50	9.2649
939932	AE1-229 E O1	6.2772	50/50	6.2772
940001	AE1-240 C O1	2.9455	50/50	2.9455
940002	AE1-240 E O1	2.1025	50/50	2.1025
940361	AE2-020 C	8.9551	Adder	10.54
940362	AE2-020 E	41.9285	Adder	49.33
940371	AE2-021 C	8.9551	Adder	10.54
940372	AE2-021 E	41.9285	Adder	49.33
940381	AE2-022 C	5.2238	Adder	6.15
940382	AE2-022 E	24.4583	Adder	28.77
940782	AE2-065 E	0.0869	Adder	0.1
942101	AE2-222 C O1	7.1569	Adder	8.42
942102	AE2-222 E O1	17.9785	Adder	21.15
942381	AE2-251 C	28.3696	Adder	33.38
942382	AE2-251 E	72.5900	Adder	85.4
943071	AE2-334 C	3.4512	50/50	3.4512
943072	AE2-334 E	1.8398	50/50	1.8398
943732	AF1-041 E	0.2010	Adder	0.24
944951	AF1-160 C	1.1330	50/50	1.1330
944952	AF1-160 E	1.1897	50/50	1.1897
945431	AF1-208 C O1	2.3046	Adder	2.71
945432	AF1-208 E O1	1.5364	Adder	1.81
945731	AF1-238 C	5.2453	50/50	5.2453
945732	AF1-238 E	7.8680	50/50	7.8680
945741	AF1-239 C	1.2342	50/50	1.2342

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945742	AF1-239 E	1.8513	50/50	1.8513
945971	AF1-262	0.0982	50/50	0.0982
999905	MARINGEN 2	0.3354	Adder	0.39
999906	PVILLEG 2	0.1424	Adder	0.17
NEWTON	NEWTON	0.0913	Confirmed LTF	0.0913
FARMERCITY	FARMERCITY	0.0046	Confirmed LTF	0.0046
NY	NY	0.3689	Confirmed LTF	0.3689
PRAIRIE	PRAIRIE	0.2118	Confirmed LTF	0.2118
O-066	O-066	5.4163	Confirmed LTF	5.4163
COFFEEN	COFFEEN	0.0170	Confirmed LTF	0.0170
CHEOAH	CHEOAH	0.0330	Confirmed LTF	0.0330
EDWARDS	EDWARDS	0.0308	Confirmed LTF	0.0308
TILTON	TILTON	0.0548	Confirmed LTF	0.0548
G-007	G-007	0.1061	Confirmed LTF	0.1061
MADISON	MADISON	0.0101	Confirmed LTF	0.0101
GIBSON	GIBSON	0.0470	Confirmed LTF	0.0470
CALDERWOOD	CALDERWOOD	0.0333	Confirmed LTF	0.0333
BLUEG	BLUEG	0.1493	Confirmed LTF	0.1493
TRIMBLE	TRIMBLE	0.0479	Confirmed LTF	0.0479
CATAWBA	CATAWBA	0.0178	Confirmed LTF	0.0178

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42251517	219754	CUTHBERT_3	PSE&G	219125	CAMDEN	PSE&G	1	PS_P2-3_CAMD_5-6_LT	breaker	792.0	100.33	102.16	AC	14.17

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
219124	CAMDEN_STG	3.1595	50/50	3.1595
219126	CAMDEN_CTG	3.8815	50/50	3.8815
219128	GLOUCSTR_26	1.6040	50/50	1.6040
219229	EAGLEPT_G3	1.9786	50/50	1.9786
219230	EAGLEPT_G1	3.7342	50/50	3.7342
219231	EAGLEPT_G2	3.7307	50/50	3.7307
219235	EAGLEPT_ST2	1.3499	50/50	1.3499
219240	MANTUACREK E	0.5472	50/50	0.5472
219258	KINSLEYDEP C	0.0848	50/50	0.0848
219259	KINSLEYDEP E	0.7807	50/50	0.7807
219340	CUTHBSP1_C	0.0050	50/50	0.0050
219344	CUTHBSP1_E	0.0456	50/50	0.0456
219345	CUTHBSP2_C	0.0053	50/50	0.0053
219347	CUTHBSP2_E	0.0488	50/50	0.0488
219350	CUTHBSP3_C	0.0057	50/50	0.0057
219351	CUTHBSP3_E	0.0521	50/50	0.0521
219357	CUTHBSP4_C	0.0043	50/50	0.0043
219379	CUTHBSP4_E	0.0393	50/50	0.0393
219384	DEPTFDSP1_C	0.0011	50/50	0.0011
219385	DEPTFDSP1_E	0.0103	50/50	0.0103
219388	DEPTFDSP2_C	0.0011	50/50	0.0011
219390	DEPTFDSP2_E	0.0103	50/50	0.0103
219393	DEPTFDSP3_C	0.0028	50/50	0.0028
219394	DEPTFDSP3_E	0.0259	50/50	0.0259
219408	DEPTFDSP4_C	0.0011	50/50	0.0011
219416	DEPTFDSP4_E	0.0103	50/50	0.0103
219623	KINSLEYBEA C	0.0612	50/50	0.0612
219624	KINSLEYBEA E	0.5826	50/50	0.5826
219662	THOROFSP1_C	0.0041	50/50	0.0041
219664	THOROFSP1_E	0.0375	50/50	0.0375
219665	THOROFSP2_C	0.0041	50/50	0.0041
219666	THOROFSP2_E	0.0378	50/50	0.0378
219683	THOSOLAR C	0.0225	50/50	0.0225
219684	THOSOLAR E	0.2075	50/50	0.2075
219706	BEAVBRSP1_C	0.0044	50/50	0.0044
219707	BEAVBRSP1_E	0.0407	50/50	0.0407
219708	BEAVBRSP2_C	0.0044	50/50	0.0044
219709	BEAVBRSP2_E	0.0406	50/50	0.0406
219718	GLOUCSP1_C	0.0087	50/50	0.0087
219719	GLOUCSP1_E	0.0802	50/50	0.0802
219726	BAT.STORG E	0.1125	Adder	0.13

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
219765	PARADISE E	0.8755	50/50	0.8755
219766	PARADISE C	0.0921	50/50	0.0921
227881	GRENWCHG	0.2487	Adder	0.29
227928	V4-067E	0.1218	Adder	0.14
228102	BLE#2 ST (Deactivation : 30/04/2019)	12.1180	Adder	14.26
228261	V4-054E	0.6489	Adder	0.76
228334	MANNMILG	1.0313	Adder	1.21
228357	V2-046E	1.5795	Adder	1.86
228400	MICK 1CT	1.4906	50/50	1.4906
228423	Q-090 2	25.7626	50/50	25.7626
228471	VALERO1	0.5067	50/50	0.5067
228472	VALERO2	0.3342	50/50	0.3342
228473	VALERO3	0.3342	50/50	0.3342
228484	VALERO4	0.2981	50/50	0.2981
228712	V2-041E	0.2203	Adder	0.26
228721	V2-035E	0.1381	Adder	0.16
228733	AB1-119 E	0.0627	Adder	0.07
902092	W1-130E	0.5789	Adder	0.68
902432	W2-030 E	0.4630	Adder	0.54
910862	OWENSCORIN E	0.3979	Adder	0.47
917471	Z2-083	2.6062	50/50	2.6062
918622	AA1-079 E	-2.5497	Adder	-3.0
918632	AA1-079 E	-2.5487	Adder	-3.0
923153	AB1-116 E	0.0780	Adder	0.09
923532	AB1-169AC	55.7213	Adder	65.55
924051	AB2-049 C	0.3526	Adder	0.41
924052	AB2-049 E	0.5752	Adder	0.68
924531	AB2-102 C	19.2971	Adder	22.7
924532	AB2-102 E	0.4288	Adder	0.5
924701	AB2-122 C	0.0526	Adder	0.06
924702	AB2-122 E	0.0902	Adder	0.11
925391	AC1-010 C (Suspended)	0.6019	50/50	0.6019
925392	AC1-010 E (Suspended)	1.0397	50/50	1.0397
925441	AC1-016 C (Withdrawn : Unavailable)	-0.2360	Adder	-0.28
930001	AB1-001 C	0.0626	Adder	0.07
930002	AB1-001 E	0.1029	Adder	0.12
933962	AD1-019 E	0.5855	Adder	0.69
936212	AD2-027 E	-0.8639	Adder	-1.02
936322	AD2-042 E	-1.1426	Adder	-1.34
936491	AD2-064 C (Withdrawn : 06/16/2020)	0.0455	Adder	0.05
936492	AD2-064 E (Withdrawn : 06/16/2020)	0.0627	Adder	0.07
936501	AD2-065 C	0.0831	Adder	0.1
936502	AD2-065 E	0.1147	Adder	0.13
937011	AD2-135 C	0.0531	Adder	0.06
937012	AD2-135 E	0.0902	Adder	0.11
938421	AE1-061 C	0.2793	Adder	0.33
938422	AE1-061 E	0.2793	Adder	0.33
938431	AE1-062 C	0.7721	Adder	0.91
938432	AE1-062 E	0.7721	Adder	0.91

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
938781	AE1-104 C O1	9.7814	Adder	11.51
938782	AE1-104 E O1	25.0255	Adder	29.44
938871	AE1-115 C	1.2406	Adder	1.46
938872	AE1-115 E	1.2406	Adder	1.46
939301	AE1-161 C	2.1568	Adder	2.54
939302	AE1-161 E	3.2352	Adder	3.81
939501	AE1-179 C O1	3.4064	Adder	4.01
939502	AE1-179 E O1	2.4039	Adder	2.83
939821	AE1-218 C O1	0.0859	Adder	0.1
939822	AE1-218 E O1	0.1288	Adder	0.15
939831	AE1-219 C O1	0.1932	Adder	0.23
939832	AE1-219 E O1	0.2790	Adder	0.33
939931	AE1-229 C O1	10.2029	Adder	12.0
939932	AE1-229 E O1	6.9128	Adder	8.13
940001	AE1-240 C O1	2.8288	Adder	3.33
940002	AE1-240 E O1	2.0192	Adder	2.38
940361	AE2-020 C	7.6993	Adder	9.06
940362	AE2-020 E	36.0489	Adder	42.41
940371	AE2-021 C	7.6993	Adder	9.06
940372	AE2-021 E	36.0489	Adder	42.41
940381	AE2-022 C	4.4913	Adder	5.28
940382	AE2-022 E	21.0285	Adder	24.74
940771	AE2-064 C	0.0372	50/50	0.0372
940772	AE2-064 E	0.2839	50/50	0.2839
940781	AE2-065 C	0.0312	50/50	0.0312
940782	AE2-065 E	0.2770	50/50	0.2770
942101	AE2-222 C O1	6.5782	Adder	7.74
942102	AE2-222 E O1	16.5248	Adder	19.44
942381	AE2-251 C	24.3914	Adder	28.7
942382	AE2-251 E	62.4106	Adder	73.42
943071	AE2-334 C	3.0833	Adder	3.63
943072	AE2-334 E	1.6437	Adder	1.93
943732	AF1-041 E	0.1839	Adder	0.22
944951	AF1-160 C	0.7721	Adder	0.91
944952	AF1-160 E	0.8107	Adder	0.95
945431	AF1-208 C O1	2.9601	Adder	3.48
945432	AF1-208 E O1	1.9734	Adder	2.32
945731	AF1-238 C	4.8188	Adder	5.67
945732	AF1-238 E	7.2282	Adder	8.5
945741	AF1-239 C	1.1338	Adder	1.33
945742	AF1-239 E	1.7007	Adder	2.0
945971	AF1-262	0.1002	Adder	0.12
945991	AF1-264	0.1724	Adder	0.2
999905	MARINGEN 2	0.3083	Adder	0.36
999906	PVILLEG 2	0.1309	Adder	0.15
NEWTON	NEWTON	0.2600	Confirmed LTF	0.2600
FARMERCITY	FARMERCITY	0.0135	Confirmed LTF	0.0135
NY	NY	0.4402	Confirmed LTF	0.4402
PRAIRIE	PRAIRIE	0.6199	Confirmed LTF	0.6199
O-066	O-066	6.2563	Confirmed LTF	6.2563
COFFEEN	COFFEEN	0.0483	Confirmed LTF	0.0483
CHEOAH	CHEOAH	0.1131	Confirmed LTF	0.1131

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
EDWARDS	EDWARDS	0.0858	Confirmed LTF	0.0858
TILTON	TILTON	0.1537	Confirmed LTF	0.1537
G-007	G-007	0.2621	Confirmed LTF	0.2621
MADISON	MADISON	0.0081	Confirmed LTF	0.0081
GIBSON	GIBSON	0.1327	Confirmed LTF	0.1327
CALDERWOOD	CALDERWOOD	0.1123	Confirmed LTF	0.1123
BLUEG	BLUEG	0.4218	Confirmed LTF	0.4218
TRIMBLE	TRIMBLE	0.1352	Confirmed LTF	0.1352
CATAWBA	CATAWBA	0.0742	Confirmed LTF	0.0742

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
42777679	219756	BEAVERBK_2	PSE&G	219110	GLOUCST R	PSE&G	1	PS_P7-1_V2274+P2242_L T	tower	740.0	111.13	112.76	AC	13.11

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
219259	KINSLEYDEP E	0.2448	Adder	0.29
219385	DEPTFDSP1_E	0.0032	Adder	0.0
219390	DEPTFDSP2_E	0.0032	Adder	0.0
219394	DEPTFDSP3_E	0.0081	Adder	0.01
219416	DEPTFDSP4_E	0.0032	Adder	0.0
219623	KINSLEYBEA C	0.1698	50/50	0.1698
219624	KINSLEYBEA E	1.6159	50/50	1.6159
219664	THOROFSP1_E	0.0133	Adder	0.02
219666	THOROFSP2_E	0.0133	Adder	0.02
219684	THOSOLAR E	0.0734	Adder	0.09
219706	BEAVBRSP1_C	0.0122	50/50	0.0122
219707	BEAVBRSP1_E	0.1130	50/50	0.1130
219708	BEAVBRSP2_C	0.0122	50/50	0.0122
219709	BEAVBRSP2_E	0.1124	50/50	0.1124
227881	GRENWCHG	0.1411	Adder	0.17
227928	V4-067E	0.1361	Adder	0.16
228102	BLE#2 ST (Deactivation : 30/04/2019)	12.6280	Adder	14.86
228201	CARL#2CT	0.5232	50/50	0.5232
228251	CARLLS#4	0.0555	50/50	0.0555
228260	V4-054C	0.1362	50/50	0.1362
228261	V4-054E	0.6276	50/50	0.6276
228334	MANNMILG	0.7156	Adder	0.84
228357	V2-046E	1.1070	Adder	1.3
228702	WEST CT	0.4499	50/50	0.4499
228712	V2-041E	0.2090	Adder	0.25
228717	S121	1.0948	50/50	1.0948
228720	V2-035C	0.0147	50/50	0.0147
228721	V2-035E	0.1352	50/50	0.1352
228727	W2-039G	1.1436	50/50	1.1436
228728	W1-130C	0.0690	50/50	0.0690
228733	AB1-119 E	0.0889	50/50	0.0889
902092	W1-130E	0.6342	50/50	0.6342
902432	W2-030 E	0.5038	Adder	0.59
918622	AA1-079 E	-2.9598	Adder	-3.48
918632	AA1-079 E	-2.9591	Adder	-3.48
923153	AB1-116 E	0.0951	50/50	0.0951
923532	AB1-169AC	61.1215	50/50	61.1215
924051	AB2-049 C	0.4712	50/50	0.4712
924052	AB2-049 E	0.7687	50/50	0.7687
924531	AB2-102 C	18.5340	Adder	21.8
924532	AB2-102 E	0.4119	Adder	0.48

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
924701	AB2-122 C	0.0588	Adder	0.07
924702	AB2-122 E	0.1008	Adder	0.12
925441	AC1-016 C (Withdrawn : Unavailable)	-0.2760	Adder	-0.32
930001	AB1-001 C	0.0684	Adder	0.08
930002	AB1-001 E	0.1124	Adder	0.13
930242	AB1-063 E	-0.0375	Adder	-0.04
933962	AD1-019 E	0.6370	Adder	0.75
936212	AD2-027 E	-0.9528	Adder	-1.12
936322	AD2-042 E	-1.2602	Adder	-1.48
936491	AD2-064 C (Withdrawn : 06/16/2020)	0.0646	50/50	0.0646
936492	AD2-064 E (Withdrawn : 06/16/2020)	0.0889	50/50	0.0889
936501	AD2-065 C	0.1180	50/50	0.1180
936502	AD2-065 E	0.1629	50/50	0.1629
937011	AD2-135 C	0.0820	50/50	0.0820
937012	AD2-135 E	0.1394	50/50	0.1394
938421	AE1-061 C	0.3060	50/50	0.3060
938422	AE1-061 E	0.3060	50/50	0.3060
938431	AE1-062 C	1.1329	50/50	1.1329
938432	AE1-062 E	1.1329	50/50	1.1329
938781	AE1-104 C O1	10.1931	Adder	11.99
938782	AE1-104 E O1	26.0789	Adder	30.68
938871	AE1-115 C	0.8265	Adder	0.97
938872	AE1-115 E	0.8265	Adder	0.97
939301	AE1-161 C	2.2876	50/50	2.2876
939302	AE1-161 E	3.4314	50/50	3.4314
939501	AE1-179 C O1	3.5665	50/50	3.5665
939502	AE1-179 E O1	2.5169	50/50	2.5169
939821	AE1-218 C O1	0.0965	50/50	0.0965
939822	AE1-218 E O1	0.1448	50/50	0.1448
939831	AE1-219 C O1	0.2172	50/50	0.2172
939832	AE1-219 E O1	0.3137	50/50	0.3137
939931	AE1-229 C O1	9.2640	50/50	9.2640
939932	AE1-229 E O1	6.2766	50/50	6.2766
940001	AE1-240 C O1	2.9452	50/50	2.9452
940002	AE1-240 E O1	2.1023	50/50	2.1023
940361	AE2-020 C	8.9533	Adder	10.53
940362	AE2-020 E	41.9200	Adder	49.32
940371	AE2-021 C	8.9533	Adder	10.53
940372	AE2-021 E	41.9200	Adder	49.32
940381	AE2-022 C	5.2228	Adder	6.14
940382	AE2-022 E	24.4534	Adder	28.77
940771	AE2-064 C	0.1031	50/50	0.1031
940772	AE2-064 E	0.7874	50/50	0.7874
940782	AE2-065 E	0.0869	Adder	0.1
942101	AE2-222 C O1	7.1561	Adder	8.42
942102	AE2-222 E O1	17.9767	Adder	21.15
942381	AE2-251 C	28.3639	Adder	33.37
942382	AE2-251 E	72.5753	Adder	85.38
943071	AE2-334 C	3.4509	50/50	3.4509
943072	AE2-334 E	1.8397	50/50	1.8397

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
943732	AF1-041 E	0.2010	Adder	0.24
944951	AF1-160 C	1.1329	50/50	1.1329
944952	AF1-160 E	1.1895	50/50	1.1895
945431	AF1-208 C O1	2.3044	Adder	2.71
945432	AF1-208 E O1	1.5363	Adder	1.81
945731	AF1-238 C	5.2448	50/50	5.2448
945732	AF1-238 E	7.8673	50/50	7.8673
945741	AF1-239 C	1.2341	50/50	1.2341
945742	AF1-239 E	1.8511	50/50	1.8511
945971	AF1-262	0.0982	50/50	0.0982
999905	MARINGEN 2	0.3354	Adder	0.39
999906	PVILLEG 2	0.1424	Adder	0.17
NEWTON	NEWTON	0.0924	Confirmed LTF	0.0924
FARMERCITY	FARMERCITY	0.0047	Confirmed LTF	0.0047
NY	NY	0.3694	Confirmed LTF	0.3694
PRAIRIE	PRAIRIE	0.2144	Confirmed LTF	0.2144
O-066	O-066	5.4230	Confirmed LTF	5.4230
COFFEEN	COFFEEN	0.0172	Confirmed LTF	0.0172
CHEOAH	CHEOAH	0.0335	Confirmed LTF	0.0335
EDWARDS	EDWARDS	0.0312	Confirmed LTF	0.0312
TILTON	TILTON	0.0554	Confirmed LTF	0.0554
G-007	G-007	0.1071	Confirmed LTF	0.1071
MADISON	MADISON	0.0101	Confirmed LTF	0.0101
GIBSON	GIBSON	0.0475	Confirmed LTF	0.0475
CALDERWOOD	CALDERWOOD	0.0338	Confirmed LTF	0.0338
BLUEG	BLUEG	0.1510	Confirmed LTF	0.1510
TRIMBLE	TRIMBLE	0.0484	Confirmed LTF	0.0484
CATAWBA	CATAWBA	0.0182	Confirmed LTF	0.0182

11.7.17 Index 17

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41041980	227900	CARDIFF	AE	219100	NEWFRDM	PSE&G	1	JC-P7-1-JCC-230-13	tower	692.0	263.11	265.21	DC	17.07

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206360	280 CRK C1	2.3601	50/50	2.3601
206361	280 CRK C2	1.5794	50/50	1.5794
227842	MARINGEN	0.2297	50/50	0.2297
227927	V4-067C	0.0651	50/50	0.0651
227928	V4-067E	0.6075	50/50	0.6075
228014	PVILLEG	0.0784	50/50	0.0784
228102	BLE#2 ST (Deactivation : 30/04/2019)	43.1904	50/50	43.1904
228202	CUMB CT	1.9431	50/50	1.9431
228203	P06	2.0623	50/50	2.0623
228261	V4-054E	0.5802	Adder	0.68
228712	V2-041E	0.3529	Adder	0.42
228721	V2-035E	0.0902	Adder	0.11
228729	W2-030 C	0.2211	50/50	0.2211
228731	V3-036	0.3323	50/50	0.3323
228732	V1-021 C	0.0834	50/50	0.0834
902092	W1-130E	0.6973	Adder	0.82
902432	W2-030 E	2.0518	50/50	2.0518
923532	AB1-169AC	68.2690	Adder	80.32
924531	AB2-102 C	38.5043	50/50	38.5043
924532	AB2-102 E	0.7273	Adder	0.86
924701	AB2-122 C	0.2625	50/50	0.2625
924702	AB2-122 E	0.4500	50/50	0.4500
930001	AB1-001 C	0.3031	50/50	0.3031
930002	AB1-001 E	0.4982	50/50	0.4982
933962	AD1-019 E	2.5923	50/50	2.5923
938421	AE1-061 C	0.3364	Adder	0.4
938422	AE1-061 E	0.3364	Adder	0.4
938781	AE1-104 C O1	34.8624	50/50	34.8624
938782	AE1-104 E O1	89.1950	50/50	89.1950
939301	AE1-161 C	1.5560	Adder	1.83
939302	AE1-161 E	2.3340	Adder	2.75
939501	AE1-179 C O1	3.8458	Adder	4.52
939502	AE1-179 E O1	2.7140	Adder	3.19
939931	AE1-229 C O1	6.0603	Adder	7.13
939932	AE1-229 E O1	4.1060	Adder	4.83
940001	AE1-240 C O1	3.2067	Adder	3.77
940002	AE1-240 E O1	2.2889	Adder	2.69
940161	AE2-000 C O1	89.8673	50/50	89.8673
940162	AE2-000 E O1	229.9395	50/50	229.9395
940361	AE2-020 C	45.0412	50/50	45.0412
940362	AE2-020 E	210.8860	50/50	210.8860

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
940371	AE2-021 C	45.0412	50/50	45.0412
940372	AE2-021 E	210.8860	50/50	210.8860
940381	AE2-022 C	26.2740	50/50	26.2740
940382	AE2-022 E	123.0168	50/50	123.0168
942101	AE2-222 C O1	29.1163	50/50	29.1163
942102	AE2-222 E O1	73.1417	50/50	73.1417
942381	AE2-251 C	142.6896	50/50	142.6896
942382	AE2-251 E	365.1025	50/50	365.1025
943732	AF1-041 E	0.8330	50/50	0.8330
944332	AF1-101 E O1	225.4324	50/50	225.4324
945431	AF1-208 C O1	2.2424	Adder	2.64
945432	AF1-208 E O1	1.4950	Adder	1.76
945731	AF1-238 C	5.8028	Adder	6.83
945732	AF1-238 E	8.7042	Adder	10.24
945741	AF1-239 C	1.3654	Adder	1.61
945742	AF1-239 E	2.0481	Adder	2.41
945971	AF1-262	0.0655	Adder	0.08
999905	MARINGEN 2	1.3650	50/50	1.3650
999906	PVILLEG 2	0.5792	50/50	0.5792
NEWTON	NEWTON	0.0774	Confirmed LTF	0.0774
FARMERCITY	FARMERCITY	0.0040	Confirmed LTF	0.0040
NY	NY	0.1123	Confirmed LTF	0.1123
PRAIRIE	PRAIRIE	0.1834	Confirmed LTF	0.1834
O-066	O-066	1.6934	Confirmed LTF	1.6934
COFFEEN	COFFEEN	0.0144	Confirmed LTF	0.0144
CHEOAH	CHEOAH	0.0345	Confirmed LTF	0.0345
EDWARDS	EDWARDS	0.0252	Confirmed LTF	0.0252
TILTON	TILTON	0.0454	Confirmed LTF	0.0454
G-007	G-007	0.2361	Confirmed LTF	0.2361
GIBSON	GIBSON	0.0393	Confirmed LTF	0.0393
CALDERWOOD	CALDERWOOD	0.0348	Confirmed LTF	0.0348
BLUEG	BLUEG	0.1250	Confirmed LTF	0.1250
TRIMBLE	TRIMBLE	0.0401	Confirmed LTF	0.0401
CATAWBA	CATAWBA	0.0238	Confirmed LTF	0.0238

11.7.18 Index 18

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42250808	227900	CARDIFF	AE	227955	CEDAR	AE	1	AE_P4-2 AE6	breaker	805.0	163.25	164.7	AC	13.86

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
227842	MARINGEN	0.1483	50/50	0.1483
227927	V4-067C	0.0429	50/50	0.0429
227928	V4-067E	0.4002	50/50	0.4002
228014	PVILLEG	0.0498	50/50	0.0498
228102	BLE#2 ST (Deactivation : 30/04/2019)	30.0695	50/50	30.0695
228202	CUMB CT	1.4838	50/50	1.4838
228203	P06	1.5807	50/50	1.5807
228206	SHRMN CT	1.3483	50/50	1.3483
228261	V4-054E	0.4926	Adder	0.58
228334	MANNMILG	0.4444	Adder	0.52
228357	V2-046E	0.6787	Adder	0.8
228711	V2-041C	0.0347	50/50	0.0347
228712	V2-041E	0.3193	50/50	0.3193
228721	V2-035E	0.0839	Adder	0.1
228727	W2-039G	1.2139	50/50	1.2139
228728	W1-130C	0.0715	50/50	0.0715
228729	W2-030 C	0.1456	50/50	0.1456
228731	V3-036	0.2134	50/50	0.2134
228732	V1-021 C	0.0598	50/50	0.0598
902092	W1-130E	0.6577	50/50	0.6577
902432	W2-030 E	1.3515	50/50	1.3515
919663	AA2-048 BAT	0.5218	Merchant Transmission	0.5218
923532	AB1-169AC	64.1520	50/50	64.1520
924531	AB2-102 C	29.5133	50/50	29.5133
924532	AB2-102 E	0.6559	50/50	0.6559
924701	AB2-122 C	0.1729	50/50	0.1729
924702	AB2-122 E	0.2964	50/50	0.2964
930001	AB1-001 C	0.1283	50/50	0.1283
930002	AB1-001 E	0.2108	50/50	0.2108
933962	AD1-019 E	1.6653	50/50	1.6653
938421	AE1-061 C	0.3173	50/50	0.3173
938422	AE1-061 E	0.3173	50/50	0.3173
938781	AE1-104 C O1	24.2715	50/50	24.2715
938782	AE1-104 E O1	62.0983	50/50	62.0983
938871	AE1-115 C	0.5403	Adder	0.64
938872	AE1-115 E	0.5403	Adder	0.64
939121	AE1-142 C O1	-0.9048	Adder	-1.06
939301	AE1-161 C	1.4290	Adder	1.68
939302	AE1-161 E	2.1435	Adder	2.52
939501	AE1-179 C O1	3.1416	Adder	3.7
939502	AE1-179 E O1	2.2171	Adder	2.61

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
939931	AE1-229 C O1	5.7085	Adder	6.72
939932	AE1-229 E O1	3.8677	Adder	4.55
940001	AE1-240 C O1	2.6139	Adder	3.08
940002	AE1-240 E O1	1.8658	Adder	2.2
940361	AE2-020 C	33.2593	50/50	33.2593
940362	AE2-020 E	155.7225	50/50	155.7225
940371	AE2-021 C	33.2593	50/50	33.2593
940372	AE2-021 E	155.7225	50/50	155.7225
940381	AE2-022 C	19.4013	50/50	19.4013
940382	AE2-022 E	90.8381	50/50	90.8381
940691	AE2-056 C	-0.0827	Adder	-0.1
940701	AE2-057 C	-0.0241	Adder	-0.03
940921	AE2-081 C	-0.0878	Adder	-0.1
942101	AE2-222 C O1	18.5771	50/50	18.5771
942102	AE2-222 E O1	46.6669	50/50	46.6669
942381	AE2-251 C	105.3649	50/50	105.3649
942382	AE2-251 E	269.5991	50/50	269.5991
943732	AF1-041 E	0.5311	50/50	0.5311
945431	AF1-208 C O1	1.9647	Adder	2.31
945432	AF1-208 E O1	1.3098	Adder	1.54
945731	AF1-238 C	4.7113	Adder	5.54
945732	AF1-238 E	8.3140	50/50	8.3140
945741	AF1-239 C	1.1085	Adder	1.3
945742	AF1-239 E	1.6628	Adder	1.96
945971	AF1-262	0.0609	Adder	0.07
999905	MARINGEN 2	0.8812	50/50	0.8812
999906	PVILLEG 2	0.3675	50/50	0.3675
WEC	WEC	0.0047	Confirmed LTF	0.0047
LGEE	LGEE	0.0097	Confirmed LTF	0.0097
CPL	CPL	0.0351	Confirmed LTF	0.0351
CBM-W2	CBM-W2	0.1802	Confirmed LTF	0.1802
NY	NY	0.6083	Confirmed LTF	0.6083
TVA	TVA	0.0378	Confirmed LTF	0.0378
O-066	O-066	11.0544	Confirmed LTF	11.0544
CBM-S2	CBM-S2	0.2601	Confirmed LTF	0.2601
CBM-S1	CBM-S1	0.2130	Confirmed LTF	0.2130
G-007	G-007	3.9426	Confirmed LTF	3.9426
MADISON	MADISON	0.0081	Confirmed LTF	0.0081
MEC	MEC	0.0286	Confirmed LTF	0.0286
CBM-W1	CBM-W1	0.1501	Confirmed LTF	0.1501

11.7.19 Index 19

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41648231	227905	SCULL#1	AE	227903	MILL #1	AE	1	AE_P4-2 AE29	breaker	306.0	127.37	131.7	AC	15.58

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
227928	V4-067E	-0.1098	Adder	-0.13
228102	BLE#2 ST (Deactivation : 30/04/2019)	95.6980	50/50	95.6980
228202	CUMB CT	1.8243	50/50	1.8243
228203	P06	1.7817	50/50	1.7817
228206	SHRMN CT	1.6207	50/50	1.6207
228261	V4-054E	0.4771	Adder	0.56
228711	V2-041C	0.0427	50/50	0.0427
228712	V2-041E	0.3937	50/50	0.3937
228732	V1-021 C	0.0862	50/50	0.0862
924531	AB2-102 C	33.2663	50/50	33.2663
924532	AB2-102 E	0.7393	50/50	0.7393
924701	AB2-122 C	-0.0474	Adder	-0.06
938781	AE1-104 C O1	77.2456	50/50	77.2456
938782	AE1-104 E O1	197.6317	50/50	197.6317
939501	AE1-179 C O1	3.4266	Adder	4.03
939502	AE1-179 E O1	2.4182	Adder	2.84
940001	AE1-240 C O1	2.8513	Adder	3.35
940002	AE1-240 E O1	2.0352	Adder	2.39
945431	AF1-208 C O1	1.6797	Adder	1.98
945432	AF1-208 E O1	1.1198	Adder	1.32
945731	AF1-238 C	6.2332	50/50	6.2332
945732	AF1-238 E	9.3498	50/50	9.3498
945741	AF1-239 C	1.4666	50/50	1.4666
945742	AF1-239 E	2.2000	50/50	2.2000
999905	MARINGEN 2	-0.3396	Adder	-0.4
999906	PVILLEG 2	-0.1490	Adder	-0.18
NEWTON	NEWTON	0.0322	Confirmed LTF	0.0322
FARMERCITY	FARMERCITY	0.0017	Confirmed LTF	0.0017
NY	NY	0.0951	Confirmed LTF	0.0951
PRAIRIE	PRAIRIE	0.0775	Confirmed LTF	0.0775
O-066	O-066	1.5994	Confirmed LTF	1.5994
COFFEEN	COFFEEN	0.0060	Confirmed LTF	0.0060
CHEOAH	CHEOAH	0.0145	Confirmed LTF	0.0145
EDWARDS	EDWARDS	0.0109	Confirmed LTF	0.0109
TILTON	TILTON	0.0195	Confirmed LTF	0.0195
G-007	G-007	0.4815	Confirmed LTF	0.4815
MADISON	MADISON	0.0020	Confirmed LTF	0.0020
GIBSON	GIBSON	0.0169	Confirmed LTF	0.0169
CALDERWOOD	CALDERWOOD	0.0144	Confirmed LTF	0.0144
BLUEG	BLUEG	0.0521	Confirmed LTF	0.0521
TRIMBLE	TRIMBLE	0.0173	Confirmed LTF	0.0173

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
CATAWBA	CATAWBA	0.0094	Confirmed LTF	0.0094

11.7.20 Index 20

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41648241	227906	SCULL#2	AE	227904	MILL #2	AE	1	AE_P4-2 AE28	breaker	306.0	128.7	133.61	AC	17.63

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
227928	V4-067E	-0.1249	Adder	-0.15
228102	BLE#2 ST (Deactivation : 30/04/2019)	88.7405	50/50	88.7405
228202	CUMB CT	2.0641	50/50	2.0641
228203	P06	2.0152	50/50	2.0152
228206	SHRMN CT	1.8337	50/50	1.8337
228261	V4-054E	0.5398	Adder	0.64
228711	V2-041C	0.0484	50/50	0.0484
228712	V2-041E	0.4454	50/50	0.4454
228732	V1-021 C	0.0970	50/50	0.0970
924531	AB2-102 C	37.6245	50/50	37.6245
924532	AB2-102 E	0.8361	50/50	0.8361
924701	AB2-122 C	-0.0540	Adder	-0.06
938781	AE1-104 C O1	71.6296	50/50	71.6296
938782	AE1-104 E O1	183.2633	50/50	183.2633
939501	AE1-179 C O1	3.8770	Adder	4.56
939502	AE1-179 E O1	2.7361	Adder	3.22
940001	AE1-240 C O1	3.2262	Adder	3.8
940002	AE1-240 E O1	2.3028	Adder	2.71
945431	AF1-208 C O1	1.9005	Adder	2.24
945432	AF1-208 E O1	1.2670	Adder	1.49
945731	AF1-238 C	7.0528	50/50	7.0528
945732	AF1-238 E	10.5792	50/50	10.5792
945741	AF1-239 C	1.6595	50/50	1.6595
945742	AF1-239 E	2.4892	50/50	2.4892
999905	MARINGEN 2	-0.3855	Adder	-0.45
999906	PVILLEG 2	-0.1685	Adder	-0.2
NEWTON	NEWTON	0.0376	Confirmed LTF	0.0376
FARMERCITY	FARMERCITY	0.0019	Confirmed LTF	0.0019
NY	NY	0.1073	Confirmed LTF	0.1073
PRAIRIE	PRAIRIE	0.0878	Confirmed LTF	0.0878
O-066	O-066	1.8077	Confirmed LTF	1.8077
COFFEEN	COFFEEN	0.0070	Confirmed LTF	0.0070
CHEOAH	CHEOAH	0.0160	Confirmed LTF	0.0160
EDWARDS	EDWARDS	0.0122	Confirmed LTF	0.0122
TILTON	TILTON	0.0220	Confirmed LTF	0.0220
G-007	G-007	0.5450	Confirmed LTF	0.5450
GIBSON	GIBSON	0.0191	Confirmed LTF	0.0191
CALDERWOOD	CALDERWOOD	0.0164	Confirmed LTF	0.0164
BLUEG	BLUEG	0.0608	Confirmed LTF	0.0608
TRIMBLE	TRIMBLE	0.0195	Confirmed LTF	0.0195
CATAWBA	CATAWBA	0.0109	Confirmed LTF	0.0109

11.7.21 Index 21

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
51072462	227913	CARDIFF	AE	227902	LEWIS #1	AE	1	AE_P4-2 AE9	breaker	478.0	121.08	123.42	AC	11.14

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
228261	V4-054E	-0.3675	Adder	-0.43
924701	AB2-122 C	-0.0297	Adder	-0.03
938423	AE1-061 BAT	0.7697	50/50	0.7697
939303	AE1-161 BAT	4.4705	Merchant Transmission	4.4705
940361	AE2-020 C	17.7510	50/50	17.7510
940362	AE2-020 E	83.1115	50/50	83.1115
940371	AE2-021 C	17.7510	50/50	17.7510
940372	AE2-021 E	83.1115	50/50	83.1115
940381	AE2-022 C	10.3547	50/50	10.3547
940382	AE2-022 E	48.4817	50/50	48.4817
942381	AE2-251 C	56.2348	50/50	56.2348
942382	AE2-251 E	143.8892	50/50	143.8892
945733	AF1-238 BAT	11.1409	Merchant Transmission	11.1409
945743	AF1-239 BAT	2.6214	Merchant Transmission	2.6214
999905	MARINGEN 2	-0.7894	Adder	-0.93
999906	PVILLEG 2	-0.3842	Adder	-0.45
WEC	WEC	0.0076	Confirmed LTF	0.0076
LGEE	LGEE	0.0131	Confirmed LTF	0.0131
CPL	CPL	0.0099	Confirmed LTF	0.0099
G-007A	G-007A	2.0595	Confirmed LTF	2.0595
VFT	VFT	2.6058	Confirmed LTF	2.6058
CBM-W2	CBM-W2	0.1802	Confirmed LTF	0.1802
TVA	TVA	0.0294	Confirmed LTF	0.0294
CBM-S2	CBM-S2	0.0983	Confirmed LTF	0.0983
CBM-S1	CBM-S1	0.1874	Confirmed LTF	0.1874
MEC	MEC	0.0365	Confirmed LTF	0.0365
CBM-W1	CBM-W1	0.3128	Confirmed LTF	0.3128

11.7.22 Index 22

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
51073203	227934	CARDIFF2	AE	227945	LEWIS #2	AE	1	AE_P7-1 AE3TOWER	tower	478.0	133.12	134.91	AC	8.54

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
227928	V4-067E	-0.0776	Adder	-0.09
924701	AB2-122 C	-0.0336	Adder	-0.04
938423	AE1-061 BAT	0.7302	50/50	0.7302
939303	AE1-161 BAT	4.0450	Merchant Transmission	4.0450
940361	AE2-020 C	16.2694	50/50	16.2694
940362	AE2-020 E	76.1743	50/50	76.1743
940371	AE2-021 C	16.2694	50/50	16.2694
940372	AE2-021 E	76.1743	50/50	76.1743
940381	AE2-022 C	9.4905	50/50	9.4905
940382	AE2-022 E	44.4350	50/50	44.4350
942381	AE2-251 C	51.5410	50/50	51.5410
942382	AE2-251 E	131.8790	50/50	131.8790
945733	AF1-238 BAT	8.5387	Merchant Transmission	8.5387
945743	AF1-239 BAT	2.0091	Merchant Transmission	2.0091
999905	MARINGEN 2	-0.7952	Adder	-0.94
999906	PVILLEG 2	-0.3874	Adder	-0.46
WEC	WEC	0.0104	Confirmed LTF	0.0104
LGEE	LGEE	0.0188	Confirmed LTF	0.0188
CPL	CPL	0.0238	Confirmed LTF	0.0238
CBM-W2	CBM-W2	0.2703	Confirmed LTF	0.2703
NY	NY	0.0321	Confirmed LTF	0.0321
TVA	TVA	0.0476	Confirmed LTF	0.0476
O-066	O-066	0.6854	Confirmed LTF	0.6854
CBM-S2	CBM-S2	0.2081	Confirmed LTF	0.2081
CBM-S1	CBM-S1	0.2897	Confirmed LTF	0.2897
G-007	G-007	0.2933	Confirmed LTF	0.2933
MEC	MEC	0.0524	Confirmed LTF	0.0524
CBM-W1	CBM-W1	0.4128	Confirmed LTF	0.4128

11.7.23 Index 23

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41648252	227955	CEDAR	AE	206302	28OYSTER C	JCP&L	1	AE_P4-2 AE6	breaker	564.0	218.93	221.38	AC	16.13

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
227842	MARINGEN	0.1908	50/50	0.1908
227927	V4-067C	0.0509	50/50	0.0509
227928	V4-067E	0.4755	50/50	0.4755
228014	PVILLEG	0.0661	50/50	0.0661
228102	BLE#2 ST (Deactivation : 30/04/2019)	35.9606	50/50	35.9606
228202	CUMB CT	1.7267	50/50	1.7267
228203	P06	1.8296	50/50	1.8296
228206	SHRMN CT	1.5696	50/50	1.5696
228261	V4-054E	0.5704	Adder	0.67
228334	MANNMILG	0.5067	Adder	0.6
228357	V2-046E	0.7758	Adder	0.91
228711	V2-041C	0.0404	50/50	0.0404
228712	V2-041E	0.3719	50/50	0.3719
228721	V2-035E	0.0980	Adder	0.12
228727	W2-039G	1.4126	50/50	1.4126
228728	W1-130C	0.0852	50/50	0.0852
228729	W2-030 C	0.1806	50/50	0.1806
228731	V3-036	0.2772	50/50	0.2772
228732	V1-021 C	0.0708	50/50	0.0708
902092	W1-130E	0.7836	50/50	0.7836
902432	W2-030 E	1.6758	50/50	1.6758
919663	AA2-048 BAT	0.5904	Merchant Transmission	0.5904
923532	AB1-169AC	76.4500	50/50	76.4500
924531	AB2-102 C	34.1595	50/50	34.1595
924532	AB2-102 E	0.7591	50/50	0.7591
924701	AB2-122 C	0.2054	50/50	0.2054
924702	AB2-122 E	0.3521	50/50	0.3521
930001	AB1-001 C	0.2897	50/50	0.2897
930002	AB1-001 E	0.4762	50/50	0.4762
933962	AD1-019 E	2.1627	50/50	2.1627
938421	AE1-061 C	0.3780	50/50	0.3780
938422	AE1-061 E	0.3780	50/50	0.3780
938781	AE1-104 C O1	29.0267	50/50	29.0267
938782	AE1-104 E O1	74.2645	50/50	74.2645
938871	AE1-115 C	0.6106	Adder	0.72
938872	AE1-115 E	0.6106	Adder	0.72
939121	AE1-142 C O1	-1.0242	Adder	-1.2
939301	AE1-161 C	1.6895	Adder	1.99
939302	AE1-161 E	2.5342	Adder	2.98
939501	AE1-179 C O1	3.6515	Adder	4.3
939502	AE1-179 E O1	2.5769	Adder	3.03

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
939931	AE1-229 C O1	6.5800	Adder	7.74
939932	AE1-229 E O1	4.4582	Adder	5.24
940001	AE1-240 C O1	3.0371	Adder	3.57
940002	AE1-240 E O1	2.1679	Adder	2.55
940361	AE2-020 C	35.3658	50/50	35.3658
940362	AE2-020 E	165.5851	50/50	165.5851
940371	AE2-021 C	35.3658	50/50	35.3658
940372	AE2-021 E	165.5851	50/50	165.5851
940381	AE2-022 C	20.6300	50/50	20.6300
940382	AE2-022 E	96.5913	50/50	96.5913
940691	AE2-056 C	-0.0935	Adder	-0.11
940701	AE2-057 C	-0.0272	Adder	-0.03
940921	AE2-081 C	-0.0994	Adder	-0.12
942101	AE2-222 C O1	24.4284	50/50	24.4284
942102	AE2-222 E O1	61.3656	50/50	61.3656
942381	AE2-251 C	112.0381	50/50	112.0381
942382	AE2-251 E	286.6740	50/50	286.6740
943732	AF1-041 E	0.6930	50/50	0.6930
945431	AF1-208 C O1	2.2656	Adder	2.67
945432	AF1-208 E O1	1.5104	Adder	1.78
945731	AF1-238 C	5.4842	Adder	6.45
945732	AF1-238 E	9.6780	50/50	9.6780
945741	AF1-239 C	1.2904	Adder	1.52
945742	AF1-239 E	1.9356	Adder	2.28
945971	AF1-262	0.0711	Adder	0.08
999905	MARINGEN 2	1.1341	50/50	1.1341
999906	PVILLEG 2	0.4882	50/50	0.4882
WEC	WEC	0.0047	Confirmed LTF	0.0047
LGEE	LGEE	0.0103	Confirmed LTF	0.0103
CPL	CPL	0.0384	Confirmed LTF	0.0384
CBM-W2	CBM-W2	0.1884	Confirmed LTF	0.1884
NY	NY	0.6874	Confirmed LTF	0.6874
TVA	TVA	0.0406	Confirmed LTF	0.0406
O-066	O-066	12.4925	Confirmed LTF	12.4925
CBM-S2	CBM-S2	0.2832	Confirmed LTF	0.2832
CBM-S1	CBM-S1	0.2300	Confirmed LTF	0.2300
G-007	G-007	4.4595	Confirmed LTF	4.4595
MADISON	MADISON	0.0081	Confirmed LTF	0.0081
MEC	MEC	0.0286	Confirmed LTF	0.0286
CBM-W1	CBM-W1	0.1501	Confirmed LTF	0.1501

11.7.24 Index 24

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41648226	228110	BLE	AE	227905	SCULL#1	AE	1	AE_P4-2 AE29	breaker	306.0	134.29	138.62	AC	15.58

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
227928	V4-067E	-0.1098	Adder	-0.13
228102	BLE#2 ST (Deactivation : 30/04/2019)	95.6980	50/50	95.6980
228202	CUMB CT	1.8243	50/50	1.8243
228203	P06	1.7817	50/50	1.7817
228206	SHRMN CT	1.6207	50/50	1.6207
228261	V4-054E	0.4771	Adder	0.56
228711	V2-041C	0.0427	50/50	0.0427
228712	V2-041E	0.3937	50/50	0.3937
228732	V1-021 C	0.0862	50/50	0.0862
924531	AB2-102 C	33.2663	50/50	33.2663
924532	AB2-102 E	0.7393	50/50	0.7393
924701	AB2-122 C	-0.0474	Adder	-0.06
938781	AE1-104 C O1	77.2456	50/50	77.2456
938782	AE1-104 E O1	197.6317	50/50	197.6317
939501	AE1-179 C O1	3.4266	Adder	4.03
939502	AE1-179 E O1	2.4182	Adder	2.84
940001	AE1-240 C O1	2.8513	Adder	3.35
940002	AE1-240 E O1	2.0352	Adder	2.39
945431	AF1-208 C O1	1.6797	Adder	1.98
945432	AF1-208 E O1	1.1198	Adder	1.32
945731	AF1-238 C	6.2332	50/50	6.2332
945732	AF1-238 E	9.3498	50/50	9.3498
945741	AF1-239 C	1.4666	50/50	1.4666
945742	AF1-239 E	2.2000	50/50	2.2000
999905	MARINGEN 2	-0.3396	Adder	-0.4
999906	PVILLEG 2	-0.1490	Adder	-0.18
NEWTON	NEWTON	0.0322	Confirmed LTF	0.0322
FARMERCITY	FARMERCITY	0.0017	Confirmed LTF	0.0017
NY	NY	0.0951	Confirmed LTF	0.0951
PRAIRIE	PRAIRIE	0.0775	Confirmed LTF	0.0775
O-066	O-066	1.5994	Confirmed LTF	1.5994
COFFEEN	COFFEEN	0.0060	Confirmed LTF	0.0060
CHEOAH	CHEOAH	0.0145	Confirmed LTF	0.0145
EDWARDS	EDWARDS	0.0109	Confirmed LTF	0.0109
TILTON	TILTON	0.0195	Confirmed LTF	0.0195
G-007	G-007	0.4815	Confirmed LTF	0.4815
MADISON	MADISON	0.0020	Confirmed LTF	0.0020
GIBSON	GIBSON	0.0169	Confirmed LTF	0.0169
CALDERWOOD	CALDERWOOD	0.0144	Confirmed LTF	0.0144
BLUEG	BLUEG	0.0521	Confirmed LTF	0.0521
TRIMBLE	TRIMBLE	0.0173	Confirmed LTF	0.0173

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
CATAWBA	CATAWBA	0.0094	Confirmed LTF	0.0094

11.7.25 Index 25

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41648236	228110	BLE	AE	227906	SCULL#2	AE	1	AE_P4-2 AE28	breaker	306.0	133.64	138.55	AC	17.63

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
227928	V4-067E	-0.1249	Adder	-0.15
228102	BLE#2 ST (Deactivation : 30/04/2019)	88.7405	50/50	88.7405
228202	CUMB CT	2.0641	50/50	2.0641
228203	P06	2.0152	50/50	2.0152
228206	SHRMN CT	1.8337	50/50	1.8337
228261	V4-054E	0.5398	Adder	0.64
228711	V2-041C	0.0484	50/50	0.0484
228712	V2-041E	0.4454	50/50	0.4454
228732	V1-021 C	0.0970	50/50	0.0970
924531	AB2-102 C	37.6245	50/50	37.6245
924532	AB2-102 E	0.8361	50/50	0.8361
924701	AB2-122 C	-0.0540	Adder	-0.06
938781	AE1-104 C O1	71.6296	50/50	71.6296
938782	AE1-104 E O1	183.2633	50/50	183.2633
939501	AE1-179 C O1	3.8770	Adder	4.56
939502	AE1-179 E O1	2.7361	Adder	3.22
940001	AE1-240 C O1	3.2262	Adder	3.8
940002	AE1-240 E O1	2.3028	Adder	2.71
945431	AF1-208 C O1	1.9005	Adder	2.24
945432	AF1-208 E O1	1.2670	Adder	1.49
945731	AF1-238 C	7.0528	50/50	7.0528
945732	AF1-238 E	10.5792	50/50	10.5792
945741	AF1-239 C	1.6595	50/50	1.6595
945742	AF1-239 E	2.4892	50/50	2.4892
999905	MARINGEN 2	-0.3855	Adder	-0.45
999906	PVILLEG 2	-0.1685	Adder	-0.2
NEWTON	NEWTON	0.0376	Confirmed LTF	0.0376
FARMERCITY	FARMERCITY	0.0019	Confirmed LTF	0.0019
NY	NY	0.1073	Confirmed LTF	0.1073
PRAIRIE	PRAIRIE	0.0878	Confirmed LTF	0.0878
O-066	O-066	1.8077	Confirmed LTF	1.8077
COFFEEN	COFFEEN	0.0070	Confirmed LTF	0.0070
CHEOAH	CHEOAH	0.0160	Confirmed LTF	0.0160
EDWARDS	EDWARDS	0.0122	Confirmed LTF	0.0122
TILTON	TILTON	0.0220	Confirmed LTF	0.0220
G-007	G-007	0.5450	Confirmed LTF	0.5450
GIBSON	GIBSON	0.0191	Confirmed LTF	0.0191
CALDERWOOD	CALDERWOOD	0.0164	Confirmed LTF	0.0164
BLUEG	BLUEG	0.0608	Confirmed LTF	0.0608
TRIMBLE	TRIMBLE	0.0195	Confirmed LTF	0.0195
CATAWBA	CATAWBA	0.0109	Confirmed LTF	0.0109

11.7.26 Index 26

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42777361	228311	CHAMBERS	AE	228312	PEDRKTWN	AE	1	JC-P7-1-JCC-230-13	tower	552.0	114.11	117.01	AC	18.82

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
227928	V4-067E	0.1595	Adder	0.19
228102	BLE#2 ST (Deactivation : 30/04/2019)	17.1664	Adder	20.2
228201	CARL#2CT	0.9667	50/50	0.9667
228261	V4-054E	0.9792	Adder	1.15
228309	CCLP NUG	20.1373	50/50	20.1373
228334	MANNMILG	1.9621	50/50	1.9621
228343	QUINTN#1 (Deactivation : 26/04/2020)	0.0782	50/50	0.0782
228351	V2-046C	0.2638	50/50	0.2638
228357	V2-046E	2.7443	50/50	2.7443
228712	V2-041E	0.3394	Adder	0.4
228720	V2-035C	0.0267	50/50	0.0267
228721	V2-035E	0.2463	50/50	0.2463
902092	W1-130E	0.5357	Adder	0.63
902432	W2-030 E	0.6097	Adder	0.72
923532	AB1-169AC	51.7242	Adder	60.85
924531	AB2-102 C	32.8261	Adder	38.62
924532	AB2-102 E	0.7295	Adder	0.86
924701	AB2-122 C	0.0689	Adder	0.08
924702	AB2-122 E	0.1182	Adder	0.14
930001	AB1-001 C	0.0847	Adder	0.1
930002	AB1-001 E	0.1392	Adder	0.16
933962	AD1-019 E	0.7555	Adder	0.89
938421	AE1-061 C	0.2585	Adder	0.3
938422	AE1-061 E	0.2585	Adder	0.3
938781	AE1-104 C O1	13.8564	Adder	16.3
938782	AE1-104 E O1	35.4513	Adder	41.71
938871	AE1-115 C	3.1721	50/50	3.1721
938872	AE1-115 E	3.1721	50/50	3.1721
939301	AE1-161 C	1.8477	Adder	2.17
939302	AE1-161 E	2.7716	Adder	3.26
939501	AE1-179 C O1	4.8177	Adder	5.67
939502	AE1-179 E O1	3.3999	Adder	4.0
939931	AE1-229 C O1	25.1478	50/50	25.1478
939932	AE1-229 E O1	17.0384	50/50	17.0384
940001	AE1-240 C O1	4.0690	Adder	4.79
940002	AE1-240 E O1	2.9044	Adder	3.42
940161	AE2-000 C O1	20.4923	Adder	24.11
940162	AE2-000 E O1	52.4328	Adder	61.69
940361	AE2-020 C	10.0327	Adder	11.8
940362	AE2-020 E	46.9737	Adder	55.26

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
940371	AE2-021 C	10.0327	Adder	11.8
940372	AE2-021 E	46.9737	Adder	55.26
940381	AE2-022 C	5.8524	Adder	6.89
940382	AE2-022 E	27.4013	Adder	32.24
942101	AE2-222 C O1	8.4420	Adder	9.93
942102	AE2-222 E O1	21.2068	Adder	24.95
942381	AE2-251 C	31.7833	Adder	37.39
942382	AE2-251 E	81.3245	Adder	95.68
942571	AE2-272	0.0301	50/50	0.0301
943732	AF1-041 E	0.2360	Adder	0.28
944332	AF1-101 E O1	51.4051	Adder	60.48
945431	AF1-208 C O1	5.7443	50/50	5.7443
945432	AF1-208 E O1	3.8295	50/50	3.8295
945731	AF1-238 C	6.3998	Adder	7.53
945732	AF1-238 E	9.5996	Adder	11.29
945741	AF1-239 C	1.5058	Adder	1.77
945742	AF1-239 E	2.2587	Adder	2.66
945971	AF1-262	0.1788	50/50	0.1788
999905	MARINGEN 2	0.3993	Adder	0.47
999906	PVILLEG 2	0.1671	Adder	0.2
WEC	WEC	0.0022	Confirmed LTF	0.0022
LGEE	LGEE	0.0046	Confirmed LTF	0.0046
CPL	CPL	0.0159	Confirmed LTF	0.0159
CBM-W2	CBM-W2	0.0819	Confirmed LTF	0.0819
NY	NY	0.1416	Confirmed LTF	0.1416
TVA	TVA	0.0182	Confirmed LTF	0.0182
O-066	O-066	2.2579	Confirmed LTF	2.2579
CBM-S2	CBM-S2	0.1214	Confirmed LTF	0.1214
CBM-S1	CBM-S1	0.1022	Confirmed LTF	0.1022
G-007	G-007	0.2974	Confirmed LTF	0.2974
MADISON	MADISON	0.0020	Confirmed LTF	0.0020
MEC	MEC	0.0127	Confirmed LTF	0.0127
CBM-W1	CBM-W1	0.0751	Confirmed LTF	0.0751

11.7.27 Index 27

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42777290	228312	PEDRKTWN	AE	228313	BRIDGPRT	AE	1	JC-P7-1-JCC-230-13	tower	552.0	130.58	133.48	AC	18.75

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
227928	V4-067E	0.1587	Adder	0.19
228102	BLE#2 ST (Deactivation : 30/04/2019)	17.0884	Adder	20.1
228261	V4-054E	0.9760	Adder	1.15
228306	PCLP STM	6.7851	50/50	6.7851
228307	PCLP GT	6.7851	50/50	6.7851
228309	CCLP NUG	20.1114	50/50	20.1114
228334	MANNMILG	1.9569	50/50	1.9569
228343	QUINTN#1 (Deactivation : 26/04/2020)	0.0781	50/50	0.0781
228351	V2-046C	0.2630	50/50	0.2630
228357	V2-046E	2.7365	50/50	2.7365
228712	V2-041E	0.3381	Adder	0.4
228720	V2-035C	0.0267	50/50	0.0267
228721	V2-035E	0.2455	50/50	0.2455
902092	W1-130E	0.5328	Adder	0.63
902432	W2-030 E	0.6066	Adder	0.71
923532	AB1-169AC	51.4437	Adder	60.52
924531	AB2-102 C	32.7095	Adder	38.48
924532	AB2-102 E	0.7269	Adder	0.86
924701	AB2-122 C	0.0686	Adder	0.08
924702	AB2-122 E	0.1176	Adder	0.14
930001	AB1-001 C	0.0843	Adder	0.1
930002	AB1-001 E	0.1385	Adder	0.16
933962	AD1-019 E	0.7516	Adder	0.88
938421	AE1-061 C	0.2570	Adder	0.3
938422	AE1-061 E	0.2570	Adder	0.3
938781	AE1-104 C O1	13.7934	Adder	16.23
938782	AE1-104 E O1	35.2902	Adder	41.52
938871	AE1-115 C	3.1661	50/50	3.1661
938872	AE1-115 E	3.1661	50/50	3.1661
939301	AE1-161 C	1.8375	Adder	2.16
939302	AE1-161 E	2.7563	Adder	3.24
939501	AE1-179 C O1	4.7996	Adder	5.65
939502	AE1-179 E O1	3.3871	Adder	3.98
939931	AE1-229 C O1	25.0935	50/50	25.0935
939932	AE1-229 E O1	17.0016	50/50	17.0016
940001	AE1-240 C O1	4.0539	Adder	4.77
940002	AE1-240 E O1	2.8937	Adder	3.4
940161	AE2-000 C O1	20.3734	Adder	23.97
940162	AE2-000 E O1	52.1286	Adder	61.33
940361	AE2-020 C	9.9775	Adder	11.74

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
940362	AE2-020 E	46.7153	Adder	54.96
940371	AE2-021 C	9.9775	Adder	11.74
940372	AE2-021 E	46.7153	Adder	54.96
940381	AE2-022 C	5.8202	Adder	6.85
940382	AE2-022 E	27.2506	Adder	32.06
942101	AE2-222 C O1	8.3977	Adder	9.88
942102	AE2-222 E O1	21.0956	Adder	24.82
942381	AE2-251 C	31.6085	Adder	37.19
942382	AE2-251 E	80.8772	Adder	95.15
942571	AE2-272	0.0301	50/50	0.0301
943732	AF1-041 E	0.2348	Adder	0.28
944332	AF1-101 E O1	51.1068	Adder	60.13
945431	AF1-208 C O1	5.7281	50/50	5.7281
945432	AF1-208 E O1	3.8187	50/50	3.8187
945731	AF1-238 C	6.3733	Adder	7.5
945732	AF1-238 E	9.5600	Adder	11.25
945741	AF1-239 C	1.4996	Adder	1.76
945742	AF1-239 E	2.2494	Adder	2.65
945971	AF1-262	0.1782	50/50	0.1782
999905	MARINGEN 2	0.3972	Adder	0.47
999906	PVILLEG 2	0.1662	Adder	0.2
NEWTON	NEWTON	0.0569	Confirmed LTF	0.0569
FARMERCITY	FARMERCITY	0.0029	Confirmed LTF	0.0029
NY	NY	0.1753	Confirmed LTF	0.1753
PRAIRIE	PRAIRIE	0.1343	Confirmed LTF	0.1343
O-066	O-066	2.6678	Confirmed LTF	2.6678
COFFEEN	COFFEEN	0.0106	Confirmed LTF	0.0106
CHEOAH	CHEOAH	0.0230	Confirmed LTF	0.0230
EDWARDS	EDWARDS	0.0189	Confirmed LTF	0.0189
TILTON	TILTON	0.0340	Confirmed LTF	0.0340
G-007	G-007	0.3609	Confirmed LTF	0.3609
MADISON	MADISON	0.0020	Confirmed LTF	0.0020
GIBSON	GIBSON	0.0289	Confirmed LTF	0.0289
CALDERWOOD	CALDERWOOD	0.0234	Confirmed LTF	0.0234
BLUEG	BLUEG	0.0920	Confirmed LTF	0.0920
TRIMBLE	TRIMBLE	0.0295	Confirmed LTF	0.0295
CATAWBA	CATAWBA	0.0144	Confirmed LTF	0.0144

11.7.28 Index 28

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42777400	228313	BRIDGPRT	AE	228401	MCKLTON	AE	1	JC-P7-1-JCC-230-13	tower	804.0	114.74	116.73	AC	18.6

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
227928	V4-067E	0.1571	Adder	0.18
228102	BLE#2 ST (Deactivation : 30/04/2019)	16.9439	Adder	19.93
228261	V4-054E	0.9700	Adder	1.14
228304	LOGAN	27.5404	50/50	27.5404
228306	PCLP STM	6.7734	50/50	6.7734
228307	PCLP GT	6.7734	50/50	6.7734
228309	CCLP NUG	20.0633	50/50	20.0633
228334	MANNMILG	1.9475	50/50	1.9475
228343	QUINTN#1 (Deactivation : 26/04/2020)	0.0777	50/50	0.0777
228351	V2-046C	0.2616	50/50	0.2616
228357	V2-046E	2.7219	50/50	2.7219
228712	V2-041E	0.3357	Adder	0.39
228720	V2-035C	0.0265	50/50	0.0265
228721	V2-035E	0.2441	50/50	0.2441
902092	W1-130E	0.5273	Adder	0.62
902432	W2-030 E	0.6008	Adder	0.71
923532	AB1-169AC	50.9154	Adder	59.9
924531	AB2-102 C	32.4934	Adder	38.23
924532	AB2-102 E	0.7221	Adder	0.85
924701	AB2-122 C	0.0679	Adder	0.08
924702	AB2-122 E	0.1164	Adder	0.14
930001	AB1-001 C	0.0834	Adder	0.1
930002	AB1-001 E	0.1371	Adder	0.16
933962	AD1-019 E	0.7443	Adder	0.88
938421	AE1-061 C	0.2544	Adder	0.3
938422	AE1-061 E	0.2544	Adder	0.3
938781	AE1-104 C O1	13.6768	Adder	16.09
938782	AE1-104 E O1	34.9919	Adder	41.17
938871	AE1-115 C	3.1548	50/50	3.1548
938872	AE1-115 E	3.1548	50/50	3.1548
939301	AE1-161 C	1.8183	Adder	2.14
939302	AE1-161 E	2.7275	Adder	3.21
939501	AE1-179 C O1	4.7659	Adder	5.61
939502	AE1-179 E O1	3.3634	Adder	3.96
939931	AE1-229 C O1	24.9930	50/50	24.9930
939932	AE1-229 E O1	16.9334	50/50	16.9334
940001	AE1-240 C O1	4.0261	Adder	4.74
940002	AE1-240 E O1	2.8738	Adder	3.38
940161	AE2-000 C O1	20.1532	Adder	23.71
940162	AE2-000 E O1	51.5651	Adder	60.66

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
940361	AE2-020 C	9.8752	Adder	11.62
940362	AE2-020 E	46.2366	Adder	54.4
940371	AE2-021 C	9.8752	Adder	11.62
940372	AE2-021 E	46.2366	Adder	54.4
940381	AE2-022 C	5.7606	Adder	6.78
940382	AE2-022 E	26.9713	Adder	31.73
942101	AE2-222 C O1	8.3157	Adder	9.78
942102	AE2-222 E O1	20.8895	Adder	24.58
942381	AE2-251 C	31.2846	Adder	36.81
942382	AE2-251 E	80.0484	Adder	94.17
942571	AE2-272	0.0299	50/50	0.0299
943732	AF1-041 E	0.2325	Adder	0.27
944332	AF1-101 E O1	50.5543	Adder	59.48
945431	AF1-208 C O1	5.6975	50/50	5.6975
945432	AF1-208 E O1	3.7984	50/50	3.7984
945731	AF1-238 C	6.3243	Adder	7.44
945732	AF1-238 E	9.4865	Adder	11.16
945741	AF1-239 C	1.4881	Adder	1.75
945742	AF1-239 E	2.2321	Adder	2.63
945971	AF1-262	0.1772	50/50	0.1772
999905	MARINGEN 2	0.3934	Adder	0.46
999906	PVILLEG 2	0.1646	Adder	0.19
NEWTON	NEWTON	0.1784	Confirmed LTF	0.1784
FARMERCITY	FARMERCITY	0.0092	Confirmed LTF	0.0092
NY	NY	0.2389	Confirmed LTF	0.2389
PRAIRIE	PRAIRIE	0.4262	Confirmed LTF	0.4262
O-066	O-066	3.4272	Confirmed LTF	3.4272
COFFEEN	COFFEEN	0.0331	Confirmed LTF	0.0331
CHEOAH	CHEOAH	0.0796	Confirmed LTF	0.0796
EDWARDS	EDWARDS	0.0585	Confirmed LTF	0.0585
TILTON	TILTON	0.1052	Confirmed LTF	0.1052
G-007	G-007	0.4784	Confirmed LTF	0.4784
MADISON	MADISON	0.0020	Confirmed LTF	0.0020
GIBSON	GIBSON	0.0906	Confirmed LTF	0.0906
CALDERWOOD	CALDERWOOD	0.0795	Confirmed LTF	0.0795
BLUEG	BLUEG	0.2882	Confirmed LTF	0.2882
TRIMBLE	TRIMBLE	0.0924	Confirmed LTF	0.0924
CATAWBA	CATAWBA	0.0539	Confirmed LTF	0.0539

11.7.29 Index 29

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
88617735	228502	MNOTLA 1	AE	228500	LANDIS	AE	1	AE_P4-2 AE46	breaker	478.0	146.12	147.94	AC	10.16

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
227842	MARINGEN	0.1128	50/50	0.1128
227927	V4-067C	0.0252	50/50	0.0252
227928	V4-067E	0.2349	50/50	0.2349
228014	PVILLEG	0.0397	50/50	0.0397
228102	BLE#2 ST (Deactivation : 30/04/2019)	24.0505	50/50	24.0505
228202	CUMB CT	1.2943	50/50	1.2943
228203	P06	1.4649	50/50	1.4649
228206	SHRMN CT	1.1055	50/50	1.1055
228711	V2-041C	0.0292	50/50	0.0292
228712	V2-041E	0.2693	50/50	0.2693
228728	W1-130C	0.4120	50/50	0.4120
228729	W2-030 C	0.1048	50/50	0.1048
228731	V3-036	0.1646	50/50	0.1646
228732	V1-021 C	0.0495	50/50	0.0495
902092	W1-130E	3.7883	50/50	3.7883
902432	W2-030 E	0.9731	50/50	0.9731
923532	AB1-169AC	362.7360	50/50	362.7360
924531	AB2-102 C	27.3510	50/50	27.3510
924532	AB2-102 E	0.6078	50/50	0.6078
924701	AB2-122 C	0.1015	50/50	0.1015
924702	AB2-122 E	0.1740	50/50	0.1740
930001	AB1-001 C	0.1380	50/50	0.1380
930002	AB1-001 E	0.2268	50/50	0.2268
933962	AD1-019 E	1.2839	50/50	1.2839
938421	AE1-061 C	1.8277	50/50	1.8277
938422	AE1-061 E	1.8277	50/50	1.8277
938781	AE1-104 C O1	19.4131	50/50	19.4131
938782	AE1-104 E O1	49.6680	50/50	49.6680
939303	AE1-161 BAT	10.4105	50/50	10.4105
939501	AE1-179 C O1	2.1372	Adder	2.51
939502	AE1-179 E O1	1.5083	Adder	1.77
940001	AE1-240 C O1	1.7723	Adder	2.09
940002	AE1-240 E O1	1.2651	Adder	1.49
940361	AE2-020 C	12.6472	50/50	12.6472
940362	AE2-020 E	59.2151	50/50	59.2151
940371	AE2-021 C	12.6472	50/50	12.6472
940372	AE2-021 E	59.2151	50/50	59.2151
940381	AE2-022 C	7.3775	50/50	7.3775
940382	AE2-022 E	34.5422	50/50	34.5422
942101	AE2-222 C O1	14.5855	50/50	14.5855
942102	AE2-222 E O1	36.6395	50/50	36.6395

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
942381	AE2-251 C	40.0661	50/50	40.0661
942382	AE2-251 E	102.5179	50/50	102.5179
943732	AF1-041 E	0.4015	50/50	0.4015
945731	AF1-238 C	3.4546	Adder	4.06
945732	AF1-238 E	5.1818	Adder	6.1
945741	AF1-239 C	0.8128	Adder	0.96
945742	AF1-239 E	1.2193	Adder	1.43
999905	MARINGEN 2	0.6704	50/50	0.6704
999906	PVILLEG 2	0.2929	50/50	0.2929
NEWTON	NEWTON	0.0247	Confirmed LTF	0.0247
FARMERCITY	FARMERCITY	0.0013	Confirmed LTF	0.0013
G-007A	G-007A	1.7933	Confirmed LTF	1.7933
VFT	VFT	1.8640	Confirmed LTF	1.8640
PRAIRIE	PRAIRIE	0.0620	Confirmed LTF	0.0620
COFFEEN	COFFEEN	0.0046	Confirmed LTF	0.0046
CHEOAH	CHEOAH	0.0130	Confirmed LTF	0.0130
EDWARDS	EDWARDS	0.0081	Confirmed LTF	0.0081
TILTON	TILTON	0.0145	Confirmed LTF	0.0145
GIBSON	GIBSON	0.0126	Confirmed LTF	0.0126
CALDERWOOD	CALDERWOOD	0.0129	Confirmed LTF	0.0129
BLUEG	BLUEG	0.0399	Confirmed LTF	0.0399
TRIMBLE	TRIMBLE	0.0128	Confirmed LTF	0.0128
CATAWBA	CATAWBA	0.0094	Confirmed LTF	0.0094

11.7.30 Index 30

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/DC	MW IMPACT
41649105	940000	AE1-240 TAP	AE	228226	SHRMAN#2	AE	1	AE_P4-2 AE46	breaker	93.0	105.37	122.96	AC	17.19

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
228200	CARL#1CT	2.0940	50/50	2.0940
228201	CARL#2CT	2.2165	50/50	2.2165
228251	CARLLS#4	0.2370	50/50	0.2370
228260	V4-054C	0.3498	50/50	0.3498
228261	V4-054E	1.6110	50/50	1.6110
228334	MANNMILG	0.6930	Adder	0.82
228343	QUINTN#1 (Deactivation : 26/04/2020)	0.0577	50/50	0.0577
228351	V2-046C	0.1319	50/50	0.1319
228357	V2-046E	1.3727	50/50	1.3727
938871	AE1-115 C	0.5038	Adder	0.59
938872	AE1-115 E	0.5038	Adder	0.59
940001	AE1-240 C O1	20.6996	50/50	20.6996
940002	AE1-240 E O1	14.7752	50/50	14.7752
942571	AE2-272	0.0151	50/50	0.0151
945431	AF1-208 C O1	8.1116	50/50	8.1116
945432	AF1-208 E O1	5.4077	50/50	5.4077
945733	AF1-238 BAT	17.1921	50/50	17.1921
945743	AF1-239 BAT	4.0452	50/50	4.0452
NEWTON	NEWTON	0.0054	Confirmed LTF	0.0054
FARMERCITY	FARMERCITY	0.0003	Confirmed LTF	0.0003
NY	NY	0.0304	Confirmed LTF	0.0304
PRAIRIE	PRAIRIE	0.0129	Confirmed LTF	0.0129
O-066	O-066	0.5510	Confirmed LTF	0.5510
COFFEEN	COFFEEN	0.0010	Confirmed LTF	0.0010
CHEOAH	CHEOAH	0.0025	Confirmed LTF	0.0025
EDWARDS	EDWARDS	0.0017	Confirmed LTF	0.0017
TILTON	TILTON	0.0031	Confirmed LTF	0.0031
G-007	G-007	0.2049	Confirmed LTF	0.2049
GIBSON	GIBSON	0.0027	Confirmed LTF	0.0027
CALDERWOOD	CALDERWOOD	0.0025	Confirmed LTF	0.0025
BLUEG	BLUEG	0.0087	Confirmed LTF	0.0087
TRIMBLE	TRIMBLE	0.0028	Confirmed LTF	0.0028
CATAWBA	CATAWBA	0.0017	Confirmed LTF	0.0017

11.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA1-060	Great Adventure 34.5kV	In Service
AA1-079	Emilie 230kV	In Service
AA2-048	Allenwood-Larrabee 34kV	Engineering and Procurement
AA2-184	Atlantic-Red Bank 34kV	In Service
AB1-000	N/A	N/A
AB1-001	Absecon 12 kV	Engineering and Procurement
AB1-063	Locust St. 13kV	Partially in Service - Under Construction
AB1-116	Egg Harbor Road 12kV	In Service
AB1-119	Tansboro Road 12kV	In Service
AB1-138	Navy 34.5kV	In Service
AB1-163	Glidden-Van Hiseville 34.5 kV	Engineering and Procurement
AB1-169A	Minotola 138 kV	Active
AB2-014	Lakewood 230kV	Partially in Service - Under Construction
AB2-049	Gloucester Township 12kV	Active
AB2-102	Cumberland 230kV	Active
AB2-122	Egg Harbor 12kV	Engineering and Procurement
AB2-139	Raceway Mall 34.5kV	In Service
AC1-010	Gloucester 26kV	Suspended
AC1-016	Burlington 26 kV	Withdrawn
AC1-017	Levittown 13 kV	In Service
AC1-030	Cinnaminson 13kV	In Service
AC1-207	Monroe 34.5kV	Engineering and Procurement
AC2-050	Tabernacle 12kV	Active
AD1-019	Ontario 23 kV	Active
AD1-059	Lakewood 230 kV	In Service
AD1-097	Linwood 230 kV II	Active
AD1-113	Raritan River 230 kV	Suspended
AD2-027	Camden Switch 26 kV	In Service
AD2-042	Camden Switch 26 kV II	In Service
AD2-064	Tansboro 12kV	Withdrawn
AD2-065	Berlin 12kV	Engineering and Procurement
AD2-135	Williamstown 12kV	Active
AE1-061	Minotola 12 kV	Active
AE1-062	Silver Lake 69 kV	Active
AE1-104	BL England 138 kV	Active
AE1-115	Churchtown 69 kV	Active
AE1-142	Manitou-Pleasant Plains 34.5 kV	Engineering and Procurement
AE1-161	Landis 138 kV	Active
AE1-179	South Millville-Newport 69 kV	Active

Queue Number	Project Name	Status
AE1-218	Glassboro 12 kV	Engineering and Procurement
AE1-219	Glassboro 12 kV II	Engineering and Procurement
AE1-229	Deepwater-Upper Pittsgrove 138 kV	Active
AE1-238	Oceanview Wind 230 kV	Active
AE1-240	Carlls Corner-Sherman Avenue 69 kV	Active
AE2-000	N/A	N/A
AE2-010	Paper Tap 69 kV	Engineering and Procurement
AE2-020	Cardiff 230 kV I	Active
AE2-021	Cardiff 230 kV II	Active
AE2-022	Cardiff 230 kV III	Active
AE2-024	Larrabee 230 kV I	Active
AE2-025	Larrabee 230 kV II	Active
AE2-056	Howell 12.47 kV Solar I	Engineering and Procurement
AE2-057	Jerseyville 12.47 kV Solar	Engineering and Procurement
AE2-064	Beaverbrook 13 kV	Under Construction
AE2-065	Deptford 13 kV	Under Construction
AE2-081	Howell 12.47 kV Solar II	Engineering and Procurement
AE2-082	Englishtown-Wyckoff 34.5 kV Solar	Active
AE2-222	Higbee 69 kV	Active
AE2-251	Cardiff 230 kV	Active
AE2-272	Woodstown 12 kV	In Service
AE2-334	Clayton-Williamstown 69 kV	Active
AF1-023	Atlantic-Red Bank 34 kV II	Active
AF1-027	Plumsted 537 Energy Storage (CIRs)	Active
AF1-041	Absecon 12.47 kV	In Service
AF1-101	Oyster Creek 230 kV III	Active
AF1-126	Cookstown 230 kV	Active
AF1-160	Silver Lake 69 kV	Active
AF1-185	Sayreville 1-2-3 230 kV	In Service
AF1-208	Quinton-Roadstown 69 kV	Active
AF1-222	Oceanview Wind 2 230 kV	Active
AF1-238	Sherman Ave. 69 kV	Active
AF1-239	Sherman Ave-Vineland 69 kV	Active
AF1-260	Allenwood-Larrabee 2 34.5 kV	Active
AF1-261	Burlington 26 kV	Active
AF1-262	Upper Pittsgrove 12 kV	Active
AF1-264	Runnemede 13 kV	Active
V1-021	Cape May County 12kV	In Service
V2-035	Pittsgrove	In Service
V2-041	Clayville 12kV	In Service
V2-046	Pilesgrove Township 12kV	In Service
V3-036	Ontario 23kV	In Service
V4-054	Fairfield Township 12kV	In Service
V4-067	Cates Road Egg Harbor Township 12kV	In Service
W1-024	Manalapan 12.5kV	In Service
W1-032	Millhurst 12.5kV	In Service
W1-112	Holmdel	In Service
W1-113	Millstone 2	In Service
W1-119	Pemberton Township 1 34.5 kV	In Service
W1-120	Pemberton Township 2 34.5 kV	In Service
W1-124	Tinton Falls 34.5kV	In Service
W1-129	Cookstown 34.5kV	In Service

Queue Number	Project Name	Status
W1-130	Vine Road 12kV	In Service
W2-019	Wrightstown 34.5kV	In Service
W2-030	Egg Harbor Township	In Service
W2-039	Clayville 69kV	In Service
W2-056	Lumberton 13kV	In Service
W2-078	Applegarth 12.5kV	In Service
W2-082	Fort Dix-McGuire 34.5kV	In Service
W3-079	Allenwood-Larabee 34.5kV	In Service
W4-025	Cookstown-Fort Dix 34.5kV	In Service
W4-060	Midland-Werner 34.5kV	In Service
X1-037	Howell	In Service
X1-085	Hornerstown-Windsor 34kV	In Service
X4-005	Raritan River 230kV	In Service
X4-015	Cookstown-Fort Dix 34kV	In Service
X4-031	Bennett-Farmingdale 34kV	In Service
Y1-057	Barbadoes 34kV	Deactivated
Y2-051	Brick-Lane Mill 34.5kV	In Service
Z1-082	Lawnside 13kV	In Service
Z2-083	Mickleton 230kV	In Service
Z2-102	Argonne-New Lisbon 34.5kV	In Service

11.9 Contingency Descriptions

Contingency Name	Contingency Definition
AE_P1-2 CUMB-DENN	CONTINGENCY 'AE_P1-2 CUMB-DENN' OPEN LINE FROM BUS 228207 TO BUS 228213 CIRCUIT 1 / END
JC-P7-1-JCC-230-7A	CONTINGENCY 'JC-P7-1-JCC-230-7A' /* SMITHBURG - LARRABEE 230 LINES DISCONNECT BRANCH FROM BUS 206309 TO BUS 206294 CKT 1 DISCONNECT BRANCH FROM BUS 206309 TO BUS 206294 CKT 2 END
AE_P7-1 AE7TOWER	CONTINGENCY 'AE_P7-1 AE7TOWER' DISCONNECT BUS 227905 /* #1 BLE TO SCULL TO MILL 138 KV DISCONNECT BUS 227929 /* #1 SCULL 12 KV DISCONNECT BUS 227906 /* #2 BLE TO SCULL TO MILL 138 KV DISCONNECT BUS 227930 /* #2 SCULL 12 KV DISCONNECT BUS 227903 /* #1 MILL TO LEWIS 138 KV DISCONNECT BUS 227904 /* #2 MILL TO LEWIS 138 KV END
PS_P7-1_V2274+P2242_LT	CONTINGENCY 'PS_P7-1_V2274+P2242_LT' /* EAGLE POINT - GLOUCESTER & DEPTFORD - GLOUCESTER DISCONNECT BUS 219757 /* DEPTFORD SECTION 2 TRIP LINE FROM BUS 219110 TO BUS 219128 CKT 1 CLOSE LINE FROM BUS 219255 TO BUS 219256 CKT Z /* DEPTFORD CLOSE LINE FROM BUS 219180 TO BUS 219181 CKT Z /* DEPTFORD DISCONNECT BUS 219760 /* EAGLE POINT SECTION 4 MOVE 8 MW LOAD FROM BUS 219180 TO BUS 219162 /* INTERSTATION TIE TRANSFER LOAD FROM DEPTFORD TO BEAVERBK T1 MOVE 8 MW LOAD FROM BUS 219181 TO BUS 219163 /* INTERSTATION TIE TRANSFER LOAD FROM DEPTFORD TO BEAVERBK T2 MOVE 8 MW LOAD FROM BUS 219255 TO BUS 219162 /* INTERSTATION TIE TRANSFER LOAD FROM DEPTFORD TO BEAVERBK T1 MOVE 8 MW LOAD FROM BUS 219256 TO BUS 219163 /* INTERSTATION TIE TRANSFER LOAD FROM DEPTFORD TO BEAVERBK T2 END

Contingency Name	Contingency Definition
PS_P2-3_CUTB_3-4_LT	CONTINGENCY 'PS_P2-3_CUTB_3-4_LT' DISCONNECT BUS 219754 /* CUTHBERTH SECTION 3 DISCONNECT BUS 219755 /* BUS SECTION 4 CUTHBERT CLOSE LINE FROM BUS 219177 TO BUS 219178 CKT Z /* CUTHBERT CLOSE LINE FROM BUS 219176 TO BUS 219179 CKT Z /* CUTBERTH MOVE 8 MW LOAD FROM BUS 219176 TO BUS 219170 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO CINN T2 MOVE 8 MW LOAD FROM BUS 219176 TO BUS 219678 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219179 TO BUS 219678 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219178 TO BUS 219678 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219177 TO BUS 219677 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219179 TO BUS 219629 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO LAWNSIDE END
PS_P2-3_CUTB_1-4_LT	CONTINGENCY 'PS_P2-3_CUTB_1-4_LT' DISCONNECT BUS 219108 /* CUTBERTH SECTION 1 DISCONNECT BUS 219755 /* BUS SECTION 4 CUTHBERT CLOSE LINE FROM BUS 219177 TO BUS 219178 CKT Z /* CUTHBERT CLOSE LINE FROM BUS 219176 TO BUS 219179 CKT Z /* CUTBERTH MOVE 8 MW LOAD FROM BUS 219176 TO BUS 219170 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO CINN T2 MOVE 8 MW LOAD FROM BUS 219176 TO BUS 219678 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219179 TO BUS 219678 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219178 TO BUS 219678 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219177 TO BUS 219677 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219179 TO BUS 219629 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO LAWNSIDE END
PECO_P1-2_5014/* \$ CHESCO \$ 5014 \$ L	CONTINGENCY 'PECO_P1-2_5014/* \$ CHESCO \$ 5014 \$ L' TRIP BRANCH FROM BUS 200065 TO BUS 200051 CKT 1 /* PCHBTM2S 500.00 ROCKSPGS 500.00 \$ CHESCO \$ 5014 \$ L END
AE_P4-2 AE28	CONTINGENCY 'AE_P4-2 AE28' /*ENGLAND TO MERION CITY BREAKER C DISCONNECT BRANCH FROM BUS 228110 TO BUS 228197 CKT 1 /*ENGLAND MERION 138 138 DISCONNECT BRANCH FROM BUS 228110 TO BUS 227905 CKT 1 /*ENGLAND #1 SCULL 138 138 END

Contingency Name	Contingency Definition
AE_P4-2 AE29	CONTINGENCY 'AE_P4-2 AE29' /*ENGLAND TO CORSON BREAKER D DISCONNECT BRANCH FROM BUS 228110 TO BUS 228111 CKT 1 /*ENGLAND MIDDLE TAP 138 138 DISCONNECT BRANCH FROM BUS 228110 TO BUS 227906 CKT 1 /*ENGLAND #2 SCULL 138 138 END
PECO_P4_CHICH045/* \$ DELCO \$ CHICH045 \$ STBK	CONTINGENCY 'PECO_P4_CHICH045/* \$ DELCO \$ CHICH045 \$ STBK' DISCONNECT BUS 213489 /* CHICHST1 230.00 \$ DELCO \$ CHICH045 \$ STBK DISCONNECT BUS 213627 /* FOULK8 230.00 \$ DELCO \$ CHICH045 \$ STBK END
PS_P2-3_CUTB_1-2_LT	CONTINGENCY 'PS_P2-3_CUTB_1-2_LT' DISCONNECT BUS 219108 /* CUTHBERTH SECTION 1 DISCONNECT BUS 219753 /* CUTHBERTH SECTION 2 CLOSE LINE FROM BUS 219176 TO BUS 219179 CKT Z /* CUTBERTH CLOSE LINE FROM BUS 219177 TO BUS 219178 CKT Z /* CUTHBERT MOVE 8 MW LOAD FROM BUS 219176 TO BUS 219170 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO CINN T2 MOVE 8 MW LOAD FROM BUS 219176 TO BUS 219678 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219179 TO BUS 219678 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219178 TO BUS 219678 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219177 TO BUS 219677 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219179 TO BUS 219629 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO LAWNSIDE END
JC-P1-2-JCC-230-031	CONTINGENCY 'JC-P1-2-JCC-230-031' /* VAN HISEVILLE TAP (B1042) 230 KV LINES DISCONNECT BRANCH FROM BUS 206318 TO BUS 206287 CKT 1 DISCONNECT BRANCH FROM BUS 206318 TO BUS 206720 CKT 1 DISCONNECT BRANCH FROM BUS 206720 TO BUS 206319 CKT 1 DISCONNECT BRANCH FROM BUS 206318 TO BUS 206294 CKT 1 DISCONNECT BUS 206318 DISCONNECT BUS 206720 END
JC-P1-2-JCC-230-019	CONTINGENCY 'JC-P1-2-JCC-230-019' /* LEISURE VILLAGE - MANITOU (A2027) 230 KV DISCONNECT BRANCH FROM BUS 206295 TO BUS 206297 CKT 1 DISCONNECT BRANCH FROM BUS 206297 TO BUS 206277 CKT 7 END

Contingency Name	Contingency Definition
JC-P1-2-JCC-230-002T	CONTINGENCY 'JC-P1-2-JCC-230-002T' /* OYSTER CREEK - CEDAR (S2045) 230 KV DISCONNECT BRANCH FROM BUS 206302 TO BUS 227955 CKT 1 /* 28OYSTER C 230 CEDAR 230 END
PS_P1-2_C-2308	CONTINGENCY 'PS_P1-2_C-2308' /* CUTHBERT TO GLOUCESTER DISCONNECT BUS 219753 /* CUTHBERTH SECTOIN 2 CLOSE LINE FROM BUS 219177 TO BUS 219178 CKT Z /* CUTHBERT END
JC-P7-1-JCC-230-10A	CONTINGENCY 'JC-P7-1-JCC-230-10A' /* LAKEWOOD - LARRABEE 230 KV LINES DISCONNECT BRANCH FROM BUS 206323 TO BUS 206294 CKT 2 DISCONNECT BRANCH FROM BUS 206323 TO BUS 206294 CKT 1 DISCONNECT BRANCH FROM BUS 206294 TO BUS 206274 CKT 3 DISCONNECT BRANCH FROM BUS 206294 TO BUS 206275 CKT 12 /* LARRABEE 12 FUTURE BREAKER AND A HALF SET BUS 206294 LOAD TO 38 MW /* LARRABEE 8 FUTURE BREAKER AND A HALF END
AE_P4-3 AE2	CONTINGENCY 'AE_P4-3 AE2' /*CARDIFF T6 BREAKER BV IN SERIES WITH CARDIFF TO LEWIS 138 KV DISCONNECT BRANCH FROM BUS 227900 TO BUS 227999 CKT 1 /*CARDIFF SVC DISCONNECT BRANCH FROM BUS 227900 TO BUS 227913 CKT 1 /*CARDIFF CARDIFF 230 138 T6 END
AE_P7-1 AE13TOWER	CONTINGENCY 'AE_P7-1 AE13TOWER' DISCONNECT BRANCH FROM BUS 228500 TO BUS 228502 CKT 1 /* LANDIS TO MINO 138 KV DISCONNECT BRANCH FROM BUS 228002 TO BUS 227900 CKT 1 /* ORCH TO CARD 230 KV END
AE_P1-3 CARD 7 XFR	CONTINGENCY 'AE_P1-3 CARD 7 XFR' OPEN LINE FROM BUS 227900 TO BUS 227934 CIRCUIT 1 / END

Contingency Name	Contingency Definition
PS_P2-3_CAMD_5-6_LT	CONTINGENCY 'PS_P2-3_CAMD_5-6_LT' TRIP LINE FROM BUS 219110 TO BUS 219125 CKT 1 /* CAMDEN TO GLOUCESTER - Y-2304 DISCONNECT BRANCH FROM BUS 219110 TO BUS 219128 CKT 2 /* REMOVE 26KV -2 DISCONNECT BUS 219129 /* REMOVE LEVITOWN BUS M-2291 CLOSE LINE FROM BUS 219196 TO BUS 219198 CKT Z /* LEVITTOWN CLOSE LINE FROM BUS 219195 TO BUS 219197 CKT Z /* LEVITOWN MOVE 8 MW LOAD FROM BUS 219195 TO BUS 219166 /* INTERSTATION TIE TRANSFER LOAD FROM LEVITN TO BUSLTN T1 MOVE 8 MW LOAD FROM BUS 219197 TO BUS 219165 /* INTERSTATION TIE TRANSFER LOAD FROM LEVITN TO BUSLTN T2 MOVE 8 MW LOAD FROM BUS 219196 TO BUS 219166 /* INTERSTATION TIE TRANSFER LOAD FROM LEVITN TO BUSLTN T1 MOVE 8 MW LOAD FROM BUS 219198 TO BUS 219165 /* INTERSTATION TIE TRANSFER LOAD FROM LEVITN TO BUSLTN T2 END
228228 SO MVLL 69.0 939500 AE1-179 TAP 69.0 1	CONTINGENCY '228228 SO MVLL 69.0 939500 AE1-179 TAP 69.0 1' OPEN BRANCH FROM BUS 228228 TO BUS 939500 CKT 1 END
AE_P1-2 SHERM-W VINE-A	CONTINGENCY 'AE_P1-2 SHERM-W VINE-A' OPEN LINE FROM BUS 228256 TO BUS 945730 CIRCUIT 1 / END
PS_P1-2_D-2282	CONTINGENCY 'PS_P1-2_D-2282' /* CUTHBERT TO GLOUCESTER DISCONNECT BUS 219755 /* BUS SECTION 4 CUTHBERT CLOSE LINE FROM BUS 219177 TO BUS 219178 CKT Z /* CUTHBERT END
AE_P1-2 ENG-MERION	CONTINGENCY 'AE_P1-2 ENG-MERION' OPEN LINE FROM BUS 228197 TO BUS 228110 CIRCUIT 1 / OPEN LINE FROM BUS 228197 TO BUS 228199 CIRCUIT 1 / CLOSE LINE FROM BUS 228198 TO BUS 228199 CIRCUIT 1 / END
AE_P4-2 AE45	CONTINGENCY 'AE_P4-2 AE45' /*ORCHARD 230 BUS BREAKER E DISCONNECT BRANCH FROM BUS 228002 TO BUS 227900 CKT 1 /* ORCHARD TO CARDIFF 230 230 DISCONNECT BRANCH FROM BUS 200063 TO BUS 228002 CKT 1 /*ORCHARD ORCHARD 500 230 T1 END
AE_P1-2 ORCHARD XF	CONTINGENCY 'AE_P1-2 ORCHARD XF' OPEN LINE FROM BUS 200063 TO BUS 228002 CIRCUIT 1 / END

Contingency Name	Contingency Definition
JC-P2-3-JCC-230-15E	CONTINGENCY 'JC-P2-3-JCC-230-15E' /* LARRABEE - VAN HISEVILLE JUNCTION (B1042) & LARRABEE NW BUS 230 KV DISCONNECT BRANCH FROM BUS 206294 TO BUS 206318 CKT 1 /* LARRABEE - VAN HISEVILLE JUNCTION DISCONNECT BRANCH FROM BUS 206287 TO BUS 206318 CKT 1 /* COOKSTOWN - VAN HISEVILLE JUNCTION DISCONNECT BUS 206720 /* MANCHESTER 230 KV BUS DISCONNECT BRANCH FROM BUS 206319 TO BUS 206281 CKT 5 /* WHITINGS #5 230/34.5 KV REMOVE LOAD 7 FROM BUS 206294 /* LARRABEE BK7 DISCONNECT BRANCH FROM BUS 206294 TO BUS 206275 CKT 9 /* LARRABEE BK9 END
JC-P2-3-JCC-230-17A	CONTINGENCY 'JC-P2-3-JCC-230-17A' /* LARRABEE-COOKSTOWN- WHITINGS (B1042) & WHITINGS-MANITOU (V2022) 230 KV DISCONNECT BRANCH FROM BUS 206294 TO BUS 206318 CKT 1 /* LARRABEE-VAN HISEVILLE TAP (B1042) DISCONNECT BRANCH FROM BUS 206318 TO BUS 206287 CKT 1 /* VAN HISEVILLE TAP - COOKSTOWN (B1042) DISCONNECT BRANCH FROM BUS 206318 TO BUS 206720 CKT 1 /* VAN HISEVILLE TAP - MANCHESTER (B1042) DISCONNECT BUS 206720 /* ISOLATE MANCHESTER DISCONNECT BRANCH FROM BUS 206319 TO BUS 206281 CKT 5 /* WHITINGS BK5 DISCONNECT BRANCH FROM BUS 206319 TO BUS 206297 CKT 1 /* WHITINGS - MANITOU (V2022) DISCONNECT BRANCH FROM BUS 206319 TO BUS 206281 CKT 4 /* WHITINGS BK4 END
AE_P7-1 W2275_O2241	CONTINGENCY 'AE_P7-1 W2275_O2241' /* DOUBLE CIRCUIT TOWER W- 2275(MICKLETON - DEPTFORD) AND O-2241(MICKLETON - THOROFARE) DISCONNECT BUS 219762 DISCONNECT BUS 219121 CLOSE LINE FROM BUS 219211 TO BUS 219212 CKT Z /*THOROFARE END
AE_P1-2 CNT-V-BUT	CONTINGENCY 'AE_P1-2 CNT-V-BUT' DISCONNECT BUS 228713 / END
PS_P1-2_2310	CONTINGENCY 'PS_P1-2_2310' /* NEW FREEDOM TO CARDIFF TIELINE TRIP LINE FROM BUS 219100 TO BUS 227900 CKT 1 /* NEW FREEDOM TO CARDIFF END

Contingency Name	Contingency Definition
JC-P7-1-JCC-230-12	CONTINGENCY 'JC-P7-1-JCC-230-12' /* LEISURE VILLAGE-MANITOU A2027 & C2029 DISCONNECT BRANCH FROM BUS 206295 TO BUS 206297 CKT 1 DISCONNECT BRANCH FROM BUS 206297 TO BUS 206277 CKT 7 DISCONNECT BRANCH FROM BUS 206296 TO BUS 206297 CKT 1 DISCONNECT BRANCH FROM BUS 206296 TO BUS 206276 CKT 3 SET BUS 206296 LOAD TO 0 MW END
JC-P7-1-JCC-230-13	CONTINGENCY 'JC-P7-1-JCC-230-13' /* MANITOU-OYSTER CREEK 230 LINES & OYSTER GEN DISCONNECT BRANCH FROM BUS 206297 TO BUS 206302 CKT 1 DISCONNECT BRANCH FROM BUS 206297 TO BUS 206302 CKT 2 DISCONNECT BRANCH FROM BUS 206302 TO BUS 206325 CKT 1 END
PS_P1-2_Y-2304	CONTINGENCY 'PS_P1-2_Y-2304' /* CAMDEN TO GLOUCESTER TRIP LINE FROM BUS 219110 TO BUS 219125 CKT 1 /* CAMDEN TO GLOUCESTER DISCONNECT BRANCH FROM BUS 219110 TO BUS 219128 CKT 2 /* REMOVE 26KV -2 END
PECO_P1-2_5038/* \$ CHESCO \$ 5038 \$ L	CONTINGENCY 'PECO_P1-2_5038/* \$ CHESCO \$ 5038 \$ L' TRIP BRANCH FROM BUS 200012 TO BUS 200028 CKT 1 /* NFREEDOM 500.00 WINDSOR 500.00 \$ CHESCO \$ 5038 \$ L END
AE_P1-2 BLE-SC-ML2	CONTINGENCY 'AE_P1-2 BLE-SC-ML2' DISCONNECT BUS 227906 / DISCONNECT BUS 227930 / CLOSE LINE FROM BUS 227929 TO BUS 227930 CIRCUIT 1 / END
PECO_P4_PEACH215/* \$ CHESCO \$ PEACH215 \$ STBK	CONTINGENCY 'PECO_P4_PEACH215/* \$ CHESCO \$ PEACH215 \$ STBK' TRIP BRANCH FROM BUS 200065 TO BUS 200051 CKT 1 /* PCHBTM2S 500.00 ROCKSPGS 500.00 \$ CHESCO \$ PEACH215 \$ STBK REMOVE MACHINE 1 FROM BUS 200034 /* PCHBTM 2 22.00 \$ CHESCO \$ PEACH215 \$ STBK END
AE_P7-1 AE3TOWER	CONTINGENCY 'AE_P7-1 AE3TOWER' DISCONNECT BRANCH FROM BUS 227900 TO BUS 227955 CKT 1 /* CARDIFF TO CEDAR 230 KV DISCONNECT BRANCH FROM BUS 227913 TO BUS 227902 CKT 1 /* CARDIFF TO LEWIS 138 KV END

Contingency Name	Contingency Definition
PECO_P2-2_CHI230B1/* \$ DELCO \$ CHI230B1 \$ B	CONTINGENCY 'PECO_P2-2_CHI230B1/* \$ DELCO \$ CHI230B1 \$ B' DISCONNECT BUS 213489 /* CHICHST1 230.00 \$ DELCO \$ CHI230B1 \$ B END
AE_P1-2 BLE-SC-ML1	CONTINGENCY 'AE_P1-2 BLE-SC-ML1' DISCONNECT BUS 227905 / DISCONNECT BUS 227929 / CLOSE LINE FROM BUS 227929 TO BUS 227930 CIRCUIT 1 / END
JC-P1-2-JCC-230-022	CONTINGENCY 'JC-P1-2-JCC-230-022' /* MANITOU - OYSTER CREEK (O1019) 230 KV DISCONNECT BRANCH FROM BUS 206297 TO BUS 206302 CKT 2 END
JC-P1-2-JCC-230-021	CONTINGENCY 'JC-P1-2-JCC-230-021' /* MANITOU - OYSTER CREEK (N1028) 230 KV DISCONNECT BRANCH FROM BUS 206297 TO BUS 206302 CKT 1 END
JC-P1-2-JCC-230-020	CONTINGENCY 'JC-P1-2-JCC-230-020' /* LEISURE VILLAGE - MANITOU (C2029) 230 KV DISCONNECT BRANCH FROM BUS 206296 TO BUS 206297 CKT 1 DISCONNECT BRANCH FROM BUS 206296 TO BUS 206276 CKT 3 SET BUS 206296 LOAD TO 0 MW END
Base Case	
AE_P1-2 BLE-ML-LEW2	CONTINGENCY 'AE_P1-2 BLE-ML-LEW2' DISCONNECT BUS 227904 / DISCONNECT BUS 227906 / DISCONNECT BUS 227930 / 227930 CIRCUIT 1 / CLOSE LINE FROM BUS 227929 TO BUS 227930 CIRCUIT 1 / END
AE_P1-3 CARD 6 XFR	CONTINGENCY 'AE_P1-3 CARD 6 XFR' OPEN LINE FROM BUS 227900 TO BUS 227913 CIRCUIT 1 / END

Contingency Name	Contingency Definition
AE_P4-2 AE7	CONTINGENCY 'AE_P4-2 AE7' /*CARDIFF TO NEW FREEDOM BREAKER V DISCONNECT BRANCH FROM BUS 219100 TO BUS 227900 CKT 1 /*NEW FREEDOM TO CARDIFF 230 230 DISCONNECT BRANCH FROM BUS 227900 TO BUS 227910 CKT 1 /*CARDIFF CARDIFF 230 69 T1 END
PS_P2-3_CUTB_2-3_LT	CONTINGENCY 'PS_P2-3_CUTB_2-3_LT' DISCONNECT BUS 219753 /* CUTHBERTH SECTION 2 DISCONNECT BUS 219754 /* CUTHBERTH SECTION 3 CLOSE LINE FROM BUS 219176 TO BUS 219179 CKT Z /* CUTBERTH CLOSE LINE FROM BUS 219177 TO BUS 219178 CKT Z /* CUTHBERT MOVE 8 MW LOAD FROM BUS 219176 TO BUS 219170 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO CINN T2 MOVE 8 MW LOAD FROM BUS 219176 TO BUS 219678 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219179 TO BUS 219678 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219178 TO BUS 219678 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219177 TO BUS 219677 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO MAPLE SHADE MOVE 8 MW LOAD FROM BUS 219179 TO BUS 219629 /* INTERSTATION TIE TRANSFER LOAD FROM CUTHBERT TO LAWNSIDE END
AE_P4-2 AE6	CONTINGENCY 'AE_P4-2 AE6' /*CARDIFF TO NEW FREEDOM BREAKER W DISCONNECT BRANCH FROM BUS 219100 TO BUS 227900 CKT 1 /*NEW FREEDOM TO CARDIFF 230 230 DISCONNECT BRANCH FROM BUS 227900 TO BUS 227911 CKT 1 /*CARDIFF CARDIFF 230 69 T2 END
AE_P4-2 AE9	CONTINGENCY 'AE_P4-2 AE9' /*CARDIFF TO ORCHARD BREAKER NEW1 DISCONNECT BRANCH FROM BUS 228002 TO BUS 227900 CKT 1 /* CARDIFF TO ORCHARD 230 230 DISCONNECT BRANCH FROM BUS 227900 TO BUS 227934 CKT 1 /*CARDIFF CARDIFF 230 138 T7 END

Contingency Name	Contingency Definition
JC-P7-1-JCC-230-11A	CONTINGENCY 'JC-P7-1-JCC-230-11A' /* LAKEWOOD - LEISURE V 230 KV LINES DISCONNECT BRANCH FROM BUS 206323 TO BUS 206295 CKT 1 DISCONNECT BRANCH FROM BUS 206323 TO BUS 206296 CKT 1 DISCONNECT BRANCH FROM BUS 206323 TO BUS 206282 CKT 6 DISCONNECT BRANCH FROM BUS 206282 TO BUS 206999 CKT 1 SET BUS 206295 LOAD TO 0 MW DISCONNECT BUS 206999 DISCONNECT BUS 207110 DISCONNECT BUS 207010 DISCONNECT BUS 207111 END
AE_P4-2 AE46	CONTINGENCY 'AE_P4-2 AE46' /* ORCHARD 230 BUS BREAKER D DISCONNECT BRANCH FROM BUS 228002 TO BUS 228310 CKT 1 /* ORCHARD TO CHURCHTOWN 230 230 DISCONNECT BRANCH FROM BUS 200063 TO BUS 228002 CKT 1 /* ORCHARD ORCHARD 500 230 T1 END
AE_P1-2 NCENT-S121	CONTINGENCY 'AE_P1-2 NCENT-S121' OPEN LINE FROM BUS 228714 TO BUS 228716 CIRCUIT 1 / END
PS_P7-1_W2275+O2241_LT	CONTINGENCY 'PS_P7-1_W2275+O2241_LT' /* MICKLETON -DEPTFORD & MICKLETON - THOROFARE DISCONNECT BUS 219121 /* THOROFARE SECTION 1 DISCONNECT BUS 219109 /* DEPTFORD BUS SECTION 1 CLOSE LINE FROM BUS 219255 TO BUS 219256 CKT Z /* DEPTFORD T3 TO T4 CLOSE LINE FROM BUS 219180 TO BUS 219181 CKT Z /* DEPTFORD T1 TO T2 DISCONNECT BUS 219762 /* THOROFARE SECTION 2 DISCONNECT BUS 219759 /* EAGLE POINT BUS SECTION 3 MOVE 8 MW LOAD FROM BUS 219180 TO BUS 219162 /* INTERSTATION TIE TRANSFER LOAD FROM DEPTFORD TO BEAVERBK T1 MOVE 8 MW LOAD FROM BUS 219181 TO BUS 219163 /* INTERSTATION TIE TRANSFER LOAD FROM DEPTFORD TO BEAVERBK T2 MOVE 8 MW LOAD FROM BUS 219255 TO BUS 219162 /* INTERSTATION TIE TRANSFER LOAD FROM DEPTFORD TO BEAVERBK T1 MOVE 8 MW LOAD FROM BUS 219256 TO BUS 219163 /* INTERSTATION TIE TRANSFER LOAD FROM DEPTFORD TO BEAVERBK T2 END

12 Light Load Analysis

The Queue Project AF1-238 was evaluated as a 127.5 MW injection at the Sherman Ave-West Vineland 69 kV line in the Vineland area. Project AF1-238 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-238 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

12.1 Generation Deliverability

(Single or N-1 contingencies)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJE CT LOADIN G %	POST PROJE CT LOADIN G %	AC D C	MW IMPAC T
71753532	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	PJM500_PS_P1-2_5015	single	1180.0	96.65	99.19	AC	30.27
71753946	214010	WANEETA2	230.0	PECO	213817	N PHILA	230.0	PECO	1	PJM500_PS_P1-2_5015	single	621.0	96.84	100.3	AC	14.74
150679445	228256	SHRMAN#1	69.0	AE	945730	AF1-238 TAP	69.0	AE	1	AE_P1-2 G10-WVINE	single	132.0	5.55	101.49	AC	127.5
150679446	228256	SHRMAN#1	69.0	AE	945730	AF1-238 TAP	69.0	AE	1	AE_P1-2 S121-G10	single	132.0	5.54	101.48	AC	127.5
150679448	228256	SHRMAN#1	69.0	AE	945730	AF1-238 TAP	69.0	AE	1	AE_P1-2 NCENT-S121	single	132.0	5.53	101.48	AC	127.5
150679450	228256	SHRMAN#1	69.0	AE	945730	AF1-238 TAP	69.0	AE	1	AE_P1-2 SHERM-W VINE-B	single	132.0	0.04	96.15	AC	127.5
150679934	228257	SHRMAN#3	69.0	AE	228703	BUTLER	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	single	129.0	10.57	86.01	AC	64.14
150679927	228703	BUTLER	69.0	AE	228713	VCLP TP	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	single	132.0	4.94	85.31	AC	77.8
150679216	228705	VINE G10	69.0	AE	228706	WEST VLD	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	6.85	130.32	AC	127.5
150679344	228705	VINE G10	69.0	AE	228716	S121TAP	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	6.84	111.58	AC	127.5
150679342	228706	WEST VLD	69.0	AE	228705	VINE G10	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	6.85	111.59	AC	127.5
150679464	228706	WEST VLD	69.0	AE	945730	AF1-238 TAP	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	single	132.0	0.04	100.24	AC	127.5
150679957	228713	VCLP TP	69.0	AE	228704	CENTRAL	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	single	132.0	5.69	84.63	AC	77.8
150679221	228714	CNTRL N	69.0	AE	228716	S121TAP	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	6.83	130.32	AC	127.5
59580241	228716	S121TAP	69.0	AE	228714	CNTRL N	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	6.83	111.58	AC	127.5
150679219	228716	S121TAP	69.0	AE	228705	VINE G10	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	6.84	130.32	AC	127.5
59580299	945730	AF1-238 TAP	69.0	AE	228256	SHRMAN#1	69.0	AE	1	AE_P1-2 NCENT-S121	single	132.0	5.53	90.35	AC	127.5
150679622	945730	AF1-238 TAP	69.0	AE	228706	WEST VLD	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	single	132.0	0.04	93.25	AC	127.5
150679624	945730	AF1-238 TAP	69.0	AE	228256	SHRMAN#1	69.0	AE	1	AE_P1-2 SHERM-W VINE-B	single	132.0	0.04	93.24	AC	127.5

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC D C	MW IMPACT
150679626	945730	AF1-238 TAP	69.0	AE	228256	SHRMAN#1	69.0	AE	1	AE_P1-2 G10-WVINE	single	132.0	5.55	90.35	AC	127.5

12.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies)

None

12.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC D C	MW IMPACT
71753582	206295	28LEISUR D	230.0	JCP&L	206323	28LAKEWOOD	230.0	JCP&L	1	JC-P1-2-JCC-230-020	single	869.0	102.57	103.58	AC	8.94
71753585	206295	28LEISUR D	230.0	JCP&L	206323	28LAKEWOOD	230.0	JCP&L	1	JC-P1-2-JCC-230-015	single	869.0	101.29	102.3	AC	8.93
71753458	206296	28LEISUR U	230.0	JCP&L	206323	28LAKEWOOD	230.0	JCP&L	1	JC-P1-2-JCC-230-019	single	817.0	112.97	114.04	AC	8.9
71753460	206296	28LEISUR U	230.0	JCP&L	206323	28LAKEWOOD	230.0	JCP&L	1	JC-P1-2-JCC-230-016	single	817.0	110.01	111.08	AC	8.94
42777150	206297	28MANITO U	230.0	JCP&L	206319	28WHITINGS	230.0	JCP&L	1	JC-P7-1-JCC-230-12	tower	817.0	128.06	128.93	AC	8.57
71753369	206297	28MANITO U	230.0	JCP&L	206296	28LEISUR U	230.0	JCP&L	1	JC-P1-2-JCC-230-019	single	817.0	119.84	120.93	AC	9.03
71753371	206297	28MANITO U	230.0	JCP&L	206296	28LEISUR U	230.0	JCP&L	1	JC-P1-2-JCC-230-016	single	817.0	119.19	120.28	AC	9.02
71753386	206297	28MANITO U	230.0	JCP&L	206295	28LEISUR D	230.0	JCP&L	1	JC-P1-2-JCC-230-020	single	817.0	119.52	120.61	AC	9.03
71753389	206297	28MANITO U	230.0	JCP&L	206295	28LEISUR D	230.0	JCP&L	1	JC-P1-2-JCC-230-015	single	817.0	112.06	113.12	AC	8.76
42557945	206302	28OYSTER C	230.0	JCP&L	206297	28MANITO U	230.0	JCP&L	1	JC-P1-2-JCC-230-022	single	817.0	180.12	181.51	AC	11.35
42557966	206302	28OYSTER C	230.0	JCP&L	206297	28MANITO U	230.0	JCP&L	2	JC-P1-2-JCC-230-021	single	869.0	169.37	170.67	AC	11.35
42777172	206318	28VANHISV L	230.0	JCP&L	206294	28LARRABE E	230.0	JCP&L	1	JC-P7-1-JCC-230-12	tower	869.0	116.23	117.22	AC	10.94
42777174	206318	28VANHISV L	230.0	JCP&L	206294	28LARRABE E	230.0	JCP&L	1	JC-P7-1-JCC-230-10A	tower	869.0	113.46	114.42	AC	10.45
42557983	206323	28LAKEWOOD	230.0	JCP&L	206294	28LARRABE E	230.0	JCP&L	2	JC-P1-2-JCC-230-014	single	817.0	128.87	129.97	AC	9.26

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPAC T
42557989	206323	28LAKEWOOD	230.0	JCP&L	206294	28LARRABEE	230.0	JCP&L	1	JC-P1-2-JCC-230-013	single	869.0	121.49	122.53	AC	9.26
49617678	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	Base Case	single	760.0	108.68	112.72	AC	23.95
41041980	227900	CARDIFF	230.0	AE	219100	NEWFRDM	230.0	PSE&G	1	JC-P7-1-JCC-230-13	tower	692.0	193.15	195.64	AC	17.13
41041982	227900	CARDIFF	230.0	AE	219100	NEWFRDM	230.0	PSE&G	1	AE_P7-1 AE15TOWER	tower	692.0	165.38	167.25	AC	12.98
41648246	227900	CARDIFF	230.0	AE	219100	NEWFRDM	230.0	PSE&G	1	AE_P4-2 AE46	breaker	692.0	152.52	157.05	AC	31.48
41648248	227900	CARDIFF	230.0	AE	219100	NEWFRDM	230.0	PSE&G	1	AE_P4-2 AE9	breaker	692.0	154.05	155.31	AC	10.24
42250807	227900	CARDIFF	230.0	AE	227955	CEDAR	230.0	AE	1	AE_P4-2 AE7	breaker	805.0	111.79	113.56	AC	13.82
42250808	227900	CARDIFF	230.0	AE	227955	CEDAR	230.0	AE	1	AE_P4-2 AE6	breaker	805.0	111.78	113.54	AC	13.82
49617588	227900	CARDIFF	230.0	AE	227955	CEDAR	230.0	AE	1	Base Case	single	650.0	107.43	109.06	AC	10.4
53846916	227900	CARDIFF	230.0	AE	227955	CEDAR	230.0	AE	1	PS_P1-2_2310	single	805.0	111.13	112.92	AC	13.99
71753011	227900	CARDIFF	230.0	AE	219100	NEWFRDM	230.0	PSE&G	1	AE_P1-2 ORCH-CARD	single	692.0	150.04	151.64	AC	11.34
71753013	227900	CARDIFF	230.0	AE	219100	NEWFRDM	230.0	PSE&G	1	JC-P1-2-JCC-230-002T	single	692.0	137.25	139.87	AC	17.65
41648251	227955	CEDAR	230.0	AE	206302	28OYSTER C	230.0	JCP&L	1	AE_P4-2 AE7	breaker	564.0	156.25	158.46	AC	16.09
41648252	227955	CEDAR	230.0	AE	206302	28OYSTER C	230.0	JCP&L	1	AE_P4-2 AE6	breaker	564.0	156.24	158.45	AC	16.09
49617532	227955	CEDAR	230.0	AE	206302	28OYSTER C	230.0	JCP&L	1	Base Case	single	464.0	148.36	150.98	AC	12.38
49617535	227955	CEDAR	230.0	AE	206302	28OYSTER C	230.0	JCP&L	1	PS_P1-2_2310	single	564.0	157.89	160.65	AC	16.19

12.4 Steady-State Voltage Requirements

None.

12.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/D C	MW IMPACT
42558154	206295	28LEISUR D	230.0	JCP&L	206323	28LAKEWOD	230.0	JCP&L	1	JC-P1-2-JCC-230-020	operation	869.0	102.54	103.4	AC	8.94
42558060	206296	28LEISUR U	230.0	JCP&L	206323	28LAKEWOD	230.0	JCP&L	1	JC-P1-2-JCC-230-019	operation	817.0	112.93	113.84	AC	8.9
42558032	206297	28MANITO U	230.0	JCP&L	206296	28LEISUR U	230.0	JCP&L	1	JC-P1-2-JCC-230-019	operation	817.0	119.8	120.72	AC	9.03
42558038	206297	28MANITO U	230.0	JCP&L	206295	28LEISUR D	230.0	JCP&L	1	JC-P1-2-JCC-230-020	operation	817.0	119.48	120.41	AC	9.03
42557940	206302	28OYSTER C	230.0	JCP&L	206297	28MANITO U	230.0	JCP&L	1	JC-P1-2-JCC-230-022	operation	817.0	180.01	181.18	AC	11.35
42557961	206302	28OYSTER C	230.0	JCP&L	206297	28MANITO U	230.0	JCP&L	2	JC-P1-2-JCC-230-021	operation	869.0	169.26	170.36	AC	11.35
42557978	206323	28LAKEWOD	230.0	JCP&L	206294	28LARRABE E	230.0	JCP&L	2	JC-P1-2-JCC-230-014	operation	817.0	128.86	129.8	AC	9.26
42557984	206323	28LAKEWOD	230.0	JCP&L	206294	28LARRABE E	230.0	JCP&L	1	JC-P1-2-JCC-230-013	operation	869.0	121.48	122.37	AC	9.26
42558345	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	Base Case	operation	760.0	105.93	109.99	AC	23.95
49617675	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	PJM500_PS_P1-2_5015	operation	1180.0	96.31	98.83	AC	30.27
41332127	227900	CARDIFF	230.0	AE	219100	NEWFRDM	230.0	PSE&G	1	AE_P1-2 ORCH-CARD	operation	692.0	149.09	150.47	AC	11.34
42558055	227900	CARDIFF	230.0	AE	227955	CEDAR	230.0	AE	1	PS_P1-2_2310	operation	805.0	111.08	112.86	AC	13.99
42558056	227900	CARDIFF	230.0	AE	227955	CEDAR	230.0	AE	1	Base Case	operation	650.0	107.37	108.73	AC	10.4
41332122	227955	CEDAR	230.0	AE	206302	28OYSTER C	230.0	JCP&L	1	PS_P1-2_2310	operation	564.0	155.65	157.9	AC	16.19
41332123	227955	CEDAR	230.0	AE	206302	28OYSTER C	230.0	JCP&L	1	Base Case	operation	464.0	145.38	147.65	AC	12.38
150679444	228256	SHRMAN#1	69.0	AE	945730	AF1-238 TAP	69.0	AE	1	AE_P1-2 G10-WVINE	operation	132.0	5.55	101.49	AC	127.5
150679447	228256	SHRMAN#1	69.0	AE	945730	AF1-238 TAP	69.0	AE	1	AE_P1-2 S121-G10	operation	132.0	5.54	101.48	AC	127.5
150679449	228256	SHRMAN#1	69.0	AE	945730	AF1-238 TAP	69.0	AE	1	AE_P1-2 NCENT-S121	operation	132.0	5.53	101.48	AC	127.5
51073058	228257	SHRMAN#3	69.0	AE	228703	BUTLER	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	129.0	10.57	86.01	AC	64.14
59580566	228703	BUTLER	69.0	AE	228713	VCLP TP	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	132.0	4.94	85.31	AC	77.8
59580341	228705	VINE G10	69.0	AE	228716	S121TAP	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	107.0	6.84	111.58	AC	127.5
150679217	228705	VINE G10	69.0	AE	228706	WEST VLD	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	107.0	6.85	130.32	AC	127.5
59580344	228706	WEST VLD	69.0	AE	228705	VINE G10	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	107.0	6.85	111.59	AC	127.5
150679465	228706	WEST VLD	69.0	AE	945730	AF1-238 TAP	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	132.0	0.04	100.24	AC	127.5
59580588	228713	VCLP TP	69.0	AE	228704	CENTRAL	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	132.0	5.69	84.63	AC	77.8
150679220	228714	CNTRL N	69.0	AE	228716	S121TAP	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	107.0	6.83	130.32	AC	127.5
51072920	228716	S121TAP	69.0	AE	228714	CNTRL N	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	107.0	6.83	111.58	AC	127.5
150679218	228716	S121TAP	69.0	AE	228705	VINE G10	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	107.0	6.84	130.32	AC	127.5

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/D C	MW IMPACT
59580296	945730	AF1-238 TAP	69.0	AE	228256	SHRMAN#1	69.0	AE	1	AE_P1-2 SHERM-W VINE-B	operation	132.0	0.04	93.24	AC	127.5
59580486	945730	AF1-238 TAP	69.0	AE	228706	WEST VLD	69.0	AE	1	AE_P1-2 SHERM-W VINE-A	operation	132.0	0.04	93.25	AC	127.5

12.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																
150679216	6	VINE G10 69.0 kV - WEST VLD 69.0 kV Ckt 1	Note: that the non-PJM Co-Op will evaluate this overload in the facility study	N/A	N/A	N/A																
71753458,71753460	17	28LEISUR U 230.0 kV - 28LAKEWOOD 230.0 kV Ckt 1	Project Id: N6428 Description: Reconductor the Leisure Village - Lakewood Ckt U 230 kV transmission line (~2.4 miles). Type : FAC Total Cost : \$11,444,599 Time Estimate : 30 Months Ratings : 815/954/954 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-251</td><td>128</td><td>35.19%</td><td>\$ 4,027,794</td></tr><tr><td>AF1-101</td><td>226.8</td><td>62.36%</td><td>\$ 7,136,747</td></tr><tr><td>AF1-238</td><td>8.9</td><td>2.45%</td><td>\$ 280,058</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-251	128	35.19%	\$ 4,027,794	AF1-101	226.8	62.36%	\$ 7,136,747	AF1-238	8.9	2.45%	\$ 280,058	\$11,444,599	\$280,058	N6428
Queue	MW	Cost %	Cost \$																			
AE2-251	128	35.19%	\$ 4,027,794																			
AF1-101	226.8	62.36%	\$ 7,136,747																			
AF1-238	8.9	2.45%	\$ 280,058																			
150679622	15	AF1-238 TAP 69.0 kV - WEST VLD 69.0 kV Ckt 1	Note: that the non-PJM Co-Op will evaluate this overload in the facility study	N/A	N/A	N/A																
49617532,41648252,49617535,41648251	28	CEDAR 230.0 kV - 28OYSTER C 230.0 kV Ckt 1	SP – See 12.6	N/A	N/A	N/A																
49617678,71753532	1	RICHMOND 230.0 kV - WANEETA3 230.0 kV Ckt 1	SP – SEE 12.6	N/A	N/A	N/A																
42777150	18	28MANITOU 230.0 kV - 28WHITINGS 230.0 kV Ckt 1	SP – SEE 12.6	N/A	N/A	N/A																
71753013,71753011,41648248,41041982,41041980,41648246	26	CARDIFF 230.0 kV - NEWFRDM 230.0 kV Ckt 1	SP – SEE 12.6	N/A	N/A	N/A																
150679934	4	SHRMAN#3 69.0 kV - BUTLER 69.0 kV Ckt 1	Note: that non-PJM Co-Op will evaluate this overload in the facility study	N/A	N/A	N/A																
150679957	10	VCLP TP 69.0 kV - CENTRAL 69.0 kV Ckt 1	Note: that the non-PJM Co-Op will evaluate this overload in the facility study	N/A	N/A	N/A																
150679219	13	S121TAP 69.0 kV - VINE G10 69.0 kV Ckt 1	Note: that the non-PJM Co-Op will evaluate this overload in the facility study	N/A	N/A	N/A																

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																
59580241	12	S121TAP 69.0 kV - CNTRL N 69.0 kV Ckt 1	Project Id: N6844 Description: Line Upgrade West Sherman Substation, including the installation of new poles, foundations, insulators, and conductor. Type : FAC Total Cost : \$3,125,000 Time Estimate : 18.0 Months Ratings : 200.0/286.0/286.0 Notes : Upgrade for City of Vineland <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AF1-238</td><td>127.5</td><td>100.00%</td><td>\$3,125,000</td></tr></table>	Queue	MW	Cost %	Cost \$	AF1-238	127.5	100.00%	\$3,125,000	\$13,031,000	\$13,031,000	N6844 N6801 N6845								
			Queue	MW	Cost %	Cost \$																
			AF1-238	127.5	100.00%	\$3,125,000																
			Project Id: N6801 Description: Ring Bus for AF1-238 & AF1-239- New 69kV class substation and associated relaying, fiber, etc. Type : CON Total Cost : \$7,405,000 Time Estimate : 34.0 Months Ratings : 200.0/286.0/286.0 Notes : Upgrade for City of Vineland <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AF1-238</td><td>127.5</td><td>100.00%</td><td>\$7,405,000</td></tr></table>	Queue	MW	Cost %	Cost \$	AF1-238	127.5	100.00%	\$7,405,000											
Queue	MW	Cost %	Cost \$																			
AF1-238	127.5	100.00%	\$7,405,000																			
Project Id: N6845 Description: Line Upgrade CN-G10, including the installation of new poles, circuit breaker, relaying, foundations, insulators, and conductor. Type : FAC Total Cost : \$2,501,000 Time Estimate : 12.0 Months Ratings : 200.0/286.0/286.0 Notes : Upgrade for City of Vineland <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td><td></td></tr><tr><td>AF1-238</td><td>127.5</td><td>100.00%</td><td>\$2,501,000</td><td></td></tr></table>	Queue	MW	Cost %	Cost \$		AF1-238	127.5	100.00%	\$2,501,000													
Queue	MW	Cost %	Cost \$																			
AF1-238	127.5	100.00%	\$2,501,000																			
71753582,71753585	16	28LEISUR D 230.0 kV - 28LAKEWOOD 230.0 kV Ckt 1	Project Id: N6429 (JCP&L-AF1-L-0003) Description : Reconductor the Leisure Village - Lakewood Ckt D 230 kV transmission line (~2.4 miles). Type : FAC Total Cost : \$11,276,790 Time Estimate : 12.0 Months Ratings : 757/920/920 <table><tr><td>Queue</td><td>MW</td><td>Cost %</td><td>Cost \$</td></tr><tr><td>AE2-251</td><td>55.8</td><td>19.07%</td><td>\$2,150,970</td></tr><tr><td>AF1-101</td><td>227.8</td><td>77.87%</td><td>\$8,781,202</td></tr><tr><td>AF1-238</td><td>8.94</td><td>3.06%</td><td>\$344,618</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-251	55.8	19.07%	\$2,150,970	AF1-101	227.8	77.87%	\$8,781,202	AF1-238	8.94	3.06%	\$344,618	\$11,276,790	\$344,618	N6429
Queue	MW	Cost %	Cost \$																			
AE2-251	55.8	19.07%	\$2,150,970																			
AF1-101	227.8	77.87%	\$8,781,202																			
AF1-238	8.94	3.06%	\$344,618																			

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																																								
150679342	8	WEST VLD 69.0 kV - VINE G10 69.0 kV Ckt 1	Note: that the non-PJM Co-Op will evaluate this overload in the facility study	N/A	N/A	N/A																																								
71753946	2	WANEETA2 230.0 kV - N PHILA 230.0 kV Ckt 1	Project Id: n6530 Description : Reconductor the North Philadelphia-Waneeta 230 kV line with Pecos conductor (Does not include permitting) Type : FAC Total Cost : \$6,335,000 Time Estimate : 36.0 Months Ratings : 516/647/647 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-222</td><td>29.23</td><td>18.48%</td><td>\$1,170,493</td></tr><tr><td>AE2-251</td><td>114.23</td><td>72.21%</td><td>\$4,574,254</td></tr><tr><td>AF1-238</td><td>14.74</td><td>9.32%</td><td>\$590,252</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-222	29.23	18.48%	\$1,170,493	AE2-251	114.23	72.21%	\$4,574,254	AF1-238	14.74	9.32%	\$590,252	\$6,335,000	\$590,252	N6530																								
Queue	MW	Cost %	Cost \$																																											
AE2-222	29.23	18.48%	\$1,170,493																																											
AE2-251	114.23	72.21%	\$4,574,254																																											
AF1-238	14.74	9.32%	\$590,252																																											
42557945	21	28OYSTER C 230.0 kV - 28MANITOU 230.0 kV Ckt 1	Project Id: N6547 Description : Reconductor the Oyster Creek - Manitou 230 kV transmission line (~11.1 miles) Type : FAC Total Cost : \$52,322,963 Time Estimate : 48.0 Months Ratings : 815/923/923 Notes: Reinforcement for Circuit #2 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-020</td><td>1.59</td><td>0.17%</td><td>\$89,245</td></tr><tr><td>AE2-021</td><td>100.3</td><td>10.78%</td><td>\$5,639,078</td></tr><tr><td>AE2-022</td><td>58.5</td><td>6.29%</td><td>\$3,288,994</td></tr><tr><td>AE2-023</td><td>55.9</td><td>6.01%</td><td>\$3,142,816</td></tr><tr><td>AE2-222</td><td>45.5</td><td>4.89%</td><td>\$2,558,106</td></tr><tr><td>AE2-232</td><td>220.8</td><td>23.73%</td><td>\$12,413,843</td></tr><tr><td>AE2-251</td><td>119.1</td><td>12.80%</td><td>\$6,696,054</td></tr><tr><td>AF1-101</td><td>317.61</td><td>34.13%</td><td>\$17,856,706</td></tr><tr><td>AF1-238</td><td>11.35</td><td>1.22%</td><td>\$638,121</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-020	1.59	0.17%	\$89,245	AE2-021	100.3	10.78%	\$5,639,078	AE2-022	58.5	6.29%	\$3,288,994	AE2-023	55.9	6.01%	\$3,142,816	AE2-222	45.5	4.89%	\$2,558,106	AE2-232	220.8	23.73%	\$12,413,843	AE2-251	119.1	12.80%	\$6,696,054	AF1-101	317.61	34.13%	\$17,856,706	AF1-238	11.35	1.22%	\$638,121	\$52,322,963	\$638,121	N6547
Queue	MW	Cost %	Cost \$																																											
AE2-020	1.59	0.17%	\$89,245																																											
AE2-021	100.3	10.78%	\$5,639,078																																											
AE2-022	58.5	6.29%	\$3,288,994																																											
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AE2-222	45.5	4.89%	\$2,558,106																																											
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AE2-251	119.1	12.80%	\$6,696,054																																											
AF1-101	317.61	34.13%	\$17,856,706																																											
AF1-238	11.35	1.22%	\$638,121																																											

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																																								
42557966	22	28OYSTER C 230.0 kV - 28MANITOU 230.0 kV Ckt 2	Project Id: N6549 Description: Reconductor the Manitou - Oyster Creek Ckt O1029 230 kV transmission line (~11.1 miles). Type : FAC Total Cost : \$52,155,154 Time Estimate : 48.0 Months Ratings : 815/923/923 Notes: Reinforcement for Circuit #2 <table><thead><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr></thead><tbody><tr><td>AE2-021</td><td>48.89</td><td>5.10%</td><td>\$2,662,265</td></tr><tr><td>AE2-022</td><td>58.54</td><td>6.11%</td><td>\$3,187,679</td></tr><tr><td>AE2-023</td><td>55.91</td><td>5.84%</td><td>\$3,044,468</td></tr><tr><td>AE2-222</td><td>45.5</td><td>4.75%</td><td>\$2,477,612</td></tr><tr><td>AE2-232</td><td>220.9</td><td>23.06%</td><td>\$12,028,671</td></tr><tr><td>AE2-251</td><td>199.1</td><td>20.79%</td><td>\$10,841,595</td></tr><tr><td>AF1-101</td><td>317.61</td><td>33.16%</td><td>\$17,294,822</td></tr><tr><td>AF1-238</td><td>11.35</td><td>1.19%</td><td>\$618,042</td></tr></tbody></table>	Queue	MW	Cost %	Cost \$	AE2-021	48.89	5.10%	\$2,662,265	AE2-022	58.54	6.11%	\$3,187,679	AE2-023	55.91	5.84%	\$3,044,468	AE2-222	45.5	4.75%	\$2,477,612	AE2-232	220.9	23.06%	\$12,028,671	AE2-251	199.1	20.79%	\$10,841,595	AF1-101	317.61	33.16%	\$17,294,822	AF1-238	11.35	1.19%	\$618,042	\$52,155,154	\$618,042	N6549				
Queue	MW	Cost %	Cost \$																																											
AE2-021	48.89	5.10%	\$2,662,265																																											
AE2-022	58.54	6.11%	\$3,187,679																																											
AE2-023	55.91	5.84%	\$3,044,468																																											
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AE2-232	220.9	23.06%	\$12,028,671																																											
AE2-251	199.1	20.79%	\$10,841,595																																											
AF1-101	317.61	33.16%	\$17,294,822																																											
AF1-238	11.35	1.19%	\$618,042																																											
42557983	24	28LAKEWOOD 230.0 kV - 28LARRABEE 230.0 kV Ckt 2	Project Id: N6411 Description: Replace substation conductor at Lakewood Gen. and Larrabee. Reconductor the Lakewood Gen. - Larrabee 230 kV line. Replace disconnect switch at Larrabee and disconnect switch at Lakewood Gen. Replace wave trap at Larrabee. Replace metering and fault switches at Lakewood Gen. and Larrabee. Type : FAC Total Cost : \$16,176,824 Time Estimate : 30.0 Months Ratings : 1306.0/1697.0/1697.0 <table><thead><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr></thead><tbody><tr><td>AE2-021</td><td>86.92</td><td>8.98%</td><td>\$1,453,353</td></tr><tr><td>AE2-022</td><td>56.8</td><td>5.87%</td><td>\$949,729</td></tr><tr><td>AE2-023</td><td>54.84</td><td>5.67%</td><td>\$916,956</td></tr><tr><td>AE2-222</td><td>44.35</td><td>4.58%</td><td>\$741,558</td></tr><tr><td>AE2-232</td><td>209.32</td><td>21.64%</td><td>\$3,499,951</td></tr><tr><td>AE2-251</td><td>193.5</td><td>20.00%</td><td>\$3,235,432</td></tr><tr><td>AE2-314</td><td>11.61</td><td>1.20%</td><td>\$194,126</td></tr><tr><td>AF1-101</td><td>300.88</td><td>31.10%</td><td>\$5,030,887</td></tr><tr><td>AF1-238</td><td>9.26</td><td>0.96%</td><td>\$154,833</td></tr></tbody></table>	Queue	MW	Cost %	Cost \$	AE2-021	86.92	8.98%	\$1,453,353	AE2-022	56.8	5.87%	\$949,729	AE2-023	54.84	5.67%	\$916,956	AE2-222	44.35	4.58%	\$741,558	AE2-232	209.32	21.64%	\$3,499,951	AE2-251	193.5	20.00%	\$3,235,432	AE2-314	11.61	1.20%	\$194,126	AF1-101	300.88	31.10%	\$5,030,887	AF1-238	9.26	0.96%	\$154,833	\$16,176,824	\$154,833	N6411
Queue	MW	Cost %	Cost \$																																											
AE2-021	86.92	8.98%	\$1,453,353																																											
AE2-022	56.8	5.87%	\$949,729																																											
AE2-023	54.84	5.67%	\$916,956																																											
AE2-222	44.35	4.58%	\$741,558																																											
AE2-232	209.32	21.64%	\$3,499,951																																											
AE2-251	193.5	20.00%	\$3,235,432																																											
AE2-314	11.61	1.20%	\$194,126																																											
AF1-101	300.88	31.10%	\$5,030,887																																											
AF1-238	9.26	0.96%	\$154,833																																											
150679927	5	BUTLER 69.0 kV - VCLP TP 69.0 kV Ckt 1	Note: that the non-PJM Co-Op will evaluate this overload in the facility study	\$0																																										

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number																				
42557989	25	28LAKEWOOD 230.0 kV - 28LARRABEE 230.0 kV Ckt 1	Project Id: N6431 Description: Reconductor the Lakewood Gen - Larrabee K2011 230 kV transmission line (~7.6 miles). Replace substation conductor at Lakewood. Replace wave trap at Larrabee. Replace wave trap at Lakewood. Type : FAC Total Cost : \$ 36,985,186 Time Estimate : 42 Months Ratings : 913/1147/1147 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-251</td><td>65.7</td><td>20.78%</td><td>\$7,685,021</td></tr><tr><td>AF1-101</td><td>241.23</td><td>76.29%</td><td>\$28,217,010</td></tr><tr><td>AF1-238</td><td>9.26</td><td>2.93%</td><td>\$1,083,155</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-251	65.7	20.78%	\$7,685,021	AF1-101	241.23	76.29%	\$28,217,010	AF1-238	9.26	2.93%	\$1,083,155	\$36,985,186	\$1,083,155	N6431				
Queue	MW	Cost %	Cost \$																							
AE2-251	65.7	20.78%	\$7,685,021																							
AF1-101	241.23	76.29%	\$28,217,010																							
AF1-238	9.26	2.93%	\$1,083,155																							
150679344	7	VINE G10 69.0 kV - S121TAP 69.0 kV Ckt 1	Note: that the non-PJM Co-Op will evaluate this overload in the facility study	N/A	N/A	N/A																				
150679464	9	WEST VLD 69.0 kV - AF1-238 TAP 69.0 kV Ckt 1	Note: that the non-PJM Co-Op will evaluate this overload in the facility study	N/A	N/A	N/A																				
71753369,7175 3371	19	28MANITOU 230.0 kV - 28LEISUR U 230.0 kV Ckt 1	Project Id: N6432 Description: Replace substation conductor at Manitou and Leisure Village. Reconductor the Manitou - Leisure Village U 230 kV transmission line (~7.1 miles). Replace wave traps at Leisure Village and Manitou. Replace metering and relaying at Leisure Village and Manitou. Replace circuit breaker at Leisure Village Type : FAC Total Cost : \$35,307,092 Time Estimate : 42.0 Months Ratings : 913/1147/1147 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-232</td><td>48.91</td><td>11.44%</td><td>\$4,038,800</td></tr><tr><td>AE2-251</td><td>139.43</td><td>32.61%</td><td>\$11,513,595</td></tr><tr><td>AF1-101</td><td>230.2</td><td>53.84%</td><td>\$19,009,034</td></tr><tr><td>AF1-238</td><td>9.03</td><td>2.11%</td><td>\$745,663</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-232	48.91	11.44%	\$4,038,800	AE2-251	139.43	32.61%	\$11,513,595	AF1-101	230.2	53.84%	\$19,009,034	AF1-238	9.03	2.11%	\$745,663	\$35,307,092	\$745,663	N6432
Queue	MW	Cost %	Cost \$																							
AE2-232	48.91	11.44%	\$4,038,800																							
AE2-251	139.43	32.61%	\$11,513,595																							
AF1-101	230.2	53.84%	\$19,009,034																							
AF1-238	9.03	2.11%	\$745,663																							
71753386,7175 3389	20	28MANITOU 230.0 kV - 28LEISUR D 230.0 kV Ckt 1	Project Id: N6433 Description: Replace substation conductor at Manitou and Leisure Village. Reconductor the Manitou - Leisure Village D 230 kV transmission line (~7.1 miles). Replace wave traps at Leisure Village and Manitou. Replace metering at Manitou Type : FAC Total Cost : \$34,233,112 Time Estimate : 42.0 Months Ratings : 913/1147/1147 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-232</td><td>49.0</td><td>11.46%</td><td>\$3,922,327</td></tr><tr><td>AE2-251</td><td>139.43</td><td>32.60%</td><td>\$11,161,022</td></tr><tr><td>AF1-101</td><td>230.2</td><td>53.83%</td><td>\$18,426,934</td></tr><tr><td>AF1-238</td><td>9.03</td><td>2.11%</td><td>\$722,829</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-232	49.0	11.46%	\$3,922,327	AE2-251	139.43	32.60%	\$11,161,022	AF1-101	230.2	53.83%	\$18,426,934	AF1-238	9.03	2.11%	\$722,829	\$34,233,112	\$722,829	N6433
Queue	MW	Cost %	Cost \$																							
AE2-232	49.0	11.46%	\$3,922,327																							
AE2-251	139.43	32.60%	\$11,161,022																							
AF1-101	230.2	53.83%	\$18,426,934																							
AF1-238	9.03	2.11%	\$722,829																							

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-238	Upgrade Number
53846916,42250807,42250808,49617588	27	CARDIFF 230.0 kV - CEDAR 230.0 kV Ckt 1	SP – SEE SECTION 12.6	N/A	N/A	N/A
150679221	11	CNTRL N 69.0 kV - S121TAP 69.0 kV Ckt 1	Note: that the non-PJM Co-Op will evaluate this overload in the facility study	N/A	N/A	N/A
42777172,42777174	23	28VANHISVL 230.0 kV - 28LARRABEE 230.0 kV Ckt 1	SP – See Section 12.6	N/A	N/A	N/A
150679445,150679446,150679448,150679450	3	SHRMAN#1 69.0 kV - AF1-238 TAP 69.0 kV Ckt 1	Note: that non-PJM Co-Op will evaluate this overload in the facility study	N/A	N/A	N/A
150679624,150679626,59580299	14	AF1-238 TAP 69.0 kV - SHRMAN#1 69.0 kV Ckt 1	Note: that non-PJM Co-Op will evaluate this overload in the facility study	N/A	N/A	N/A
			TOTAL COST	\$269,267,720	\$18,208,571	

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

12.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

1.1.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
49617678	213922	RICHMOND	PECO	214012	WANEETA3	PECO	1	Base Case	single	760.0	108.68	112.72	AC	23.95

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	280 C GEN (Deactivation : 17/09/2018)	25.5822	80/20	25.5822
937563	AD2-077 BAT	14.0360	80/20	14.0360
938231	AE1-037 FTIR (Withdrawn : 01/24/2020)	59.4960	Merchant Transmission	59.4960
938421	AE1-061 C	0.5515	80/20	0.5515
938422	AE1-061 E	0.5515	80/20	0.5515
938431	AE1-062 C	2.1187	80/20	2.1187
938432	AE1-062 E	2.1187	80/20	2.1187
938781	AE1-104 C O1	16.8756	80/20	16.8756
938782	AE1-104 E O1	43.1759	80/20	43.1759
938871	AE1-115 C	2.1471	80/20	2.1471
938872	AE1-115 E	2.1471	80/20	2.1471
939301	AE1-161 C	4.1554	80/20	4.1554
939302	AE1-161 E	6.2331	80/20	6.2331
939981	AE1-238 C	14.5782	80/20	14.5782
939982	AE1-238 E	38.2921	80/20	38.2921
940161	AE2-000 C O1	20.8755	80/20	20.8755
940162	AE2-000 E O1	53.4132	80/20	53.4132
940361	AE2-020 C	14.3413	80/20	14.3413
940362	AE2-020 E	67.1470	80/20	67.1470
940371	AE2-021 C	14.3413	80/20	14.3413
940372	AE2-021 E	67.1470	80/20	67.1470
940381	AE2-022 C	8.3658	80/20	8.3658
940382	AE2-022 E	39.1691	80/20	39.1691
940401	AE2-024 C O1	11.5603	80/20	11.5603
940402	AE2-024 E O1	54.1240	80/20	54.1240
940411	AE2-025 C	5.8356	80/20	5.8356
940412	AE2-025 E	27.3193	80/20	27.3193
942101	AE2-222 C O1	11.7914	80/20	11.7914
942102	AE2-222 E O1	29.6206	80/20	29.6206
942381	AE2-251 C	45.4330	80/20	45.4330
942382	AE2-251 E	116.2502	80/20	116.2502
943561	AF1-027	0.2650	80/20	0.2650
944332	AF1-101 E O1	52.3662	80/20	52.3662
944611	AF1-126 C O1	8.1965	80/20	8.1965
944612	AF1-126 E O1	32.7860	80/20	32.7860
944951	AF1-160 C	2.1187	80/20	2.1187
944952	AF1-160 E	2.1187	80/20	2.1187
945201	AF1-185 1	0.2623	80/20	0.2623
945211	AF1-185 2	0.0583	80/20	0.0583
945431	AF1-208 C O1	0.0002	80/20	0.0002

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945432	AF1-208 E O1	2.0203	80/20	2.0203
945571	AF1-222 C	9.0871	80/20	9.0871
945572	AF1-222 E	23.9568	80/20	23.9568
945721	AF1-237 C	10.6780	80/20	10.6780
945722	AF1-237 E	0.0001	80/20	0.0001
945731	AF1-238 C	9.5788	80/20	9.5788
945732	AF1-238 E	14.3682	80/20	14.3682
945741	AF1-239 C	2.2538	80/20	2.2538
945742	AF1-239 E	3.3808	80/20	3.3808

1.1.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
71753946	214010	WANEETA2	PECO	213817	N PHILA	PECO	1	PJM500_PS_P1-2_5015	single	621.0	96.84	100.3	AC	14.74

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	280 C GEN (Deactivation : 17/09/2018)	18.0357	80/20	18.0357
938231	AE1-037 FTIR (Withdrawn : 01/24/2020)	55.7760	Merchant Transmission	55.7760
938421	AE1-061 C	0.3302	80/20	0.3302
938422	AE1-061 E	0.3302	80/20	0.3302
938431	AE1-062 C	1.2866	80/20	1.2866
938432	AE1-062 E	1.2866	80/20	1.2866
938781	AE1-104 C O1	10.6793	80/20	10.6793
938782	AE1-104 E O1	27.3229	80/20	27.3229
938871	AE1-115 C	1.2386	80/20	1.2386
938872	AE1-115 E	1.2386	80/20	1.2386
939301	AE1-161 C	2.4566	80/20	2.4566
939302	AE1-161 E	3.6849	80/20	3.6849
939981	AE1-238 C	11.3886	80/20	11.3886
939982	AE1-238 E	29.9141	80/20	29.9141
940161	AE2-000 C O1	14.7174	80/20	14.7174
940162	AE2-000 E O1	37.6568	80/20	37.6568
940361	AE2-020 C	9.1768	80/20	9.1768
940362	AE2-020 E	42.9666	80/20	42.9666
940371	AE2-021 C	9.1768	80/20	9.1768
940372	AE2-021 E	42.9666	80/20	42.9666
940381	AE2-022 C	5.3532	80/20	5.3532
940382	AE2-022 E	25.0639	80/20	25.0639
940401	AE2-024 C O1	8.6631	80/20	8.6631
940402	AE2-024 E O1	40.5596	80/20	40.5596
940411	AE2-025 C	4.3731	80/20	4.3731
940412	AE2-025 E	20.4726	80/20	20.4726
942101	AE2-222 C O1	7.4657	80/20	7.4657
942102	AE2-222 E O1	18.7543	80/20	18.7543
942381	AE2-251 C	29.0720	80/20	29.0720
942382	AE2-251 E	74.3872	80/20	74.3872
943561	AF1-027	0.1803	80/20	0.1803
944332	AF1-101 E O1	36.9186	80/20	36.9186
944611	AF1-126 C O1	5.2980	80/20	5.2980
944612	AF1-126 E O1	21.1920	80/20	21.1920
944951	AF1-160 C	1.2866	80/20	1.2866
944952	AF1-160 E	1.2866	80/20	1.2866
945201	AF1-185 1	0.2280	80/20	0.2280
945211	AF1-185 2	0.0507	80/20	0.0507
945431	AF1-208 C O1	0.0001	80/20	0.0001
945432	AF1-208 E O1	1.2001	80/20	1.2001

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945571	AF1-222 C	7.0989	80/20	7.0989
945572	AF1-222 E	18.7153	80/20	18.7153
945721	AF1-237 C	8.1480	80/20	8.1480
945722	AF1-237 E	0.0001	80/20	0.0001
945731	AF1-238 C	5.8946	80/20	5.8946
945732	AF1-238 E	8.8419	80/20	8.8419
945741	AF1-239 C	1.3870	80/20	1.3870
945742	AF1-239 E	2.0804	80/20	2.0804

1.1.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
150679448	228256	SHRMAN#1	AE	945730	AF1-238 TAP	AE	1	AE_P1-2 NCENT-S121	single	132.0	5.53	101.48	AC	127.5

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945733	AF1-238 BAT	127.5000	80/20	127.5000
945743	AF1-239 BAT	30.0000	80/20	30.0000

1.1.4 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
150679934	228257	SHRMAN#3	AE	228703	BUTLER	AE	1	AE_P1-2 SHERM-W VINE-A	single	129.0	10.57	86.01	AC	64.14

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
938781	AE1-104 C O1	3.3780	Adder	3.97
938782	AE1-104 E O1	8.6426	Adder	10.17
940361	AE2-020 C	1.0799	Adder	1.27
940362	AE2-020 E	5.0562	Adder	5.95
940371	AE2-021 C	1.0799	Adder	1.27
940372	AE2-021 E	5.0562	Adder	5.95
942381	AE2-251 C	3.4211	Adder	4.02
942382	AE2-251 E	8.7536	Adder	10.3
945733	AF1-238 BAT	64.1401	80/20	64.1401
945743	AF1-239 BAT	15.0918	80/20	15.0918

1.1.5 Index 5

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/DC	MW IMPACT
150679927	228703	BUTLER	AE	228713	VCLP TP	AE	1	AE_P1-2 SHERM-W VINE-A	single	132.0	4.94	85.31	AC	77.8

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
938781	AE1-104 C O1	3.5250	Adder	4.15
938782	AE1-104 E O1	9.0186	Adder	10.61
940361	AE2-020 C	1.1327	Adder	1.33
940362	AE2-020 E	5.3035	Adder	6.24
940371	AE2-021 C	1.1327	Adder	1.33
940372	AE2-021 E	5.3035	Adder	6.24
942381	AE2-251 C	3.5885	Adder	4.22
942382	AE2-251 E	9.1819	Adder	10.8
945431	AF1-208 C O1	0.0001	80/20	0.0001
945432	AF1-208 E O1	1.0186	80/20	1.0186
945733	AF1-238 BAT	77.8031	80/20	77.8031
945743	AF1-239 BAT	18.3066	80/20	18.3066

1.1.6 Index 6

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
150679216	228705	VINE G10	AE	228706	WEST VLD	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	6.85	130.32	AC	127.5

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945733	AF1-238 BAT	127.5000	80/20	127.5000
945743	AF1-239 BAT	30.0000	80/20	30.0000

1.1.7 Index 7

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
150679344	228705	VINE G10	AE	228716	S121TAP	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	6.84	111.58	AC	127.5

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945731	AF1-238 C	51.0000	80/20	51.0000
945732	AF1-238 E	76.5000	80/20	76.5000
945741	AF1-239 C	12.0000	80/20	12.0000
945742	AF1-239 E	18.0000	80/20	18.0000

1.1.8 Index 8

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
150679342	228706	WEST VLD	AE	228705	VINE G10	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	6.85	111.59	AC	127.5

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945731	AF1-238 C	51.0000	80/20	51.0000
945732	AF1-238 E	76.5000	80/20	76.5000
945741	AF1-239 C	12.0000	80/20	12.0000
945742	AF1-239 E	18.0000	80/20	18.0000

1.1.9 Index 9

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
150679464	228706	WEST VLD	AE	945730	AF1-238 TAP	AE	1	AE_P1-2 SHERM-W VINE-A	single	132.0	0.04	100.24	AC	127.5

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945733	AF1-238 BAT	127.5000	80/20	127.5000
945743	AF1-239 BAT	30.0000	80/20	30.0000

1.1.10 Index 10

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/DC	MW IMPACT
150679957	228713	VCLP TP	AE	228704	CENTRAL	AE	1	AE_P1-2 SHERM-W VINE-A	single	132.0	5.69	84.63	AC	77.8

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
938781	AE1-104 C O1	3.5250	Adder	4.15
938782	AE1-104 E O1	9.0186	Adder	10.61
940361	AE2-020 C	1.1327	Adder	1.33
940362	AE2-020 E	5.3035	Adder	6.24
940371	AE2-021 C	1.1327	Adder	1.33
940372	AE2-021 E	5.3035	Adder	6.24
942381	AE2-251 C	3.5885	Adder	4.22
942382	AE2-251 E	9.1819	Adder	10.8
945431	AF1-208 C O1	0.0001	80/20	0.0001
945432	AF1-208 E O1	1.0186	80/20	1.0186
945733	AF1-238 BAT	77.8031	80/20	77.8031
945743	AF1-239 BAT	18.3066	80/20	18.3066

1.1.11 Index 11

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
150679221	228714	CNTRL N	AE	228716	S121TAP	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	6.83	130.32	AC	127.5

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945733	AF1-238 BAT	127.5000	80/20	127.5000
945743	AF1-239 BAT	30.0000	80/20	30.0000

1.1.12 Index 12

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
59580241	228716	S121TAP	AE	228714	CNTRL N	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	6.83	111.58	AC	127.5

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945731	AF1-238 C	51.0000	80/20	51.0000
945732	AF1-238 E	76.5000	80/20	76.5000
945741	AF1-239 C	12.0000	80/20	12.0000
945742	AF1-239 E	18.0000	80/20	18.0000

1.1.13 Index 13

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
150679219	228716	S121TAP	AE	228705	VINE G10	AE	1	AE_P1-2 SHERM-W VINE-A	single	107.0	6.84	130.32	AC	127.5

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945733	AF1-238 BAT	127.5000	80/20	127.5000
945743	AF1-239 BAT	30.0000	80/20	30.0000

1.1.14 Index 14

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
150679624	945730	AF1-238 TAP	AE	228256	SHRMAN#1	AE	1	AE_P1-2 SHERM-W VINE-B	single	132.0	0.04	93.24	AC	127.5

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945731	AF1-238 C	51.0000	80/20	51.0000
945732	AF1-238 E	76.5000	80/20	76.5000
945741	AF1-239 C	12.0000	80/20	12.0000
945742	AF1-239 E	18.0000	80/20	18.0000

1.1.15 Index 15

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
150679622	945730	AF1-238 TAP	AE	228706	WEST VLD	AE	1	AE_P1-2 SHERM-W VINE-A	single	132.0	0.04	93.25	AC	127.5

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945731	AF1-238 C	51.0000	80/20	51.0000
945732	AF1-238 E	76.5000	80/20	76.5000
945741	AF1-239 C	12.0000	80/20	12.0000
945742	AF1-239 E	18.0000	80/20	18.0000

1.1.16 Index 16

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
71753582	206295	28LEISUR D	JCP&L	206323	28LAKEWOOD	JCP&L	1	JC-P1-2-JCC-230-020	single	869.0	102.57	103.58	AC	8.94

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	111.3232	80/20	111.3232
938421	AE1-061 C	0.2100	80/20	0.2100
938422	AE1-061 E	0.2100	80/20	0.2100
938781	AE1-104 C O1	11.5592	80/20	11.5592
938782	AE1-104 E O1	29.5741	80/20	29.5741
939301	AE1-161 C	1.2032	80/20	1.2032
939302	AE1-161 E	1.8048	80/20	1.8048
940161	AE2-000 C O1	90.8413	80/20	90.8413
940162	AE2-000 E O1	232.4318	80/20	232.4318
940361	AE2-020 C	13.1245	80/20	13.1245
940362	AE2-020 E	61.4498	80/20	61.4498
940371	AE2-021 C	13.1245	80/20	13.1245
940372	AE2-021 E	61.4498	80/20	61.4498
940381	AE2-022 C	7.6559	80/20	7.6559
940382	AE2-022 E	35.8457	80/20	35.8457
942101	AE2-222 C O1	9.7269	80/20	9.7269
942102	AE2-222 E O1	24.4347	80/20	24.4347
942381	AE2-251 C	41.5781	80/20	41.5781
942382	AE2-251 E	106.3867	80/20	106.3867
944332	AF1-101 E O1	227.8758	80/20	227.8758
945431	AF1-208 C O1	0.0001	80/20	0.0001
945432	AF1-208 E O1	0.5769	80/20	0.5769
945731	AF1-238 C	3.5777	80/20	3.5777
945732	AF1-238 E	5.3665	80/20	5.3665
945741	AF1-239 C	0.8418	80/20	0.8418
945742	AF1-239 E	1.2627	80/20	1.2627

1.1.17 Index 17

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
71753458	206296	28LEISUR U	JCP&L	206323	28LAKEWOOD	JCP&L	1	JC-P1-2-JCC-230-019	single	817.0	112.97	114.04	AC	8.9

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	110.7972	80/20	110.7972
938421	AE1-061 C	0.2089	80/20	0.2089
938422	AE1-061 E	0.2089	80/20	0.2089
938781	AE1-104 C O1	11.5019	80/20	11.5019
938782	AE1-104 E O1	29.4275	80/20	29.4275
939301	AE1-161 C	1.1966	80/20	1.1966
939302	AE1-161 E	1.7949	80/20	1.7949
940161	AE2-000 C O1	90.4121	80/20	90.4121
940162	AE2-000 E O1	231.3335	80/20	231.3335
940361	AE2-020 C	13.0598	80/20	13.0598
940362	AE2-020 E	61.1468	80/20	61.1468
940371	AE2-021 C	13.0598	80/20	13.0598
940372	AE2-021 E	61.1468	80/20	61.1468
940381	AE2-022 C	7.6182	80/20	7.6182
940382	AE2-022 E	35.6690	80/20	35.6690
942101	AE2-222 C O1	9.6791	80/20	9.6791
942102	AE2-222 E O1	24.3145	80/20	24.3145
942381	AE2-251 C	41.3731	80/20	41.3731
942382	AE2-251 E	105.8621	80/20	105.8621
944332	AF1-101 E O1	226.7991	80/20	226.7991
945431	AF1-208 C O1	0.0001	80/20	0.0001
945432	AF1-208 E O1	0.5739	80/20	0.5739
945731	AF1-238 C	3.5588	80/20	3.5588
945732	AF1-238 E	5.3382	80/20	5.3382
945741	AF1-239 C	0.8374	80/20	0.8374
945742	AF1-239 E	1.2560	80/20	1.2560

1.1.18 Index 18

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42777150	206297	28MANITOU	JCP&L	206319	28WHITINGS	JCP&L	1	JC-P7-1-JCC-230-12	tower	817.0	128.06	128.93	AC	8.57

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	121.0593	50/50	121.0593
938421	AE1-061 C	0.1716	Adder	0.2
938422	AE1-061 E	0.1716	Adder	0.2
938781	AE1-104 C O1	11.7991	50/50	11.7991
938782	AE1-104 E O1	30.1878	50/50	30.1878
939301	AE1-161 C	0.9409	Adder	1.11
939302	AE1-161 E	1.4114	Adder	1.66
940161	AE2-000 C O1	98.7861	50/50	98.7861
940162	AE2-000 E O1	252.7597	50/50	252.7597
940361	AE2-020 C	13.6328	50/50	13.6328
940362	AE2-020 E	63.8299	50/50	63.8299
940371	AE2-021 C	13.6328	50/50	13.6328
940372	AE2-021 E	63.8299	50/50	63.8299
940381	AE2-022 C	7.9525	50/50	7.9525
940382	AE2-022 E	37.2341	50/50	37.2341
942101	AE2-222 C O1	10.0495	50/50	10.0495
942102	AE2-222 E O1	25.2449	50/50	25.2449
942381	AE2-251 C	43.1886	50/50	43.1886
942382	AE2-251 E	110.5074	50/50	110.5074
944332	AF1-101 E O1	247.8054	50/50	247.8054
945431	AF1-208 C O1	0.0000	Adder	0.0
945432	AF1-208 E O1	0.4542	Adder	0.53
945731	AF1-238 C	2.9144	Adder	3.43
945732	AF1-238 E	4.3716	Adder	5.14
945741	AF1-239 C	0.6857	Adder	0.81
945742	AF1-239 E	1.0286	Adder	1.21

1.1.19 Index 19

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
71753371	206297	28MANITOU	JCP&L	206296	28LEISUR U	JCP&L	1	JC-P1-2-JCC-230-016	single	817.0	119.19	120.28	AC	9.02

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	112.4382	80/20	112.4382
938421	AE1-061 C	0.2118	80/20	0.2118
938422	AE1-061 E	0.2118	80/20	0.2118
938781	AE1-104 C O1	11.6690	80/20	11.6690
938782	AE1-104 E O1	29.8549	80/20	29.8549
939301	AE1-161 C	1.2134	80/20	1.2134
939302	AE1-161 E	1.8201	80/20	1.8201
940161	AE2-000 C O1	91.7512	80/20	91.7512
940162	AE2-000 E O1	234.7598	80/20	234.7598
940361	AE2-020 C	13.2505	80/20	13.2505
940362	AE2-020 E	62.0398	80/20	62.0398
940371	AE2-021 C	13.2505	80/20	13.2505
940372	AE2-021 E	62.0398	80/20	62.0398
940381	AE2-022 C	7.7295	80/20	7.7295
940382	AE2-022 E	36.1899	80/20	36.1899
942101	AE2-222 C O1	9.8199	80/20	9.8199
942102	AE2-222 E O1	24.6681	80/20	24.6681
942381	AE2-251 C	41.9774	80/20	41.9774
942382	AE2-251 E	107.4082	80/20	107.4082
944332	AF1-101 E O1	230.1582	80/20	230.1582
945431	AF1-208 C O1	0.0001	80/20	0.0001
945432	AF1-208 E O1	0.5819	80/20	0.5819
945731	AF1-238 C	3.6098	80/20	3.6098
945732	AF1-238 E	5.4147	80/20	5.4147
945741	AF1-239 C	0.8494	80/20	0.8494
945742	AF1-239 E	1.2740	80/20	1.2740

1.1.20 Index 20

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
71753386	206297	28MANITOU	JCP&L	206295	28LEISUR D	JCP&L	1	JC-P1-2-JCC-230-020	single	817.0	119.52	120.61	AC	9.03

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	112.4607	80/20	112.4607
938421	AE1-061 C	0.2119	80/20	0.2119
938422	AE1-061 E	0.2119	80/20	0.2119
938781	AE1-104 C O1	11.6709	80/20	11.6709
938782	AE1-104 E O1	29.8598	80/20	29.8598
939301	AE1-161 C	1.2138	80/20	1.2138
939302	AE1-161 E	1.8207	80/20	1.8207
940161	AE2-000 C O1	91.7695	80/20	91.7695
940162	AE2-000 E O1	234.8067	80/20	234.8067
940361	AE2-020 C	13.2531	80/20	13.2531
940362	AE2-020 E	62.0518	80/20	62.0518
940371	AE2-021 C	13.2531	80/20	13.2531
940372	AE2-021 E	62.0518	80/20	62.0518
940381	AE2-022 C	7.7309	80/20	7.7309
940382	AE2-022 E	36.1969	80/20	36.1969
942101	AE2-222 C O1	9.8219	80/20	9.8219
942102	AE2-222 E O1	24.6733	80/20	24.6733
942381	AE2-251 C	41.9854	80/20	41.9854
942382	AE2-251 E	107.4290	80/20	107.4290
944332	AF1-101 E O1	230.2043	80/20	230.2043
945431	AF1-208 C O1	0.0001	80/20	0.0001
945432	AF1-208 E O1	0.5821	80/20	0.5821
945731	AF1-238 C	3.6103	80/20	3.6103
945732	AF1-238 E	5.4154	80/20	5.4154
945741	AF1-239 C	0.8495	80/20	0.8495
945742	AF1-239 E	1.2742	80/20	1.2742

1.1.21 Index 21

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42557945	206302	28OYSTER C	JCP&L	206297	28MANITOU	JCP&L	1	JC-P1-2-JCC-230-022	single	817.0	180.12	181.51	AC	11.35

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	155.1615	80/20	155.1615
938421	AE1-061 C	0.2671	80/20	0.2671
938422	AE1-061 E	0.2671	80/20	0.2671
938781	AE1-104 C O1	15.3654	80/20	15.3654
938782	AE1-104 E O1	39.3120	80/20	39.3120
938871	AE1-115 C	0.5422	80/20	0.5422
938872	AE1-115 E	0.5422	80/20	0.5422
939301	AE1-161 C	1.4820	80/20	1.4820
939302	AE1-161 E	2.2230	80/20	2.2230
940161	AE2-000 C O1	126.6140	80/20	126.6140
940162	AE2-000 E O1	323.9617	80/20	323.9617
940361	AE2-020 C	17.6742	80/20	17.6742
940362	AE2-020 E	82.7517	80/20	82.7517
940371	AE2-021 C	17.6742	80/20	17.6742
940372	AE2-021 E	82.7517	80/20	82.7517
940381	AE2-022 C	10.3099	80/20	10.3099
940382	AE2-022 E	48.2718	80/20	48.2718
942101	AE2-222 C O1	13.0460	80/20	13.0460
942102	AE2-222 E O1	32.7724	80/20	32.7724
942381	AE2-251 C	55.9914	80/20	55.9914
942382	AE2-251 E	143.2662	80/20	143.2662
944332	AF1-101 E O1	317.6116	80/20	317.6116
945431	AF1-208 C O1	0.0001	80/20	0.0001
945432	AF1-208 E O1	0.7142	80/20	0.7142
945731	AF1-238 C	4.5395	80/20	4.5395
945732	AF1-238 E	6.8093	80/20	6.8093
945741	AF1-239 C	1.0681	80/20	1.0681
945742	AF1-239 E	1.6022	80/20	1.6022

1.1.22 Index 22

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42557966	206302	28OYSTER C	JCP&L	206297	28MANITOU	JCP&L	2	JC-P1-2-JCC-230-021	single	869.0	169.37	170.67	AC	11.35

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	155.1907	80/20	155.1907
938421	AE1-061 C	0.2671	80/20	0.2671
938422	AE1-061 E	0.2671	80/20	0.2671
938781	AE1-104 C O1	15.3683	80/20	15.3683
938782	AE1-104 E O1	39.3195	80/20	39.3195
938871	AE1-115 C	0.5423	80/20	0.5423
938872	AE1-115 E	0.5423	80/20	0.5423
939301	AE1-161 C	1.4822	80/20	1.4822
939302	AE1-161 E	2.2233	80/20	2.2233
940161	AE2-000 C O1	126.6378	80/20	126.6378
940162	AE2-000 E O1	324.0227	80/20	324.0227
940361	AE2-020 C	17.6767	80/20	17.6767
940362	AE2-020 E	82.7636	80/20	82.7636
940371	AE2-021 C	17.6767	80/20	17.6767
940372	AE2-021 E	82.7636	80/20	82.7636
940381	AE2-022 C	10.3114	80/20	10.3114
940382	AE2-022 E	48.2788	80/20	48.2788
942101	AE2-222 C O1	13.0488	80/20	13.0488
942102	AE2-222 E O1	32.7792	80/20	32.7792
942381	AE2-251 C	55.9995	80/20	55.9995
942382	AE2-251 E	143.2869	80/20	143.2869
944332	AF1-101 E O1	317.6715	80/20	317.6715
945431	AF1-208 C O1	0.0001	80/20	0.0001
945432	AF1-208 E O1	0.7143	80/20	0.7143
945731	AF1-238 C	4.5405	80/20	4.5405
945732	AF1-238 E	6.8108	80/20	6.8108
945741	AF1-239 C	1.0684	80/20	1.0684
945742	AF1-239 E	1.6025	80/20	1.6025

1.1.23 Index 23

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42777172	206318	28VANHISVL	JCP&L	206294	28LARRABEE	JCP&L	1	JC-P7-1-JCC-230-12	tower	869.0	116.23	117.22	AC	10.94

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	88.2565	50/50	88.2565
919663	AA2-048 BAT	1.0057	50/50	1.0057
938421	AE1-061 C	0.2165	Adder	0.25
938422	AE1-061 E	0.2165	Adder	0.25
938431	AE1-062 C	0.6592	Adder	0.78
938432	AE1-062 E	0.6592	Adder	0.78
938781	AE1-104 C O1	11.7370	50/50	11.7370
938782	AE1-104 E O1	30.0288	50/50	30.0288
938871	AE1-115 C	0.5736	Adder	0.67
938872	AE1-115 E	0.5736	Adder	0.67
939301	AE1-161 C	1.3829	Adder	1.63
939302	AE1-161 E	2.0744	Adder	2.44
940161	AE2-000 C O1	72.0185	50/50	72.0185
940162	AE2-000 E O1	184.2707	50/50	184.2707
940361	AE2-020 C	12.5420	50/50	12.5420
940362	AE2-020 E	58.7228	50/50	58.7228
940371	AE2-021 C	12.5420	50/50	12.5420
940372	AE2-021 E	58.7228	50/50	58.7228
940381	AE2-022 C	7.3162	50/50	7.3162
940382	AE2-022 E	34.2549	50/50	34.2549
942101	AE2-222 C O1	9.4755	50/50	9.4755
942102	AE2-222 E O1	23.8029	50/50	23.8029
942381	AE2-251 C	39.7330	50/50	39.7330
942382	AE2-251 E	101.6655	50/50	101.6655
943561	AF1-027	0.4823	50/50	0.4823
944332	AF1-101 E O1	180.6588	50/50	180.6588
944611	AF1-126 C O1	20.0190	50/50	20.0190
944612	AF1-126 E O1	80.0760	50/50	80.0760
944951	AF1-160 C	0.6592	Adder	0.78
944952	AF1-160 E	0.6592	Adder	0.78
945431	AF1-208 C O1	0.0001	Adder	0.0
945432	AF1-208 E O1	0.6525	Adder	0.77
945731	AF1-238 C	3.7212	Adder	4.38
945732	AF1-238 E	5.5817	Adder	6.57
945741	AF1-239 C	0.8756	Adder	1.03
945742	AF1-239 E	1.3134	Adder	1.55

1.1.24 Index 24

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42557983	206323	28LAKEWOOD	JCP&L	206294	28LARRABEE	JCP&L	2	JC-P1-2-JCC-230-014	single	817.0	128.87	129.97	AC	9.26

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	117.8334	80/20	117.8334
938421	AE1-061 C	0.2179	80/20	0.2179
938422	AE1-061 E	0.2179	80/20	0.2179
938781	AE1-104 C O1	12.0963	80/20	12.0963
938782	AE1-104 E O1	30.9482	80/20	30.9482
939301	AE1-161 C	1.2404	80/20	1.2404
939302	AE1-161 E	1.8606	80/20	1.8606
940161	AE2-000 C O1	96.1537	80/20	96.1537
940162	AE2-000 E O1	246.0244	80/20	246.0244
940361	AE2-020 C	13.7819	80/20	13.7819
940362	AE2-020 E	64.5276	80/20	64.5276
940371	AE2-021 C	13.7819	80/20	13.7819
940372	AE2-021 E	64.5276	80/20	64.5276
940381	AE2-022 C	8.0394	80/20	8.0394
940382	AE2-022 E	37.6411	80/20	37.6411
942101	AE2-222 C O1	10.2039	80/20	10.2039
942102	AE2-222 E O1	25.6329	80/20	25.6329
942381	AE2-251 C	43.6607	80/20	43.6607
942382	AE2-251 E	111.7154	80/20	111.7154
944332	AF1-101 E O1	241.2021	80/20	241.2021
945431	AF1-208 C O1	0.0001	80/20	0.0001
945432	AF1-208 E O1	0.5941	80/20	0.5941
945731	AF1-238 C	3.7046	80/20	3.7046
945732	AF1-238 E	5.5570	80/20	5.5570
945741	AF1-239 C	0.8717	80/20	0.8717
945742	AF1-239 E	1.3075	80/20	1.3075

1.1.25 Index 25

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42557989	206323	28LAKEWOOD	JCP&L	206294	28LARRABEE	JCP&L	1	JC-P1-2-JCC-230-013	single	869.0	121.49	122.53	AC	9.26

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	117.8447	80/20	117.8447
938421	AE1-061 C	0.2179	80/20	0.2179
938422	AE1-061 E	0.2179	80/20	0.2179
938781	AE1-104 C O1	12.0982	80/20	12.0982
938782	AE1-104 E O1	30.9532	80/20	30.9532
939301	AE1-161 C	1.2406	80/20	1.2406
939302	AE1-161 E	1.8609	80/20	1.8609
940161	AE2-000 C O1	96.1629	80/20	96.1629
940162	AE2-000 E O1	246.0479	80/20	246.0479
940361	AE2-020 C	13.7836	80/20	13.7836
940362	AE2-020 E	64.5356	80/20	64.5356
940371	AE2-021 C	13.7836	80/20	13.7836
940372	AE2-021 E	64.5356	80/20	64.5356
940381	AE2-022 C	8.0404	80/20	8.0404
940382	AE2-022 E	37.6458	80/20	37.6458
942101	AE2-222 C O1	10.2053	80/20	10.2053
942102	AE2-222 E O1	25.6363	80/20	25.6363
942381	AE2-251 C	43.6661	80/20	43.6661
942382	AE2-251 E	111.7292	80/20	111.7292
944332	AF1-101 E O1	241.2251	80/20	241.2251
945431	AF1-208 C O1	0.0001	80/20	0.0001
945432	AF1-208 E O1	0.5941	80/20	0.5941
945731	AF1-238 C	3.7052	80/20	3.7052
945732	AF1-238 E	5.5577	80/20	5.5577
945741	AF1-239 C	0.8718	80/20	0.8718
945742	AF1-239 E	1.3077	80/20	1.3077

1.1.26 Index 26

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41041980	227900	CARDIFF	AE	219100	NEWFRDM	PSE&G	1	JC-P7-1-JCC-230-13	tower	692.0	193.15	195.64	AC	17.13

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
938421	AE1-061 C	0.3972	50/50	0.3972
938422	AE1-061 E	0.3972	50/50	0.3972
938781	AE1-104 C O1	27.9385	50/50	27.9385
938782	AE1-104 E O1	71.4802	50/50	71.4802
938871	AE1-115 C	0.5413	Adder	0.64
938872	AE1-115 E	0.5413	Adder	0.64
939301	AE1-161 C	1.5643	Adder	1.84
939302	AE1-161 E	2.3465	Adder	2.76
940161	AE2-000 C O1	71.9837	50/50	71.9837
940162	AE2-000 E O1	184.1816	50/50	184.1816
940361	AE2-020 C	36.0746	50/50	36.0746
940362	AE2-020 E	168.9042	50/50	168.9042
940371	AE2-021 C	36.0746	50/50	36.0746
940372	AE2-021 E	168.9042	50/50	168.9042
940381	AE2-022 C	21.0435	50/50	21.0435
940382	AE2-022 E	98.5274	50/50	98.5274
942101	AE2-222 C O1	23.3265	50/50	23.3265
942102	AE2-222 E O1	58.5975	50/50	58.5975
942381	AE2-251 C	114.2838	50/50	114.2838
942382	AE2-251 E	292.4202	50/50	292.4202
944332	AF1-101 E O1	180.5714	50/50	180.5714
945431	AF1-208 C O1	0.0001	Adder	0.0
945432	AF1-208 E O1	0.8347	Adder	0.98
945731	AF1-238 C	6.8524	50/50	6.8524
945732	AF1-238 E	10.2785	50/50	10.2785
945741	AF1-239 C	1.6123	50/50	1.6123
945742	AF1-239 E	2.4185	50/50	2.4185

1.1.27 Index 27

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42250808	227900	CARDIFF	AE	227955	CEDAR	AE	1	AE_P4-2 AE6	breaker	805.0	111.78	113.54	AC	13.82

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
938421	AE1-061 C	0.3166	50/50	0.3166
938422	AE1-061 E	0.3166	50/50	0.3166
938781	AE1-104 C O1	19.3910	50/50	19.3910
938782	AE1-104 E O1	49.6115	50/50	49.6115
938871	AE1-115 C	0.5380	Adder	0.63
938872	AE1-115 E	0.5380	Adder	0.63
939301	AE1-161 C	1.4244	Adder	1.68
939302	AE1-161 E	2.1366	Adder	2.51
940361	AE2-020 C	26.5845	50/50	26.5845
940362	AE2-020 E	124.4704	50/50	124.4704
940371	AE2-021 C	26.5845	50/50	26.5845
940372	AE2-021 E	124.4704	50/50	124.4704
940381	AE2-022 C	15.5076	50/50	15.5076
940382	AE2-022 E	72.6077	50/50	72.6077
942101	AE2-222 C O1	14.8433	50/50	14.8433
942102	AE2-222 E O1	37.2871	50/50	37.2871
942381	AE2-251 C	84.2191	50/50	84.2191
942382	AE2-251 E	215.4929	50/50	215.4929
945431	AF1-208 C O1	0.0001	Adder	0.0
945432	AF1-208 E O1	0.7254	Adder	0.85
945731	AF1-238 C	5.5289	50/50	5.5289
945732	AF1-238 E	8.2934	50/50	8.2934
945741	AF1-239 C	1.3009	50/50	1.3009
945742	AF1-239 E	1.9514	50/50	1.9514

1.1.28 Index 28

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
49617535	227955	CEDAR	AE	206302	28OYSTER C	JCP&L	1	PS_P1-2_2310	single	564.0	157.89	160.65	AC	16.19

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
938421	AE1-061 C	0.3792	80/20	0.3792
938422	AE1-061 E	0.3792	80/20	0.3792
938781	AE1-104 C O1	23.3962	80/20	23.3962
938782	AE1-104 E O1	59.8588	80/20	59.8588
938871	AE1-115 C	0.7169	80/20	0.7169
938872	AE1-115 E	0.7169	80/20	0.7169
939301	AE1-161 C	1.9912	80/20	1.9912
939302	AE1-161 E	2.9868	80/20	2.9868
940361	AE2-020 C	28.1308	80/20	28.1308
940362	AE2-020 E	131.7106	80/20	131.7106
940371	AE2-021 C	28.1308	80/20	28.1308
940372	AE2-021 E	131.7106	80/20	131.7106
940381	AE2-022 C	16.4096	80/20	16.4096
940382	AE2-022 E	76.8312	80/20	76.8312
942101	AE2-222 C O1	19.7443	80/20	19.7443
942102	AE2-222 E O1	49.5989	80/20	49.5989
942381	AE2-251 C	89.1179	80/20	89.1179
942382	AE2-251 E	228.0277	80/20	228.0277
944613	AF1-126 BAT	12.6300	80/20	12.6300
945431	AF1-208 C O1	0.0001	80/20	0.0001
945432	AF1-208 E O1	0.9890	80/20	0.9890
945731	AF1-238 C	6.4765	80/20	6.4765
945732	AF1-238 E	9.7147	80/20	9.7147
945741	AF1-239 C	1.5239	80/20	1.5239
945742	AF1-239 E	2.2858	80/20	2.2858

12.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA2-048	Allenwood-Larrabee 34kV	Engineering and Procurement
AD2-077	Buxmont 69 kV	Active
AE1-037	Deans 500kV	Withdrawn
AE1-061	Minotola 12 kV	Active
AE1-062	Silver Lake 69 kV	Active
AE1-104	BL England 138 kV	Active
AE1-115	Churchtown 69 kV	Active
AE1-161	Landis 138 kV	Active
AE1-238	Oceanview Wind 230 kV	Active
AE2-000	N/A	N/A
AE2-020	Cardiff 230 kV I	Active
AE2-021	Cardiff 230 kV II	Active
AE2-022	Cardiff 230 kV III	Active
AE2-024	Larrabee 230 kV I	Active
AE2-025	Larrabee 230 kV II	Active
AE2-222	Higbee 69 kV	Active
AE2-251	Cardiff 230 kV	Active
AF1-027	Plumsted 537 Energy Storage (CIRs)	Active
AF1-101	Oyster Creek 230 kV III	Active
AF1-126	Cookstown 230 kV	Active
AF1-160	Silver Lake 69 kV	Active
AF1-185	Sayreville 1-2-3 230 kV	In Service
AF1-208	Quinton-Roadstown 69 kV	Active
AF1-222	Oceanview Wind 2 230 kV	Active
AF1-237	Mercer 230 kV	Active
AF1-238	Sherman Ave. 69 kV	Active
AF1-239	Sherman Ave-Vineland 69 kV	Active

12.9 Contingency Descriptions

Contingency Name	Contingency Definition
AE_P7-1 AE15TOWER	CONTINGENCY 'AE_P7-1 AE15TOWER' DISCONNECT BRANCH FROM BUS 227901 TO BUS 227949 CKT 1 /* DOR TO LEW 138 KV DISCONNECT BRANCH FROM BUS 228002 TO BUS 227900 CKT 1 /* ORCH TO CARD 230 KV END
PJM500_PS_P1-2_5015	CONTINGENCY 'PJM500_PS_P1-2_5015' /* HOPE CREEK TO RED LION TRIP LINE FROM BUS 200027 TO BUS 200029 CKT 1 /* HOPE CREEK TO RED LION END
JC-P1-2-JCC-230-019	CONTINGENCY 'JC-P1-2-JCC-230-019' /* LEISURE VILLAGE - MANITOU (A2027) 230 KV DISCONNECT BRANCH FROM BUS 206295 TO BUS 206297 CKT 1 DISCONNECT BRANCH FROM BUS 206297 TO BUS 206277 CKT 7 END
JC-P1-2-JCC-230-016	CONTINGENCY 'JC-P1-2-JCC-230-016' /* LAKEWOOD - LEISURE VILLAGE (D2030) 230 KV DISCONNECT BRANCH FROM BUS 206323 TO BUS 206295 CKT 1 DISCONNECT BRANCH FROM BUS 206295 TO BUS 206276 CKT 4 SET BUS 206295 LOAD TO 0 MW END
JC-P1-2-JCC-230-014	CONTINGENCY 'JC-P1-2-JCC-230-014' /* FUTURE BREAKERS LAKEWOOD - LARRABEE (Z2026) 230 KV & LAR BK 8 & 12 DISCONNECT BRANCH FROM BUS 206323 TO BUS 206294 CKT 1 DISCONNECT BRANCH FROM BUS 206294 TO BUS 206275 CKT 12 SET BUS 206294 LOAD TO 38 MW END
JC-P1-2-JCC-230-015	CONTINGENCY 'JC-P1-2-JCC-230-015' /* LAKEWOOD - LEISURE VILLAGE (U2021) 230 KV DISCONNECT BRANCH FROM BUS 206323 TO BUS 206296 CKT 1 DISCONNECT BRANCH FROM BUS 206323 TO BUS 206282 CKT 6 DISCONNECT BUS 206999 DISCONNECT BUS 207110 DISCONNECT BUS 207010 DISCONNECT BUS 207111 END
JC-P1-2-JCC-230-013	CONTINGENCY 'JC-P1-2-JCC-230-013' /* LAKEWOOD - LARRABEE (K2011) 230 KV DISCONNECT BRANCH FROM BUS 206323 TO BUS 206294 CKT 2 END

Contingency Name	Contingency Definition
AE_P1-2 G10-WVINE	CONTINGENCY 'AE_P1-2 G10-WVINE' OPEN LINE FROM BUS 228705 TO BUS 228706 CIRCUIT 1 / END
JC-P1-2-JCC-230-002T	CONTINGENCY 'JC-P1-2-JCC-230-002T' /** OYSTER CREEK - CEDAR (S2045) 230 KV DISCONNECT BRANCH FROM BUS 206302 TO BUS 227955 CKT 1 /* 28OYSTER C 230 CEDAR 230 END
AE_P1-2 NCENT-S121	CONTINGENCY 'AE_P1-2 NCENT-S121' OPEN LINE FROM BUS 228714 TO BUS 228716 CIRCUIT 1 / END
AE_P1-2 SHERM-W VINE-A	CONTINGENCY 'AE_P1-2 SHERM-W VINE-A' OPEN LINE FROM BUS 228256 TO BUS 945730 CIRCUIT 1 / END
AE_P1-2 SHERM-W VINE-B	CONTINGENCY 'AE_P1-2 SHERM-W VINE-B' OPEN LINE FROM BUS 945730 TO BUS 228706 CIRCUIT 1 / END
PS_P1-2_2310	CONTINGENCY 'PS_P1-2_2310' /* NEW FREEDOM TO CARDIFF TIELINE TRIP LINE FROM BUS 219100 TO BUS 227900 CKT 1 /* NEW FREEDOM TO CARDIFF END
JC-P7-1-JCC-230-12	CONTINGENCY 'JC-P7-1-JCC-230-12' /* LEISURE VILLAGE-MANITOU A2027 & C2029 DISCONNECT BRANCH FROM BUS 206295 TO BUS 206297 CKT 1 DISCONNECT BRANCH FROM BUS 206297 TO BUS 206277 CKT 7 DISCONNECT BRANCH FROM BUS 206296 TO BUS 206297 CKT 1 DISCONNECT BRANCH FROM BUS 206296 TO BUS 206276 CKT 3 SET BUS 206296 LOAD TO 0 MW END
JC-P7-1-JCC-230-13	CONTINGENCY 'JC-P7-1-JCC-230-13' /* MANITOU-OYSTER CREEK 230 LINES & OYSTER GEN DISCONNECT BRANCH FROM BUS 206297 TO BUS 206302 CKT 1 DISCONNECT BRANCH FROM BUS 206297 TO BUS 206302 CKT 2 DISCONNECT BRANCH FROM BUS 206302 TO BUS 206325 CKT 1 END
JC-P1-2-JCC-230-022	CONTINGENCY 'JC-P1-2-JCC-230-022' /* MANITOU - OYSTER CREEK (O1019) 230 KV DISCONNECT BRANCH FROM BUS 206297 TO BUS 206302 CKT 2 END

Contingency Name	Contingency Definition
JC-P1-2-JCC-230-021	CONTINGENCY 'JC-P1-2-JCC-230-021' /* MANITOU - OYSTER CREEK (N1028) 230 KV DISCONNECT BRANCH FROM BUS 206297 TO BUS 206302 CKT 1 END
JC-P1-2-JCC-230-020	CONTINGENCY 'JC-P1-2-JCC-230-020' /* LEISURE VILLAGE - MANITOU (C2029) 230 KV DISCONNECT BRANCH FROM BUS 206296 TO BUS 206297 CKT 1 DISCONNECT BRANCH FROM BUS 206296 TO BUS 206276 CKT 3 SET BUS 206296 LOAD TO 0 MW END
Base Case	
JC-P7-1-JCC-230-10A	CONTINGENCY 'JC-P7-1-JCC-230-10A' /* LAKEWOOD - LARRABEE 230 KV LINES DISCONNECT BRANCH FROM BUS 206323 TO BUS 206294 CKT 2 DISCONNECT BRANCH FROM BUS 206323 TO BUS 206294 CKT 1 DISCONNECT BRANCH FROM BUS 206294 TO BUS 206274 CKT 3 DISCONNECT BRANCH FROM BUS 206294 TO BUS 206275 CKT 12 /* LARRABEE 12 FUTURE BREAKER AND A HALF SET BUS 206294 LOAD TO 38 MW /* LARRABEE 8 FUTURE BREAKER AND A HALF END
AE_P1-2 S121-G10	CONTINGENCY 'AE_P1-2 S121-G10' OPEN LINE FROM BUS 228716 TO BUS 228705 CIRCUIT 1 / END
AE_P1-2 ORCH-CARD	CONTINGENCY 'AE_P1-2 ORCH-CARD' OPEN LINE FROM BUS 228002 TO BUS 227900 CIRCUIT 1 / END
AE_P4-2 AE7	CONTINGENCY 'AE_P4-2 AE7' /*CARDIFF TO NEW FREEDOM BREAKER V DISCONNECT BRANCH FROM BUS 219100 TO BUS 227900 CKT 1 /*NEW FREEDOM TO CARDIFF 230 230 DISCONNECT BRANCH FROM BUS 227900 TO BUS 227910 CKT 1 /*CARDIFF CARDIFF 230 69 T1 END
AE_P4-2 AE6	CONTINGENCY 'AE_P4-2 AE6' /*CARDIFF TO NEW FREEDOM BREAKER W DISCONNECT BRANCH FROM BUS 219100 TO BUS 227900 CKT 1 /*NEW FREEDOM TO CARDIFF 230 230 DISCONNECT BRANCH FROM BUS 227900 TO BUS 227911 CKT 1 /*CARDIFF CARDIFF 230 69 T2 END

Contingency Name	Contingency Definition
AE_P4-2 AE9	CONTINGENCY 'AE_P4-2 AE9' /*CARDIFF TO ORCHARD BREAKER NEW1 DISCONNECT BRANCH FROM BUS 228002 TO BUS 227900 CKT 1 /* CARDIFF TO ORCHARD 230 230 DISCONNECT BRANCH FROM BUS 227900 TO BUS 227934 CKT 1 /*CARDIFF CARDIFF 230 138 T7 END
AE_P4-2 AE46	CONTINGENCY 'AE_P4-2 AE46' /*ORCHARD 230 BUS BREAKER D DISCONNECT BRANCH FROM BUS 228002 TO BUS 228310 CKT 1 /* ORCHARD TO CHURCHTOWN 230 230 DISCONNECT BRANCH FROM BUS 200063 TO BUS 228002 CKT 1 /*ORCHARD ORCHARD 500 230 T1 END

13 Short Circuit Analysis

The following Breakers are overdutied

None

14 Stability and Reactive Power

To be determined in the Facilities Study Phase.

15 Affected Systems

None

16 Attachment 1: One Line Diagram

The Interconnection Customer is responsible for contacting the Vineland Municipal Electric Utility directly for a single line drawing.

